

A torsion-free counterexample to reduced Baum–Connes injectivity

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Abstract

We disprove rational injectivity of the coefficient-free reduced Baum–Connes assembly map for torsion-free groups. Specifically, we construct a finitely generated torsion-free discrete group whose degree-zero assembly map has a kernel class of infinite order.

1 Introduction

The Baum–Connes assembly map relates the topology of a discrete group to the operator algebra generated by its regular representation. For a torsion-free discrete group G_{inj} , its coefficient-free form is

$$\mu_{G_{\text{inj}}}^r : K_*(BG_{\text{inj}}) \longrightarrow K_*(C_r^*(G_{\text{inj}})).$$

Here BG_{inj} is a classifying space, $K_*(BG_{\text{inj}})$ is complex topological K -homology with compact support, namely the directed limit over finite subcomplexes, and $C_r^*(G_{\text{inj}})$ is the reduced group C^* -algebra. The Baum–Connes conjecture asserts that this map is an isomorphism [5]. In this formulation the coefficient algebra is $\mathbb{C}$ with its trivial action; the more general conjecture with coefficients allows a G_{inj} -algebra A and has target $K_*(A \rtimes_r G_{\text{inj}})$.

Theorem 1.1. *There exist a finitely generated torsion-free discrete group G_{inj} , an injective homomorphism $i : \mathbb{Z}^2 \rightarrow G_{\text{inj}}$, and a class $c \in H^2(BG_{\text{inj}}; \mathbb{Z})$ such that*

$$\langle (Bi)^*c, [T^2] \rangle = 1, \quad i_*(\beta) = 0 \quad \text{in } K_0(C_r^*G_{\text{inj}}),$$

where β is a rank-zero Bott generator of $K_0(C_r^*\mathbb{Z}^2)$. In particular, $h = (Bi)_*[D_{T^2}]$ has infinite order in $K_0(BG_{\text{inj}})$, while its reduced analytic assembly image is zero. Here $[D_{T^2}]$ is the spin Dirac K -homology class of the oriented torus.

The theorem gives an integral reduced assembly kernel class of infinite order, and hence disproves rational injectivity of coefficient-free reduced assembly. Its two assertions have different roles. The cohomology class extends the orientation class of the embedded torus and detects its Dirac class by an index pairing. The vanishing of the corresponding rank-zero Bott class is proved by a separate operator-algebra construction. The group is finitely generated; finite presentation is not part of the conclusion.

1.1 The distinction from earlier counterexamples

Baum and Connes introduced this index-theoretic comparison in a preprint circulated in 1982 and published in 2000 [4, Section 1 and editors' note]. The formulation using the classifying space for

proper actions was developed with Higson [5, Conjecture 3.15]. In the torsion-free case that space gives the assembly map displayed above.

The conjecture is known for substantial classes of groups. Higson and Kasparov prove it with separable coefficients for groups acting properly by affine isometries on Hilbert space, including countable amenable groups [23, Theorem 9.1 and Corollary 9.2]. The coefficient-free hyperbolic case was established by Mineyev–Yu using Lafforgue’s Banach- KK methods [29, Theorem 20]; Lafforgue subsequently proved the version with coefficients [27, Theorem 0.4]. Guentner, Higson, and Weinberger prove split injectivity with coefficients for countable linear groups [20, Theorem 3]. These results describe major settings where the kernel constructed here cannot occur.

Higson, Lafforgue, and Skandalis constructed counterexamples in several assembly settings [24]. For discrete groups, their examples concern failure of surjectivity with nontrivial commutative coefficients; their paper explicitly distinguishes this from the conjecture for groups alone. Its groupoid examples also have failures of injectivity, but the groupoid and coefficient-free group assertions have different domains and targets. Theorem 1.1 concerns injectivity with coefficient algebra $\mathbb{C}$.

Gromov’s graph-labeled small-cancellation strategy connects expander geometry to group constructions [16, §4.8]. Graphical small-cancellation methods subsequently gave further group constructions with coefficient obstructions. For example, Osajda’s graphical constructions, together with the coefficient result of Willett and Yu, give a suitable coefficient algebra for which reduced assembly is injective and not surjective [33, Corollary 3.4][36, Corollary 1.7]. This provides relevant geometric precedent without supplying the coefficient-free kernel required here.

The exact crossed-product reformulation of Baum, Guentner, and Willett was designed to address the failures involving coefficients [6]. It replaces the reduced crossed product by the minimal exact Morita-compatible crossed product. That reformulation, the reduced coefficient-free conjecture, and the maximal assembly map must be kept distinct when interpreting the present construction.

1.2 Maximal assembly and the Novikov conjecture

Write $\mu_{G_{\text{inj}}}^{\max}$ for assembly with target $K_*(C_{\max}^*(G_{\text{inj}}))$, and let $\lambda_{G_{\text{inj}}}$ be the regular quotient from $C_{\max}^*(G_{\text{inj}})$ to $C_r^*(G_{\text{inj}})$. By the completion comparison [28, Remark 2.6],

$$\mu_{G_{\text{inj}}}^r = (\lambda_{G_{\text{inj}}})_* \mu_{G_{\text{inj}}}^{\max}.$$

Hanke and Schick prove maximal rational nonvanishing for every discrete group whenever a K -homology class is detected by the subring of rational cohomology generated in degrees at most two [22, Theorem 1.2]. Consequently the class h in Theorem 1.1 satisfies

$$\mu_{G_{\text{inj}}}^{\max}(h) \otimes 1 \neq 0, \quad (\lambda_{G_{\text{inj}}})_*(\mu_{G_{\text{inj}}}^{\max}(h)) = 0.$$

The name *strong Novikov conjecture* is used for both maximal and reduced rational injectivity in the literature [5, Section 7][6, Section 8.3, footnote 6]. Theorem 1.1 disproves the reduced formulation, while maximal assembly detects the same class h . The classical Novikov conjecture concerns oriented homotopy invariance of higher signatures. Our result disproves neither maximal rational injectivity nor this classical assertion; in particular, higher signatures associated to low-degree cohomology already have the homotopy invariance established in [22, Corollary 1.3].

Antonini, Buss, Engel, and Siebenand extend the low-degree nonvanishing result to suitable exotic group C^* -algebras [1, Theorem B]. Their corrected statement assumes a *finitely presented* group and applies, in particular, to the group algebra of the minimal exact Morita-compatible crossed-product functor. For exact groups this algebra agrees with the reduced group algebra.

The asserted agreement for all groups was affected by the gap recorded in the erratum of Buss, Echterhoff, and Willett [10]. Accordingly we use the corrected version of [1], arXiv:1905.07730v3, rather than its original reduced-algebra claim. Theorem 1.1 asserts finite generation; it supplies neither finite presentation nor exactness.

A further distinction concerns projections. For a countable torsion-free group, a nontrivial projection in the scalar algebra $C_r^*(G_{\text{inj}})$ would obstruct surjectivity of coefficient-free assembly [26, Corollary 1.6]. Such a projection alone does not identify an assembly kernel class. Here matrix-valued Bott representatives are used to prove a specific vanishing while the integral detector proves that the corresponding geometric class survives.

1.3 The construction

In the operator model, localized rank-zero Bott representatives occupy coordinates indexed by heights $i \geq 0$. The analytic aim is to cancel the class at height zero by moving bounded families of these representatives. Rotations and rescalings on the disjoint adjacent pairs $(0, 1), (2, 3), \dots$ move the even family to the odd family. The pairs $(1, 2), (3, 4), \dots$ then move the odd family to the positive-even family. These two homotopies identify the even family with the family obtained by removing its height-zero term. Finite additivity therefore forces that term to vanish. The families must be actual bounded operators, their homotopies must belong to a reduced-algebra norm quotient, and the resulting equality must yield a finite witness in one group algebra. These requirements determine the construction below.

The proof connects three ingredients. Its graph-labeling stage is related to the finite-alphabet separation method of Osajda [33, §2]; the layered graph, the sequential expansion constraints, and the simultaneous spectral requirements below are additional features. The voltage-cover language follows Gross and Gross–Tucker [17, 18]. Neither reference supplies the analytic realization needed here.

First, a layered graph is labeled by interleaved finite streams. Two incident labels recover successive blocks of a vertex state. The recovery procedure gives uniqueness for sufficiently long immersed words, while the random tables can be selected with a uniform expansion gap. Products of these incidences and their adjoints will later isolate the normalized constant coordinates on which the Bott families act. Pruning removes short coherent returns and configurations that would obstruct the required overlap bounds. The central issue is to retain enough independent constraints after conditioning on expansion. Proposition 2.6, Proposition 3.4, and Proposition 4.14 give the labeling, selection law, and counting estimates with compatible parameters.

Second, the graph is given voltages in the integral Heisenberg group

$$\mathbf{H} = \langle x, y, z \mid [x, y] = z, [x, z] = [y, z] = 1 \rangle.$$

A voltage assigns a group element h_e to an oriented base edge $v \rightarrow w$; its lift sends (v, f) to (w, fh_e) , and the reverse edge uses h_e^{-1} . Short circuits surviving pruning are not laws in $\mathbf{H}$. The voltage choice simultaneously separates these circuits and approximates the required spectral averages. Continuous matrix-valued symbols spread the Bott class across spectral bands, and finite sampling is controlled in the full regular operator norm. Proposition 5.2 supplies this analytic selection. The Heisenberg group provides a torsion-free source for the voltage cover and also contains the commuting pair (z, x) used for the final torus.

Third, the geometric realization keeps the Heisenberg subgroup embedded and makes its degree-two detector extend to the ambient group. A *sheet* is a translate of the embedded cover in the Cayley graph. Integral cycle decompositions in a complex built from these sheets control both torsion and restriction in cohomology; see Proposition 6.4 and Proposition 7.11. Disjoint domains

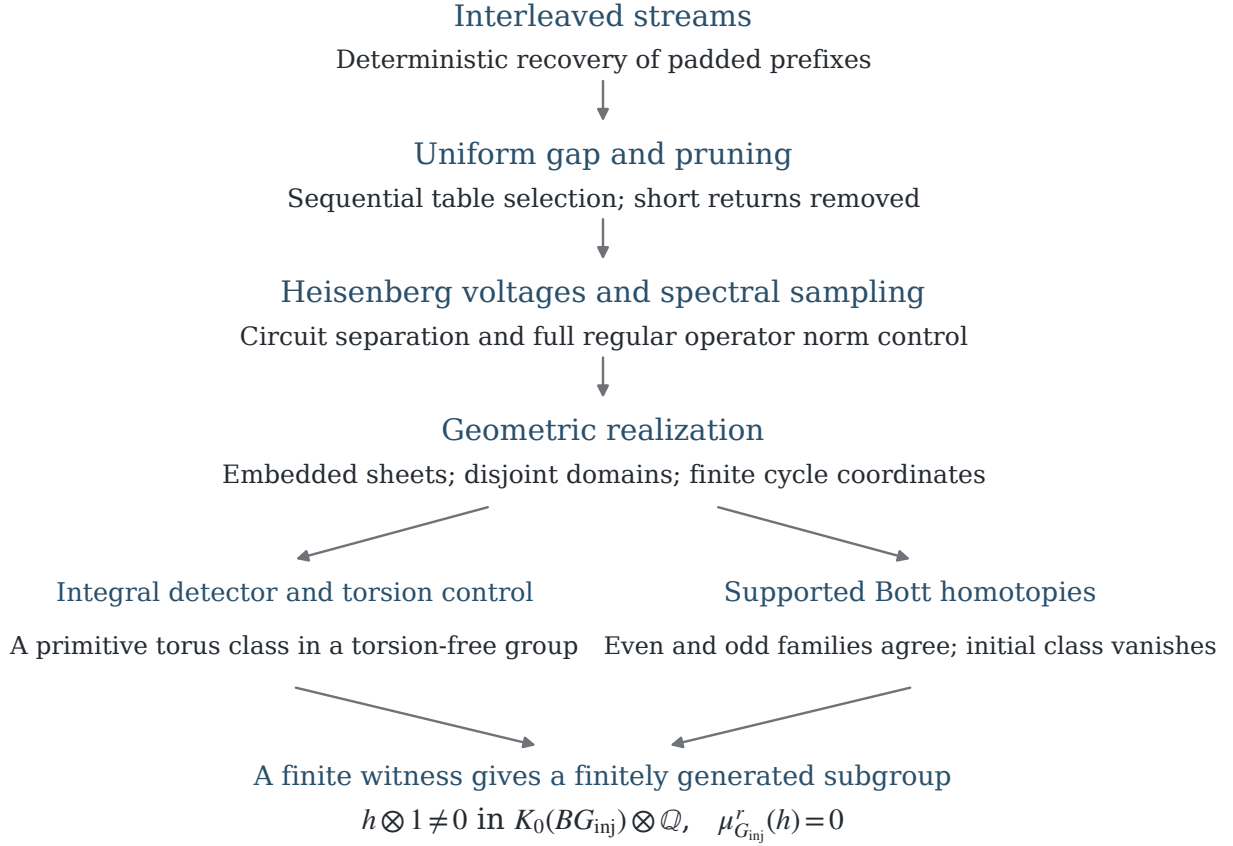


Figure 1: Principal dependencies in the construction. The stream rules, sequential selection, and pruning precede the joint voltage and spectral choice. Geometric realization supports both the integral detector and the reduced Bott cancellation. A finite witness retains both properties in a finitely generated subgroup.

in the sheets allow the spectral operators to act on the group’s regular Hilbert space. The number of excluded coordinates in each sheet is bounded, and the spectral support projections have traces small enough relative to these bounds that the loss on the supported operators is small in norm. The remaining Cayley edges form a forest, whose contribution is controlled by the ratio of the square root of the label count to the incidence normalizer. This is why the finite alphabet must retain a sharp margin throughout the graph construction. These estimates realize the bounded Bott families and their homotopies in the reduced-algebra norm quotient. Proposition 8.8 then obtains integral Bott vanishing in an actual reduced group algebra. The finite-witness argument retains this equality in the group generated by the finitely many group elements in polynomial approximations to that witness, together with the Heisenberg generators. This finitely generated subgroup is the group in Theorem 1.1.

Figure 1 records the main dependencies.

1.4 Reading the construction

There are two indices: height runs through one infinite graph tower, while accuracy indexes a sequence of increasingly precise constructions. The graph degree is constant across heights in any

one construction and may increase with accuracy. Section 9 records the complete order of parameter choices after the required objects have been defined.

Sections 2–5 construct and sample the labeled voltage tower. Sections 6–7 realize it in a group and prove the cohomological detector and torsion claims. Section 8 establishes the reduced Bott cancellation and extracts a finite witness. The final section puts these conclusions together with compatible parameters and identifies the detected torus Dirac class and its assembly image.

Throughout, all group C^* -algebras and all K -groups are complex. Commutators mean $[a, b] = aba^{-1}b^{-1}$. The graph construction uses binary logarithms. Matrix dimensions remain fixed along each approximation sequence in which quotient K -theory is used; finite generation is extracted only after an exact finite witness has been obtained.

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2 A finite alphabet with deterministic recovery

We construct a labeled graph in layers, with the aim that every sufficiently long immersed word has only one occurrence. The alphabet must also remain small relative to the incidence degrees. The finite-alphabet separation objective is shared with Osajda's construction [33, §§2.1–2.3, Theorem 2.7]; here the graph and its labels are built together, and uniqueness will follow from explicit recovery of vertex coordinates.

The local recovery step is elementary. Over a finite field, subtracting two evaluations $E_1 = v + c_1 u$ and $E_2 = v + c_2 u$ with $c_1 \neq c_2$ determines u ; either equation then determines v . We apply this observation block by block. Labels and previously recovered neighboring blocks will supply the evaluations. The rules below make this process compatible with partial terminal blocks and with comparisons between different heights.

A *slot* is a named candidate incidence at a vertex. Once the coefficient pools are fixed, the incidence, recovery, and vertex-deletion assertions in this section hold for every choice of permutation tables. Deletion fractions are counted using uniform vertex coordinates for each fixed table realization. The probability law on the tables is a separate choice, made in Section 3. All logarithms in this section have base two.

The padded-stream construction and recovery scheme adapt [32, Sections 4 and 6]. The exponents, cutoff requirements, and gated corrections needed for the present interfaces are established below; the projection theorem is not an existence input.

2.1 States, families, and extra coordinates

Fix a positive integer b . Let s be a sufficiently large multiple of 20000 and $5b$, and put

$$r = \frac{3}{5}s, \quad d = 2^{40005s}, \quad D = 2^{60005s}, \quad q = 2^{50005s}. \quad (2.1)$$

The numbers d, D will be the two incidence degrees in a fixed slice, and $q = \sqrt{dD}$ their geometric mean. The exponent choices allow both distinct coefficients for recovery and an alphabet whose square root is small compared with q ; the precise alphabet estimate is (2.20) below. These parameters stay fixed as the height i varies. They may change when a later construction asks for a smaller approximation error. Choose an integer $n_0 \geq 10^6$, to be increased later. For $i \geq 0$ write $bi = 2sa_i + t_i$, where $0 \leq t_i < 2s$, and define

$$\begin{aligned} \ell_A(i) &= r + n_0s + a_i s + \max\{t_i - s, 0\}, \\ \ell_B(i) &= n_0s + a_i s + \min\{t_i, s\}, \\ \mathcal{A}_i &= \mathbb{F}_2^{\ell_A(i)} \times \mathbb{F}_2^{r+\ell_B(i)}, \quad \mathcal{B}_i = \mathbb{F}_2^{\ell_B(i)} \times \mathbb{F}_2^{s-r+\ell_A(i)}. \end{aligned} \quad (2.2)$$

The two coordinates are denoted (u, v) . Regard each as an infinite bit stream by adjoining zero coordinates. Both coordinates on a side use that side's block convention: an A stream has an initial block A_0 of length r and blocks $A_1, A_2, \dots$ of length s ; a B stream has blocks $B_1, B_2, \dots$, all of length s . Index their interleaving by

$$A_0 \ (0), \ B_1 \ (1), \ A_1 \ (2), \ B_2 \ (3), \ A_2 \ (4), \dots \quad (2.3)$$

For the pair of spaces at height i , the active coordinates of u_A and u_B together form an initial segment of this interleaving. Its length and the logarithmic state size are

$$\begin{aligned} L_i &= r + 2n_0s + bi, \quad H_i = L_i + s - r = (2n_0 + 1)s + bi, \\ |\mathcal{B}_i| &= 2^{H_i}, \quad |\mathcal{A}_i| = 2^{H_i + 2r - s}. \end{aligned} \quad (2.4)$$

Set $m_i = H_i / \log q$. We require, by increasing s ,

$$a := \frac{b}{\log q} \leq 10^{-6}. \quad (2.5)$$

Let $\mathcal{T}$ be any fixed finite set of incidence types. A type τ specifies an offset $\delta_\tau \in \{-1, 0, 1\}$ and occurs between $\mathcal{A}_i$ and $\mathcal{B}_{i+\delta_\tau}$ whenever both indices are nonnegative. Put $T = |\mathcal{T}|$ and

$$M = 2 \cdot 10^6 s. \quad (2.6)$$

The family of this pair is $f = (\tau, i \bmod M)$. Thus there are at most TM families. Its two endpoint residues, its smaller height $h = \min\{i, i + \delta_\tau\}$ modulo M , and the phase of its cutoff are determined by f .

For an adjacent pair, truncate the larger endpoint to the coordinate lengths at height h . Exactly b coordinates are discarded: they belong to its u coordinate if that side's u stream grows from h to $h+1$, and otherwise to its v coordinate. They form one interval of b bits, since s, r and every active length are multiples of b . Denote their value by $e \in \mathbb{F}_2^b$. For an equal-height pair let e be the empty string. In a fixed e slice, both truncated endpoint spaces are exactly the spaces at height h in (2.2).

At an A endpoint a slot has serial number $j \in \{1, \dots, d\}$; at a B endpoint it has, in addition, a bucket $c \in \mathbb{F}_2^{2r-s}$. The type is always part of a slot name. At the smaller endpoint of an adjacent pair, e is also part of its slot name. At the larger endpoint it is read from the state. Every positive incidence is oriented from its $\mathcal{A}$ endpoint to its $\mathcal{B}$ endpoint. A positive edge label will be

$$\lambda = (f, j, e, h_0), \quad h_0 = c \| h'_0, \quad |h_0| = r, \quad |h'_0| = s - r. \quad (2.7)$$

Negative labels are formal inverses. The notation h_0 here is a label head, not a height.

For each family, mark its common u cutoff L_h at every actual occurrence. Let κ be the source-block index of the last active common u block and $t \in \{b, 2b, \dots, s\}$ its active prefix length. Also mark the interval of extra coordinates at each adjacent occurrence. For consecutive occurrences of one family both side lengths increase by $Mb/2$, and the indices κ increase by

$$K = Mb/s = 2 \cdot 10^6 b. \quad (2.8)$$

The marked prefix length and its side therefore remain the same within a family. The raw u data of its actual endpoint pair extend at most one source block beyond its common cutoff. An extra v interval lies at most two source blocks beyond that cutoff, when its coordinate blocks are indexed by (2.3).

2.2 Finite coefficient pools

An A coefficient name is (f, j, e) , and a B coefficient name is (f, j, c, e) . We include every legal value, including values of e read from the larger state. In particular an A name does not contain the bucket, which has not yet been read when its label head is computed. Let $\mathcal{N}_A, \mathcal{N}_B$ be these two finite sets. Their sizes satisfy

$$|\mathcal{N}_A| \leq TM2^b d, \quad |\mathcal{N}_B| \leq TM2^b D. \quad (2.9)$$

Choose an injection of $\mathcal{N}_A$ into $\mathbb{F}_{2^r}$ and an injection of $\mathcal{N}_A \sqcup \mathcal{N}_B$ into $\mathbb{F}_{2^s}$. Identify these fields linearly with their bit-coordinate spaces. The first injection acts on block A_0 ; the second acts on every full s -block. For a name ν , write K_ν for the block-diagonal map that multiplies each block by its assigned field element, expressed in bit coordinates. The same coefficients are used on every

repeated block. The finite fields used here introduce no group elements: for each h , the roots of $z^{2^h} - z$ in a splitting field over $\mathbb{F}_2$ form a field of 2^h elements, since the polynomial has derivative -1 and its roots are closed under the field operations.

Distinct coefficients make the two-evaluation recovery step possible. Expansion requires a second property: after restricting to an active prefix, no output functional should give the same input functional on too many slots. The next lemma supplies this multiplicity bound as well as distinctness; Section 3 uses it to estimate the mean incidence operator.

Lemma 2.1 (Coefficient choice). *For sufficiently large s , the injections can be chosen so that, within every fixed family and extra slice and on either side, the following holds. If n is its number of slots, ζ is any nonzero output linear functional on an ordinary s -block, and $b \leq t \leq s$, then each restricted input functional*

$$(\zeta K_\nu)|_{\mathbb{F}_2^t}$$

occurs on at most $2^{1-b}n$ slots. All distinct names have distinct coefficients in every field in which they occur. The analogous bound holds on the initial A field for every prefix length between b and r .

Proof. The injections exist because the polynomial factor TM and fixed factor 2^b in (2.9) are dominated by the positive exponent gaps $r - .40005s$ and $s - .60005s$. Choose both injections uniformly. For fixed nonzero ζ , multiplication identifies the field coefficients bijectively with all output-to-input functionals. On any specified set of n names these are therefore sampled without replacement. Restriction to t input bits has fibers of proportion $2^{-t} \leq 2^{-b}$.

Here is the required tail estimate. For n zero-one samples without replacement, with sum S and mean success proportion p , one has

$$\mathbb{E} \exp(u(S - np)) \leq \exp(nu^2/8).$$

Indeed the exponential moment before centering is an elementary symmetric mean in the population entries $1, e^u$. Replacing two unequal nonnegative entries by their average increases every elementary symmetric sum: the only changing term is their product multiplied by a nonnegative symmetric sum in the other entries. Iteration compares the moment with sampling with replacement. For a centered Bernoulli variable, the second derivative of the logarithmic moment generating function is a tilted variance, at most $1/4$; integrating twice proves the displayed bound. Exponential Markov inequality gives

$$\Pr\{S - np \geq 2^{-b}n\} \leq \exp(-2^{1-2b}n).$$

There are at most $s2^{2s}$ choices of an output functional, prefix length, and restricted value. Multiplying by the polynomial number of families and the fixed number of extra slices and sides still gives $2^{O(s)}$ tests. Each has $n \geq d = 2^{.40005s}$. Their total failure probability tends to zero. Fix a choice for which none fails. $\square$

2.3 Universal corrections and permutation rules

The edge equations will couple each affine evaluation to a permuted input block at the opposite endpoint. To compare occurrences at different heights, we use one system of equations on arbitrary padded streams. The corrections remove the extra coordinates of an adjacent-height pair, while the gates and identity blocks preserve the padded tail. Their dependence on earlier raw inputs is what will let recovery proceed one block at a time.

We now specify these rules on arbitrary padded raw u_A, u_B streams, not only on actual endpoint states. A pair of full raw blocks means (B_j, A_j) for $j \geq 1$. For a source-block index $k > 20$, put

$J(k) = \lfloor (k-5)/2 \rfloor$ and let

$$g_k = 1 \iff (u_{B,J(k)-3}, u_{A,J(k)-3}, \dots, u_{B,J(k)}, u_{A,J(k)}) \neq 0. \quad (2.10)$$

For $k \leq 20$ put $g_k = 1$. Thus every noninitial gate reads exactly eight full raw blocks and reads no source-block index greater than $k-5$. In particular a correction to an evaluation block cannot consult the source block that will be recovered from that evaluation.

Fix a positive label (2.7). At every marked extra interval of its family, XOR e into that raw coordinate interval if $g_k = 1$, where k is the coordinate-block index containing the interval. There are no corrections for an equal-height family. Denote the resulting streams by

$$\bar{u}_C = u_C + C_{C,\lambda}^u, \quad \bar{v}_C = v_C + C_{C,\lambda}^v, \quad C \in \{A, B\}. \quad (2.11)$$

The sums are XOR sums; each finite prefix uses finitely many disjoint marked intervals. Every correction offset in a block is determined by the label and raw u blocks strictly before both blocks coupled by the edge equation below. This extra lag is deliberate.

For every family, label, source-block index k , and tuple of all raw u blocks strictly earlier than k in (2.3), provide a permutation table. At an unmarked index it is a permutation of the full block; at a marked cutoff with active length t it is a permutation of $\mathbb{F}_2^t$. These are distinct variables for distinct complete keys. There are finitely many variables at each source-block index.

The transformation at a source block is defined as follows.

1. At each marked cutoff use its prefix table on the first t bits and the identity on the suffix, if $g_k = 1$; if $g_k = 0$, use the identity on the entire block.
2. At the twenty source blocks immediately following every marked cutoff, use the identity unconditionally.
3. At every other source block use its full-block table if $g_k = 1$, and otherwise the identity.

The marked and following intervals do not overlap by (2.8). Tables act on corrected $\bar{u}$ blocks, while their keys and gates read raw preceding u blocks. Write $P_{A,\lambda}(\bar{u}_A)$ and $P_{B,\lambda}(\bar{u}_B)$ for the two transformed streams. For a fixed slot write $Q_C = \bar{v}_C + K_\nu \bar{u}_C$ for its corrected evaluation; the slot and label subscripts will usually be suppressed. Its difference from the raw evaluation $v_C + K_\nu u_C$ is the known correction offset $C_{C,\lambda}^v + K_\nu C_{C,\lambda}^u$. The universal edge equations are

$$\boxed{\bar{v}_A + K_{(f,j,e)} \bar{u}_A = h_0 \| P_{B,\lambda}(\bar{u}_B), \quad \bar{v}_B + K_{(f,j,c,e)} \bar{u}_B = h'_0 \| P_{A,\lambda}(\bar{u}_A).} \quad (2.12)$$

They are equations of infinite padded streams.

For the auxiliary incidence at an actual pair, modify these rules at that occurrence only: force its extra correction on, if there is one, and force its current final-prefix permutation on. All lower rules retain their gates. Denote this auxiliary prescription by a superscript 0. We shall delete states for which either forced choice was necessary. The sparse unconditional identity zones replace any need for a post-cutoff gate that could inadvertently read the raw extra bits.

2.4 Auxiliary incidences and uniform coordinates

The auxiliary prescription gives a finite incidence at every height pair. We first verify that its equations preserve the required supports, then solve them in either direction. The same inversion will identify the uniform coordinates used to count gate failures.

Lemma 2.2 (Inversion and uniform coordinates). *For every choice of tables the auxiliary prescription has exactly one edge at every specified endpoint and slot. In a fixed extra slice it is biregular of degrees d, D . A uniform endpoint with a fixed slot, after fixing its extra value when necessary, is parametrized bijectively by the free label head and all free raw u_A, u_B coordinates at the common cutoff. Consequently those raw coordinates are independent uniform bits.*

Proof. Preservation of the common support. At the actual larger endpoint the forced correction cancels its extra bits. Lower correction intervals lie in active coordinates. Higher correction gates vanish: their eight preceding raw blocks are wholly beyond the actual endpoint support, because consecutive marks are separated by $K > 20$ source blocks. Thus corrected streams have exactly the common permitted lengths.

The transformed corrected u streams preserve these lengths. At the last common block the suffix is the identity. The next twenty blocks are identity, and their corrected inputs are zero. Raw u support extends by at most one block beyond the common cutoff. At every later ordinary index the gate window (2.10) is entirely beyond that support, so its gate is closed. The same holds at all higher marked rules. This proves preservation of the padded tail, including an extra interval in the larger endpoint's u coordinate.

For clarity, the two possible common terminal shapes are

	ℓ_A	ℓ_B	$ v_A $	$ v_B $	
last B prefix t	$r + ns$	$ns + t$	$r + ns + t$	$(n + 1)s$	
last A prefix t	$r + ns + t$	$(n + 1)s$	$r + (n + 1)s$	$(n + 1)s + t$	(2.13)

In both rows $1 \leq t \leq s$ and the output of each block-diagonal K fits in the displayed v length. If an extra interval is in v , it lies beyond this output: its same-side direct block is zero there. Thus computing v from the equations cannot leak a Ku term into the prescribed extra interval.

Inversion from either endpoint. Given an A endpoint and slot, its extra value is known, either from the state or from the slot. Its first r evaluation bits determine h_0 ; there are no marked corrections in the initial blocks. Hence c and the B coefficient name are known. Proceed through the B blocks. To compute its next corrected evaluation from the A endpoint, all correction gates use already known raw blocks by their five-index lag. The B permutation key uses only earlier B blocks and the preceding A block. Invert that known block permutation, then undo the known correction to recover the next raw B u block. The other equation now determines its raw v block. Continue in order. The support argument just given shows that all recovered streams have the required lengths.

Conversely a B slot gives c . Its first $s - r$ evaluation bits give h'_0 , and hence h_0 . Inverting successive A block permutations recovers raw u_A and then v_A . These two procedures solve the same triangular equations and are inverse. The lag is important here: a correction in $Q_{B,j}$ does not read the unknown $u_{A,j-1}$, and one in $Q_{A,j}$ does not read the unknown $u_{B,j}$.

Uniform coordinates. First freely specify the raw u blocks on both sides, omitting only the larger endpoint's fixed extra interval if that interval is in u . Specify h_0 for an A slot, or h'_0 for a B slot whose bucket is fixed. Every gate, correction, and permutation value is now determined. Equations (2.12) determine both $\bar{v}$ streams, and undoing the corrections determines both raw v streams. A forced extra interval in v is exactly e . The number of free coordinates is

$$\ell_A(h) + \ell_B(h) + r = \log |\mathcal{A}_h| \quad \text{or} \quad \ell_A(h) + \ell_B(h) + s - r = \log |\mathcal{B}_h|,$$

respectively. The preceding inversion proves bijectivity, not merely equality of dimensions. Uniform measure therefore becomes product uniform measure in these coordinates. $\square$

2.5 Deleting gate failures

Let X^0 be the union of the auxiliary incidences over all types and height pairs. At a slot, evaluate the natural gates on its auxiliary edge, even where the auxiliary prescription forced them on. If its common last source-block index is κ , delete an endpoint state if any of the following holds for any incident slot:

1. One of $g_\kappa, g_{\kappa-1}, g_{\kappa-2}$ is zero.
2. The natural gate at its actual extra correction interval is zero. This test is omitted for equal-height pairs.
3. Among the full active A source blocks other than A_0 , more than $1/250$ of their gates are closed; or the same holds among the full active B source blocks. These are separate tests for the two parities.

The last test counts ordinary gate closures; deterministic marked and following identity blocks will be accounted for separately.

Lemma 2.3 (Uniform gate budget). *For every fixed set of tables, the fraction deleted for gate reasons from either state space at any height is at most*

$$\varepsilon_{\text{gate}} = 600T2^bD2^{-8s}. \quad (2.14)$$

On a surviving edge both auxiliary forced choices were unnecessary, so the edge satisfies the universal padded equations (2.12). At each surviving slot, separately among its full A source blocks and its full B source blocks, fewer than .005 of the updates are unavailable because of a closed gate or a marked or following deterministic identity rule. The last three source gates and the actual prefix gate are open.

Proof. Fix a slot and, where necessary, condition on its actual extra value. Every gate tested at or before the common cutoff, and the gate of the extra interval at most two blocks later, reads eight full free raw u blocks. They are independent uniform by Lemma 2.2. Thus each noninitial gate is closed with probability exactly 2^{-8s} . Initial gates are never closed. Each parity in the third test has at least n_0 full active blocks. Linearity of expectation and Markov's inequality give probability at most 2502^{-8s} for each of its two failures. The first two tests add at most 42^{-8s} . There are at most $T2^bD$ slots at either endpoint, since $D \geq d$. A union bound proves (2.14), with room in the constant. Averaging over the extra value gives the same bound at a larger endpoint.

The first two tests refer only to raw coordinates obtained before the forced choices can matter. If they pass, the auxiliary and universal rules coincide along this edge; the padding argument of Lemma 2.2 applies to both ends. Restricting to edges with two surviving endpoints therefore gives a subgraph satisfying the universal equations.

For either source parity, the fraction of full active blocks belonging to marked or following twenty-block zones is at most

$$\frac{22}{K} + \frac{22}{n_0} < 10^{-4}.$$

Indeed these zones have at most eleven indices of each parity, are separated by K interleaved indices, and there is at most one boundary zone in the count. The displayed larger constants cover both endpoints of the counting interval. Together with the separate $1/250$ gate bound this is less than .005. A marked partial block is counted as unavailable for a full- s -bit test even if its prefix gate is open. The current cutoff and its preceding two indices are not in a previous marked zone, by (2.8). $\square$

2.6 Recovery of arbitrary padded prefixes

The gate deletion has made every surviving edge satisfy the same universal equations. We can now compare two occurrences of one label word, even when their vertex states have different active lengths. The distance of a position from the word's ends determines how much of its padded state can be recovered.

For an A position put

$$\delta_A(j) = (2j+1)s \quad (j \geq 0), \quad \delta_B(j) = 2js \quad (j \geq 0). \quad (2.15)$$

Depth $\delta_A(j)$ means both raw A streams through $r+js$ bits; depth $\delta_B(j)$ means both raw B streams through js bits. In particular B depth zero contains no data. The available common full depth of a state, using only blocks in which both streams are active, belongs to $[H_i - 2s, H_i]$. These conventions have the useful exact identities

$$\delta_A(j) - \delta_B(j) = s \quad (j \geq 0), \quad \delta_B(j) - \delta_A(j-1) = s \quad (j \geq 1). \quad (2.16)$$

No r -dependent rounding error occurs when a coherent junction is trimmed by one edge at each end.

The radius of a position on a path is its smaller distance in edges to the two endpoints. A path is immersed if it does not immediately traverse an edge and then its reverse.

Lemma 2.4 (Padded-prefix recovery). *Let two immersed paths in the gate-surviving graph have the same oriented label word. At corresponding A positions of radius at least $2j+1$, both raw streams agree through $r+js$ bits. At corresponding B positions of radius at least $2j$, both raw streams agree through js bits. These statements compare padded coordinates, so the positions may have arbitrarily different heights.*

Proof. The label orientation identifies the side, and the complete label identifies the family, both coefficient names, the extras, and every universal rule. At an interior vertex the two incident coefficient names are distinct. Otherwise, at A their family, serial and actual extra agree, and at B their bucket agrees as well; they are the same slot. The unique-edge assertion in Lemma 2.2 would then make the path backtrack.

For either incident name ν , rearrange its current corrected evaluation block as

$$v + K_\nu u = R_\nu + C_\nu^v + K_\nu C_\nu^u. \quad (2.17)$$

If the right sides and correction offsets are known for the two names, their difference recovers the entire raw u block because the difference of the field coefficients is nonzero. Then either equation recovers the raw v block. The argument applies to full padded blocks even when an actual endpoint has only a partial active block.

At an A position of radius one, the two first evaluations are precisely the two label heads, and the initial correction offsets vanish. This recovers A_0 . Suppose next that $j \geq 1$. At a B position of radius $2j$, each neighbor has A radius at least $2j-1$, so its raw streams are known through block A_{j-1} by induction. The central B streams are already known through B_{j-1} . To compute the required right side of the second equation in (2.12) through js bits, one needs only the A transformed stream through

$$js - (s - r) = r + s(j - 1).$$

Every key for its last needed source block uses A blocks strictly before A_{j-1} and central B blocks through B_{j-1} . Its argument is the already recovered full corrected A_{j-1} block. Its gates and all correction offsets use still earlier raw blocks by (2.10). Thus the two right sides are known, and (2.17) recovers B_j .

raw block	A_0	B_1	A_1	B_2	A_2	...
width	r	s	s	s	s	
source index	0	1	2	3	4	...
path radius	1	2	3	4	5	...

$r=3s/5$; blocks shown to scale in width

Recover B_j at radius $2j$

$$A_{j-1} \xrightarrow{P_{A,j-1}} B_j$$

raw key: $A_{<j-1}, B_{\leq j-1}$

Recover A_j at radius $2j+1$

$$B_j \xrightarrow{P_{B,j}} A_j$$

raw key: $A_{\leq j-1}, B_{<j}$

Each key also fixes the family, label, and source index.

Arguments use known corrected $\bar{u}$; gates and offsets read earlier raw u .

Two distinct incident coefficients give two known evaluations:

$$v + K_\nu u = R_\nu + C_\nu^\nu + K_\nu C_\nu^u \implies \text{recover } u, \text{ then } v.$$

Figure 2: Interleaving and the recovery induction of Lemma 2.4. The radius shown suffices to recover both raw streams through that side's indicated block. Each arrow supplies an evaluation from a recovered neighbor block; two incident coefficients recover the receiver's u, v . Key abbreviations denote raw u blocks. At the first B step, h'_0 supplies the initial $s - r$ evaluation bits. Tables act on corrected arguments; keys and gates read earlier raw inputs, with the lag in (2.10).

At an A position of radius $2j + 1$, both neighbors have B radius $2j$. The first edge equation through $r + js$ uses the B transformed stream through js . Its last argument is the already recovered corrected B_j block, and its keys use central A blocks only through A_{j-1} and earlier B blocks. Again the correction lag makes every offset known. Subtraction recovers A_j .

This induction is an induction on individual source blocks, not on whole A–B pairs. At each step both table keys and table arguments are known before any current output is used. The universal rules are identical in the two occurrences, including at marked cutoffs lying beyond one actual state's support. Padding therefore causes no exception to the induction. $\square$

Figure 2 summarizes this induction.

2.7 Separating all heights

Recovery will determine the complete state at a midpoint once a word is long enough. It does not yet identify the height: a shorter state might agree with the padded prefix of a longer one. The following deletion excludes precisely the zero extensions needed for that ambiguity.

Put $\eta_{\text{sep}} = 1/2500 = .0004$. For a vertex on side C at height i , and each $i' < i$ with $i' \equiv i \pmod{M}$, let

$$\Delta = \ell_C(i) - \ell_C(i') = (i - i')b/2, \quad q(i, i') = \lfloor \min\{\eta_{\text{sep}}H_{i'}, \Delta\} \rfloor. \quad (2.18)$$

Delete the vertex if its $q(i, i')$ raw u_C bits immediately following the smaller cutoff $\ell_C(i')$ are all zero. These are active bits at height i . Denote by X^{good} the graph induced from X^0 after the gate and separation deletions.

Lemma 2.5 (Uniform separation budget). *The fraction deleted for separation from either side at any height is at most*

$$\varepsilon_{\text{sep}} = 2 \left(\frac{2^{-Mb/2}}{1 - 2^{-Mb/2}} + \frac{2^{-\eta_{\text{sep}}H_0}}{1 - 2^{-\eta_{\text{sep}}b}} \right). \quad (2.19)$$

Proof. Uniform raw vertex coordinates are independent bits. The probability of the test for i' is $2^{-q(i, i')}$. The elementary inequality

$$2^{-\lfloor \min(x, y) \rfloor} \leq 2(2^{-x} + 2^{-y})$$

splits its bound into a term depending on the smaller height and one depending on the height difference. Summing the latter over positive multiples of M , and the former even over all nonnegative smaller heights, gives (2.19). Both sums converge uniformly in the larger height. $\square$

Proposition 2.6 (The deterministic labeled tower). *For every fixed finite type set and every sufficiently large s as above, the construction has the following properties.*

1. *The auxiliary state sizes are (2.4). In a fixed extra slice an incidence has degrees d, D . Without slicing, its smaller side acquires the extra factor 2^b and its larger side does not. The total degrees are at most $T2^bd$ on A and $T2^bD$ on B .*
2. *The labeling is folded: no two outgoing oriented edges at a vertex have the same label. Its signed alphabet satisfies*

$$|\mathcal{S}| \leq 2TM2^bd2^r, \quad \frac{\sqrt{|\mathcal{S}|}}{q} \leq 2^{-.000025s + O(\log s) + O_{T,b}(1)} \longrightarrow 0. \quad (2.20)$$

3. *At every height the fraction of removed vertices on each side is at most $\varepsilon_{\text{gate}} + \varepsilon_{\text{sep}}$. This bound can be made arbitrarily small, uniformly over heights and all table choices. The separate-parity and terminal gate conclusions of Lemma 2.3 hold.*
4. *Every immersed path in X^{good} starting at height i and having length at least $\lceil 1.002m_i \rceil$ is the unique immersed path with that oriented word anywhere in X^{good} . This is a global assertion over all heights and components.*

Proof. The degree statements follow from Lemma 2.2. At a larger endpoint, a fixed state specifies its extra value; at a smaller endpoint that value ranges over its slots. At an A endpoint equal labels specify equal serial, family, and extra, hence the same slot. At B they also specify the bucket. Uniqueness of the edge at a slot proves foldedness, and restriction to an induced subgraph preserves it. Counting (2.7) gives (2.20); its exponent uses $\log d + r = 1.00005s$, not a coarser word-counting bound. The deletion assertions are Lemmas 2.3 and 2.5.

For fixed b and incidence types, the label count has logarithm at most $1.00005s + O(\log s)$, whereas $\log_2(q^2) = 1.0001s$. This strict margin persists after fixed finite refinements of the labels. It will make the weighted forest error tend to zero in Lemma 8.4.

It remains to prove global uniqueness. Let $N = \lceil 1.002m_i \rceil$ and compare the first N edges of two paths with the same oriented word. Their midpoint radius is at least $\lfloor N/2 \rfloor$. If the first midpoint has height j , then

$$H_j \leq H_i + bN \leq (1 + 1.002a)H_i + b.$$

By Lemma 2.4, their common recovered aligned depth δ satisfies

$$\delta \geq s(N/2 - 2) \geq \frac{1.002}{1.0001}H_i - 2s \geq 1.001H_j.$$

The last inequality follows from $a \leq 10^{-6}$, $H_i \geq (2 \cdot 10^6 + 1)s$, and $b/s = .50005a$.

For all heights, (2.2) gives

$$\begin{aligned} \ell_A &\leq H/2 + .1s, & |v_A| &\leq H/2 + .6s, \\ \ell_B &\leq H/2, & |v_B| &\leq H/2 + .5s. \end{aligned}$$

An aligned A depth δ recovers $\delta/2 + .1s$ bits of each stream, and a B depth recovers $\delta/2$ bits. Since $H_j \geq 1000s$, the displayed recovery covers the entire first midpoint state.

The labels give equal height residues at the two midpoint positions. If their heights differ, write $i_- < i_+$ for them. Their own- u length difference is the Δ of (2.18). The smaller state's padding is zero on the interval tested there at the larger state. Its endpoint is at most

$$(1/2 + \eta_{\text{sep}})H_{i_-} + .1s \quad \text{on A}, \quad (1/2 + \eta_{\text{sep}})H_{i_-} \quad \text{on B}.$$

Because $H_{i_-} \leq H_j$ and $2\eta_{\text{sep}} < .001$, the recovered prefix contains that whole interval, even if i_+ is arbitrarily large. The larger state would therefore have failed its separation test, a contradiction. The two heights are equal. Their entire states agree by the recovery already proved. Foldedness now identifies the paths in both directions from this midpoint, and uniquely continues any longer common word. $\square$

2.8 The interface with probabilistic selection

Remark 2.7 (Terminal dimensions). The two rows of (2.13) give the exact terminal dimension split needed later. In the side containing a partial final u block, its final pair (u, v) has dimensions (t, s) ; the other side has a final pair of dimensions $(0, t)$. Two independent incident evaluations impose respectively $s - t$ and t compatibility bits. The first assertion follows because multiplication by a nonzero coefficient difference is injective on the active t -dimensional subspace. The second is simply equality of the two t -bit evaluations. Thus the sum is exactly s . For adjacent incidences first move each known affine correction $C^v + KC^u$ intact to the other side of its evaluation equation. A change in one input bit can affect all output coordinates under field multiplication; these known shifts cost no random dimension. Only the width of the common active random prefix can cost compatibility bits. Projection to such a prefix of width w' gives an image of dimension at most $w' + t$ in $2w'$ dimensions, hence codimension at least $\max\{0, w' - t\}$. The adjacent-cutoff and block-boundary cases, including a newly deterministic fragment, are proved in Lemma 4.5; they lose at most $4b$ compatibility bits. No injectivity of a truncated coefficient difference is asserted. The probabilistic use of these dimensions still requires distinct current table variables; that issue is addressed in Proposition 4.14.

Remark 2.8 (Inputs to the probabilistic stages). The uniform-coordinate statement is an assertion for each fixed table realization. Proposition 3.4 may therefore choose the tables successively while retaining all deletion budgets. The current tables are coordinatized in Lemma 3.1 below; they are independent under the product reference law used there. In reconstruction of a fixed labeled candidate, endpoint free u bits are enumeration data fixed before sampling. At update B_j the queried argument of $P_{A,j-1}$ is already known; at the following update A_j the argument of $P_{B,j}$ is already known. Current untested outputs can affect only later arguments. Gates for XOR corrections use the stronger lag (2.10); ordinary preceding-key dependence alone would not suffice for those corrections.

3 Expansion and the sequential probability law

We now choose the permutation tables to make every auxiliary incidence expand. The choice must also retain a quantitative probability estimate for pruning: at one source-block update, conditioning on expansion may increase the probability of a joint test of k current tables by at most $e^{\varepsilon_s k}$, where $\varepsilon_s \rightarrow 0$ as $s \rightarrow \infty$. Proposition 3.4 supplies both conclusions for one law on the full tower. The expansion bound concerns the auxiliary incidences before vertex deletions.

Throughout the finite estimates, fix a family at a marked cutoff, all its earlier tables, and one extra-bit slice. Only its current active-prefix tables are random, and the reference law makes them independent uniform permutations. The auxiliary prescription forces the current gate open. We first bound the mean incidence operator, then select tables for which all cuts expand, controlling the change of law at the same time. Every bound is uniform in the earlier history and the height; this uniformity will permit successive choices along the tower.

Put $\theta = 2^{1-b}$. Choose the fixed integer b sufficiently large that $\sqrt{5\theta} < 3/5$. All averages on either vertex side are probability averages. For one fixed-extra slice, write A, B for its common-cutoff coordinate spaces and define

$$(Nf)(a) = \frac{1}{d} \sum_{e:e^- = a} f(e^+), \quad (N^*g)(b') = \frac{1}{D} \sum_{e:e^+ = b'} g(e^-).$$

The sums count slots, including parallel edges. Since $d|A| = D|B|$, these are adjoint contractions that preserve constants. Let $M = \mathbb{E}N$, where the expectation is only over the current tables. The same statements hold for M .

3.1 The current matching variables

The triangular coordinates from Section 2 identify exactly what the current random permutations match. Slots are the objects in these matchings; different slots of one physical vertex remain distinct.

Lemma 3.1 (Fresh matching coordinates). *Fix a family occurrence, an extra slice, and every table before its last common source block. Under the independent uniform current-table law, its auxiliary incidence is randomized by independent perfect matchings between pairs of clusters of 2^t slot incidences, one matching for each full final-prefix table key. Each possible edge in a cluster pair is permitted. No later table affects this incidence.*

Proof. Fix the serial number, full label head and preceding raw u blocks. If the last common block is B , its remaining B data are exactly the t bits of raw u_B in that block; the known correction merely translates them. Its remaining A data are exactly the corresponding t evaluation bits. All other coordinates are determined by Lemma 2.2. The forced-on current prefix table is an arbitrary

bijection between these two 2^t -element sets. Different complete keys specify disjoint clusters of slot incidences. The A-terminal case reverses the roles. The padding proof in Lemma 2.2 shows that higher tables never act on this incidence. $\square$

3.2 A conditional Fourier estimate

The coefficient multiplicity bound gives cancellation when we average the evaluations $V + K_a U$ over slots. We use the following conditional form so that earlier raw prefixes and their correction offsets may be held fixed.

Lemma 3.2. *Conditional on data C , let $U \in \mathbb{F}_2^t$ and $V \in \mathbb{F}_2^w$ be independent uniform vectors, independent also of auxiliary data W . For a in a set of q_0 slots let $K_a : \mathbb{F}_2^t \rightarrow \mathbb{F}_2^w$ be linear, let $o_a = o_a(C, W)$, and let $f_a(C, z)$ have no separate dependence on U, V, W . Suppose every class of slots with a common value of ζK_a , for any nonzero functional $\zeta \in (\mathbb{F}_2^w)^*$, has at most θq_0 members. Set*

$$F = \frac{1}{q_0} \sum_a f_a(C, V + K_a U + o_a), \quad \mu(C) = \frac{1}{q_0} \sum_a 2^{-w} \sum_z f_a(C, z).$$

Then $\mathbb{E}(F \mid C, W, U) = \mu(C)$ and

$$\mathbb{E}|F - \mu(C)|^2 \leq \theta \mathbb{E} \left[\frac{1}{q_0} \sum_a 2^{-w} \sum_z |f_a(C, z)|^2 \right]. \quad (3.1)$$

Proof. Translation of uniform V proves the mean assertion. Fix C, W and write $\hat{f}_a(\zeta) = 2^{-w} \sum_z f_a(C, z)(-1)^{\zeta z}$. Orthogonality, first in V and then in U , gives

$$\mathbb{E}_{U,V}|F - \mu(C)|^2 = \frac{1}{q_0^2} \sum_{\zeta \neq 0} \sum_{\mathcal{C}} \left| \sum_{a \in \mathcal{C}} (-1)^{\zeta o_a} \hat{f}_a(\zeta) \right|^2,$$

where $\mathcal{C}$ ranges over classes with equal ζK_a . Cauchy–Schwarz bounds each square by $|\mathcal{C}| \sum_{a \in \mathcal{C}} |\hat{f}_a(\zeta)|^2$. The multiplicity assumption and Parseval’s identity prove the result after integration over C, W . In particular, integration over W introduces no variance of the conditional mean. $\square$

The coefficient pools of Section 2 give this multiplicity bound on every full block and every active initial part of length at least b . The first A block has length r ; all subsequent blocks have length s . The divisibility choices make every nonempty terminal initial part have length at least b .

We record why the right side of (3.1) is bounded by the original input norm in its applications. A uniform vertex with a uniform incident slot has a uniform neighbor, counted with the appropriate multiplicity. Consequently the slot average of squared neighbor values is exactly the squared input norm. Averaging any still random table values can only decrease this quantity, by Jensen’s inequality. Finally V makes the evaluation argument uniform conditional on all other data. Thus the individual Fourier norms in the Lemma are precisely bounded by these slot norms; no factor depending on the number of blocks is introduced.

3.3 Two interacting prefix bands

To apply the Fourier estimate to the whole incidence, decompose a function into orthogonal increments as longer state prefixes are revealed. Triangular inversion couples only two neighboring levels of these prefix decompositions. The only additional estimate is at the terminal block, where the current prefix permutation supplies the missing average.

Let P_h^A be expectation onto the first $r + hs$ bits of both raw A coordinates, and let P_j^B be expectation onto the first js bits of both raw B coordinates. Padded coordinates are deterministic zero. The filtrations therefore terminate at the identities. In particular $P_0^B = C_B$, the projection onto constants. Write C_A for the corresponding projection on A .

For a fixed A slot, a function in $\text{ran } P_j^B$ can be evaluated at its neighbor using only

$$u_A[0, r + (j - 1)s), \quad Q_A[0, r + js). \quad (3.2)$$

The first r evaluation bits give the label head. Successive inversion of the triangular permutations then gives u_B through its j th block; the equation for Q_B uses u_A only through its $(j - 1)$ st block to give v_B through length js . In the reverse direction, a fixed B slot evaluates a function in $\text{ran } P_h^A$ using only

$$u_B[0, hs), \quad Q_B[0, (h + 1)s). \quad (3.3)$$

Here the bucket is already specified by the slot. The remaining head and then u_A are recovered from the evaluation bits, and the equation for Q_A gives the required v_A bits.

These descriptions also apply to the corrected coordinates. Their extra bits are fixed in the slice. Every correction gate ends before both coupled blocks, so its offset is determined by the preceding raw prefix. The same is true of an ordinary permutation gate. A current gate is forced open, hence constant. No unknown current raw input enters a key for its own source block. The identity suffix at a mark and the prescribed subsequent identity blocks make the assertions valid beyond an actual cutoff as well. These are the reasons for using raw prefixes in the keys and the lag specified in Section 2.

Since multiplication is block diagonal, there is no spill into a following evaluation block. Equations (3.2)–(3.3) imply

$$\text{ran}(MP_j^B) \subseteq \text{ran } P_j^A, \quad \text{ran}(M^*P_h^A) \subseteq \text{ran } P_{h+1}^B. \quad (3.4)$$

They hold before as well as after averaging current tables. For every full interval that is active, Lemma 3.2 also gives

$$\|(1 - P_{j-1}^A)MP_j^B\|^2 \leq \theta, \quad (3.5)$$

$$\|(1 - P_h^B)M^*P_h^A\|^2 \leq \theta. \quad (3.6)$$

For (3.5), condition on the preceding A prefix. The new data in (3.2) occur solely as $V + K_a U + o_a$, with U the new A input block and V its evaluation block of v_A bits. The direct input in that equation stops before U . The A coefficient does not depend on the unknown bucket in the head. For (3.6), condition on P_h^B and use the next B input and evaluation blocks. This time the direct input in (3.3) again stops before U ; the slot already fixes the bucket, and its coefficient does not depend on the unknown suffix of the head. Thus neither application has a separate dependence on the fresh U . The slot-norm argument above completes both estimates, including $h = 0$.

Lemma 3.3 (The conditional mean). *For every fixed earlier-table history, every marked cutoff, and every extra-bit slice,*

$$\|M : 1_B^\perp \longrightarrow 1_A^\perp\| \leq \eta := \sqrt{5\theta} < 3/5.$$

The same bound holds for the complete adjacent-height incidence.

Proof. The full blocks. Use the orthogonal difference projections

$$E_0^A = P_0^A - C_A, \quad E_h^A = P_h^A - P_{h-1}^A \quad (h \geq 1), \quad E_j^B = P_j^B - P_{j-1}^B \quad (j \geq 1).$$

Equation (3.4) implies $E_h^A M E_j^B = 0$ unless $h \leq j \leq h+1$. The band $h = j$ is bounded by (3.5), and the band $j = h+1$ by the adjoint of (3.6), whenever the blocks in question are full. Within either band the input spaces are mutually orthogonal, as are the output spaces. Each band therefore has norm at most $\sqrt{\theta}$, and their sum at most $2\sqrt{\theta}$.

The terminal block. There are two possible terminal forms. If the final input block is a B block, write

$$\ell_A = r + ns, \quad \ell_B = ns + t, \quad |v_A| = r + ns + t, \quad |v_B| = (n+1)s, \quad b \leq t \leq s.$$

For the input differences with $1 \leq j \leq n$, all blocks required in both bands are full, and consequently

$$\|M(P_n^B - C_B)\| \leq 2\sqrt{\theta}. \quad (3.7)$$

Fix a B slot. Inversion through the preceding blocks determines all of u_A and all but the last t bits of v_A using $u_B[0, ns)$ and $Q_B[0, (n+1)s)$. The current prefix permutation uniformly averages the remaining t bits of Q_A . Its coefficient term $K_A u_A$ ends at $r + ns$, so these are exactly the remaining v_A bits, up to a known correction. The matching key contains only the label and preceding raw inputs. After this average, its slot value has no separate dependence on the last t raw u_B bits. Conditional on P_n^B , it is therefore a function of $V + K_a U + o_a$, with $U \in \mathbb{F}_2^t$ and $V \in \mathbb{F}_2^s$. Lemma 3.2 gives

$$\|(1 - P_n^B)M^*\|^2 \leq \theta.$$

Combining this orthogonally with the adjoint of (3.7) gives $\|M^*g\|^2 \leq 5\theta\|g\|^2$ for centered g .

If the final input block is an A block, the lengths are instead

$$\ell_B = (n+1)s, \quad \ell_A = r + ns + t, \quad |v_A| = r + (n+1)s, \quad |v_B| = (n+1)s + t.$$

The full-block argument on E_h^A , $0 \leq h \leq n$, gives $\|M^*(P_n^A - C_A)\| \leq 2\sqrt{\theta}$. The current permutation averages the final t evaluation bits of Q_B , on which $K_B u_B$ is zero. Conditional on P_n^A , the remaining slot function depends on the final t input bits of A only through their full s -bit evaluation. Hence $\|(1 - P_n^A)M\|^2 \leq \theta$, and the same orthogonal combination proves the assertion. Both arguments include $t = s$; the initial r -block is never terminal because the tower starts after many complete pairs.

Combining the extra-bit slices. If the larger side is A , its space is $A_0 \times \mathbb{F}_2^b$ and the complete operator has components M_e . For centered f on B , $\|Mf\|^2 = \mathbb{E}_e \|M_e f\|^2 \leq \eta^2 \|f\|^2$. If the larger side is B , use the adjoint version of this argument. Equivalently, write an input as $f_e = c_e + f_e^0$; its overall mean zero says $\mathbb{E}_e c_e = 0$, so the constants cancel before Jensen's inequality is applied to $\mathbb{E}_e M_e f_e^0$. $\square$

3.4 Cut concentration

The mean estimate gives a lower bound for every expected cut. The matching coordinates now let us turn this into a failure-probability bound for each fixed cut.

We now condition only on earlier tables and one extra-bit slice. Lemma 3.1 identifies each current table with a uniform bijection between two fibers of slots of size 2^t . Its key fixes all other endpoint coordinates. Distinct keys give disjoint such fibers of slots, and the current tables are independent. A physical vertex may occur in several fibers through its different slots; this does not identify their permutation variables.

Write $\deg(v) = d$ on A and D on B , and $\text{vol}(S) = \sum_{v \in S} \deg(v)$. The total volume is $V_0 = 2d|A|$. Let $c(S)$ count edges crossing from S to its complement, with multiplicity. The mean random-walk

operator on the union of the two sides is the self-adjoint block operator with entries M, M^* . On the orthogonal complement of global constants its spectrum has upper bound η ; its additional bipartition eigenvector has eigenvalue -1 . Applying its quadratic form to the centered indicator of S gives

$$\mathbb{E}c(S) \geq (1 - \eta) \text{vol}(S) \left(1 - \frac{\text{vol}(S)}{V_0}\right) \geq \frac{1}{5} \text{vol}(S) \quad (3.8)$$

whenever $\text{vol}(S) \leq V_0/2$.

For a current table a , let Z_a count its matched pairs whose two endpoints belong to S . If n_a is the fiber size, u_a the number of its A slots in S , and v_a the corresponding number of B slots, then Z_a is hypergeometric with these parameters. Moreover

$$c(S) = \text{vol}(S) - 2 \sum_a Z_a, \quad \sum_a u_a \leq \text{vol}(S).$$

For completeness, sampling u_a distinct entries from a population of v_a ones and $n_a - v_a$ zeros gives, for every real λ ,

$$\mathbb{E}e^{\lambda(Z_a - \mathbb{E}Z_a)} \leq e^{\lambda^2 u_a / 8}.$$

Indeed the average of products of u_a distinct positive numbers is at most the u_a th power of their arithmetic mean: this follows by repeatedly replacing two unequal numbers by their average in the elementary symmetric polynomial, which increases it. Apply this to the numbers $1, e^\lambda$. The resulting bound is the moment generating function for sampling with replacement. For one Bernoulli variable its centered log moment generating function has value and first derivative zero at zero and second derivative at most $1/4$ everywhere, proving the displayed bound. Independence and exponential Markov's inequality therefore yield

$$\mathbb{P}\{c(S) < .05 \text{vol}(S)\} \leq \exp\{-.01125 \text{vol}(S)\} \leq \exp\{-c_0 d_{\min} |S|\}, \quad c_0 = 10^{-3}, \quad (3.9)$$

where $d_{\min} = \min(d, D)$. The numerical constant uses $\sum_a Z_a - \mathbb{E} \sum_a Z_a > .075 \text{vol}(S)$ and the choice $\lambda = .3$.

3.5 The finite selection lemma and its conditioning bound

Join two physical vertices when some choice of a current table could join them. This *possible graph* has maximum degree $\Delta \leq 2^{C_s}$, where C may depend on the fixed number of types and on b . Each current table meets at most 2^{C_s} vertices; the number of tables meeting a vertex has the same bound. These bounds do not depend on the height. It suffices to avoid the cut failures in (3.9) for nonempty connected sets S in the possible graph satisfying $\text{vol}(S) \leq V_0/2$. In fact any failed cut of at most half the volume has a failed connected component, since both volume and crossing add over those components.

Let A_S be such a cut-failure event. It depends only on tables meeting S . Two events with no common table are independent, jointly with any collection of other events using disjoint tables. If their table sets meet, some vertex of one is within distance two of a vertex of the other in the possible graph. A connected j -set containing a specified vertex is encoded by a walk of length $2(j-1)$ obtained from a deterministically chosen rooted spanning tree. Thus the number of neighboring j -set events is at most

$$|S| \Delta^{2j+C_1} \quad (3.10)$$

for an absolute enlargement C_1 .

Assign weights $x_S = \exp(-c_0 d_{\min} |S|/2)$. Equations (3.10) and the exponential growth of $d_{\min} = 2^{40005s}$ imply, uniformly in S ,

$$\sum_{T \sim S} x_T \leq |S| \sum_{j \geq 1} \Delta^{2j+C_1} e^{-c_0 d_{\min} j/2} = |S| \delta_s, \quad \delta_s \rightarrow 0.$$

For sufficiently large s , all weights are below $1/2$ and $2\delta_s \leq c_0 d_{\min}/2$. Consequently

$$\mathbb{P}(A_S) \leq x_S \prod_{T \sim S} (1 - x_T). \quad (3.11)$$

This is the finite asymmetric Lovász local lemma in its product form [30, Theorem 1.1]; the original symmetric lemma is due to Erdős and Lovász [13, §2]. We include the conditional-probability induction because its quantitative conditioning bound is also needed. For a finite family satisfying (3.11), induction simultaneously proves positivity of avoidance probabilities and the following conditional bound. At a given size, positivity follows from the smaller-size bounds by the product rule; the empty avoidance event has probability one. For any A_i the bound is

$$\mathbb{P}\left(A_i \mid \bigcap_{j \in J} A_j^c\right) \leq x_i \quad (i \notin J).$$

Split J into the neighbors J_1 of i and its remaining set J_2 . The numerator after conditioning on avoidance in J_2 is at most $\mathbb{P}(A_i)$, by independence. If J_1 is empty, independence proves the desired bound immediately. Otherwise the denominator for also avoiding J_1 is, by the product rule and the induction hypothesis, at least $\prod_{j \in J_1} (1 - x_j)$. The criterion now proves the induction step. The same product rule proves that avoidance of every event has strictly positive probability.

We also need the standard conditional-distribution estimate for an additional event; see [21, Theorem 2.1]. Let B involve a specified set of k current tables. Let J_1 be the cut events sharing one of these tables and J_2 all the others. Independence gives

$$\mathbb{P}\left(B \mid \bigcap_S A_S^c\right) \leq \mathbb{P}(B) \prod_{S \in J_1} (1 - x_S)^{-1}.$$

For each specified table, its at most 2^{C_s} vertices and the connected-set count give

$$\sum_{S \in J_1} x_S \leq k 2^{C_s} \sum_{j \geq 1} \Delta^{2j+C_1} e^{-c_0 d_{\min} j/2}.$$

It follows that, for a number $\varepsilon_s \rightarrow 0$ independent of height and earlier history,

$$\mathbb{P}(B \mid \text{all current cuts succeed}) \leq e^{\varepsilon_s k} \mathbb{P}(B). \quad (3.12)$$

The same inequality applies to any nonnegative function of those k tables, by integration of its level sets. It concerns a joint test of current variables. It does not assert a bound for the next query after conditioning on arbitrary earlier answers from the same block.

3.6 A gap in every chosen incidence

The selected cut bounds give a spectral gap by the degree-weighted Cheeger argument; compare [35, Lemma 3.3] and Chung [11, Section 3, Theorem 1]. We include the argument in the bipartite form needed here, keeping its normalization and edge multiplicities. If all the selected cuts satisfy

$c(S) \geq h_0 \text{vol}(S)$ with $h_0 = .05$, the ordinary weighted spectral gap is at least $h_0^2/2$. For a real function f of degree-weighted mean zero, choose a weighted median m and put $g_+ = (f - m)_+$ and $g_- = (m - f)_+$. Each has support of volume at most half. Integration over the level sets of g^2 , for either one, yields

$$\sum_{\{u,v\}} |g(u)^2 - g(v)^2| \geq h_0 \sum_v \deg(v) g(v)^2.$$

Cauchy–Schwarz and $\sum_{\{u,v\}} (g(u) + g(v))^2 \leq 2 \sum_v \deg(v) g(v)^2$ give

$$\sum_{\{u,v\}} (g(u) - g(v))^2 \geq \frac{h_0^2}{2} \sum_v \deg(v) g(v)^2.$$

The energy of f is at least the sum of the energies of g_+ and g_- , while $\sum_v \deg(v) (f(v) - m)^2 \geq \sum_v \deg(v) f(v)^2$. This proves the claimed gap. Apply the argument to real and imaginary parts for complex functions. Bipartite symmetry sends an eigenvalue to its negative; the only exceptional modes are the global constant and bipartition functions. Hence the normalized incidence has centered singular norm at most

$$\rho = 1 - h_0^2/2 < 1. \quad (3.13)$$

The slice-to-whole argument in Lemma 3.3 also transfers this bound to adjacent-height incidences.

3.7 Choosing the tables along the tower

At one marked cutoff we have now obtained both a nonempty set of expanding table choices and the bound (3.12) on conditioning to that set. Future tables do not affect this incidence. We can therefore apply the same finite choice successively at all marked cutoffs.

Proposition 3.4 (Sequential selection). *There is a probability law on all the permutation tables such that every auxiliary incidence has centered singular norm at most the fixed number $\rho < 1$ in (3.13). At each individual source-block update, conditional on all earlier updates, any joint test of k specified current tables has its probability increased by at most $e^{\varepsilon_s k}$ relative to independent uniform current tables. The bound is uniform in the earlier history and the height, with $\varepsilon_s \rightarrow 0$.*

Proof. Within each of the finitely many families order the source blocks as $A_0, B_1, A_1, B_2, \dots$. Interleave the families by increasing source-block index and a fixed family order. At an unmarked update choose all its active permutation variables uniformly. At a marked update use the finite conditional distribution given avoidance of its current cut failures, separately in the extra-bit slices. Conditional on all earlier source-block indices, the current kernels in different families are independent, since a family’s cut conditions use only its own tables. Thus grouping the families into one source-block update gives the same bound, with k the total number of tested tables. The mark separation ensures at most one actual cutoff per family at an update. Lemma 3.3 and (3.11) show that this distribution exists for every possible preceding history, and (3.12) proves its test bound. Blocks prescribed to be identities have no random variables.

Each update has finitely many possible outcomes. An independent uniform number in $[0, 1]$ can choose among them by consecutive intervals of their conditional probabilities. Iterating on the countable sequence of updates constructs the asserted law. A future update does not change an earlier incidence: the prefix and padding properties of Section 2 guarantee this. Therefore all earlier expansion conditions persist.

When a test whose current-table choices and input arguments are measurable from earlier updates is evaluated, all later conditional distributions integrate to one. Thus its probability is governed by the current kernel just described, without an additional conditioning on future expansion successes. This observation is the form of sequential conditioning used in the pruning argument below. $\square$

4 Pruning coherent returns and bicycles

We work with the random universal tables of Proposition 3.4. The coefficient pools have already been fixed. The graph in this Section has only old vertices and old edges; the private completion is made later. Write X^{good} for the graph induced on the vertices surviving the gate and separation deletions of Proposition 2.6. These deletions are defined for every realization of the tables. A statement that a path is in X^{good} will be used as a condition on the objects being counted, never as additional conditioning of the table distribution.

All lengths in this Section count edges. A bicycle is a finite connected subgraph with first Betti number two and with no vertex of valence one. After vertices of valence two are suppressed, it is a theta, a figure-eight, or a barbell. We need to exclude embedded bicycles; their edge labels are automatically immersed because the ambient labeling is folded.

Our goal, stated precisely in Proposition 4.14, is to remove an arbitrarily small fraction of every layer while excluding short cycles and bicycles. We first count repeated-prefix returns, which yield the $.93m_i$ girth bound, and then use that bound to count the remaining short thetas. Removing bicycles below $1.52m_i$ finally ensures that nonempty cyclically reduced closed paths of length at most $6.04m_i$ are not Heisenberg laws, the separation input needed for the voltage choice.

4.1 Prefix depth, fixed margins, and admissible returns

We use the aligned depths of (2.15), writing $d_A = \delta_A$ and $d_B = \delta_B$. Thus, for an A-vertex, agreement to depth

$$d_A(j) = (2j + 1)s \quad (j \geq 0)$$

means agreement of both raw streams through length $r + js$. For a B-vertex, agreement to depth

$$d_B(j) = 2js \quad (j \geq 0)$$

means agreement of both raw streams through length js . Depth is measured in state bits; on the A-side it subtracts the constant $2r - s$ from the number of specified raw bits. This is why the common scale in both cases is H_i .

Let $D(v)$ be the largest such depth for which both specified stream prefixes are active at v . The two possible terminal cutoffs give

$$H_i - 2s \leq D(v) \leq H_i \quad \text{for } v \in \mathcal{A}_i \sqcup \mathcal{B}_i. \quad (4.1)$$

For example, if the active cutoff is inside B-block j , the B-input is partial while its evaluation block is full; the largest common full B-prefix ends at block $j - 1$. Its missing depth is between s and $2s$. On the other side the input has ended and the next evaluation block is partial; the missing depth is between zero and s . The case of a final A-input is the same with the sides reversed.

Two equal state prefixes followed through one common outward label propagate as

$$A_j \mapsto B_j, \quad B_j \mapsto A_{j-1}. \quad (4.2)$$

Here A_j denotes both A-prefixes through $r + js$, and B_j denotes both B-prefixes through js . Each propagation therefore loses exactly s of aligned depth. To verify the first assertion, the equal A-evaluations through $r + js$, after removing their common label head, determine the B-input through block j by successive inversion. The B-evaluation through block j uses only A-input blocks through $j - 1$, already known. The second assertion follows from the other equation: the B-evaluation through js determines the A-input through block $j - 1$, and its A-evaluation uses earlier B-inputs.

The correction offsets are equal because the label is equal and their raw gate windows are earlier than both coupled blocks. Adjacent cutoffs do not invalidate the stated full prefixes: the cutoff moves by at most $b < s$, whereas one propagation has removed one whole interleaved block. These observations prove (4.2) also when one of the original vertices is at its last complete prefix.

We make the margin independent of the particular subpath. For $k \geq 0$ put

$$T_k = H_0(11/10)^k, \quad I_k = [5T_k/6, 6T_k/5], \quad M_k = T_k/25.$$

A vertex belongs to band k when its height scale belongs to I_k . Each height belongs to at most a fixed number of bands, independent of H_0 and s , and

$$\frac{1}{30} \leq \frac{M_k}{H_i} \leq \frac{6}{125} < .05 \quad (H_i \in I_k). \quad (4.3)$$

We shall use the weaker interval $[.03, .05]$ in numerical estimates. If a finite old-edge subgraph through height i has total length at most $7m_i$, its scale variation is at most $7bm_i$. For sufficiently small b/s , choose k with $T_k \leq H_i < T_{k+1}$. All its vertex scales then lie strictly inside I_k . Thus every later short-path application fits one band. Unlike a disjoint partition into bands, this choice has room at every endpoint.

Definition 4.1 (Coherent return). In band k , a coherent return is a positive-length nonbacktracking path

$$v_0, e_1, v_1, \dots, e_p, v_p$$

in X^{good} , entirely in that band, whose endpoints are on the same side and agree through an available aligned depth

$$h \geq ps/2 + M_k, \quad h \leq \min\{D(v_0), D(v_p)\}. \quad (4.4)$$

It is minimal if no proper positive-length contiguous subpath is a coherent return for the same band and the same M_k . An exact closed nonbacktracking path is included whenever it satisfies the available-depth condition.

The endpoint side makes p even. For a fixed length, side, and band, we always use the least aligned h satisfying the first inequality of (4.4); it differs from its right-hand side by less than $2s$. This loses no return and removes h as an extra enumeration parameter. Every coherent return contains a minimal one, by choosing a shortest eligible contiguous subpath.

The threshold is chosen to make the return count smaller than a layer. At the leading exponential scale, a length- p pattern at initial scale $H = H_i$ costs ps bits of label choices and $H - h$ free endpoint bits, while compatibility at the final active blocks imposes $ps/2$ bits of constraints. The inequality $h \geq ps/2 + M_k$ therefore leaves an exponent $H - M_k$, below the layer exponent H . The estimates below retain the label-rate slack and control the losses from unequal heights, terminal cutoffs, and the conditioned table law.

Lemma 4.2 (The two junction coefficients are different). *At the identified endpoints of a minimal coherent return, the two outward incident coefficient names are different.*

Proof. The pools are injective on the full finite collection of family, type, slot, and permitted extra-value names, rather than only on the slots at a single vertex. Hence equality of the two coefficient names gives the same family and the same slot data. At an A-endpoint the label head h_{lab} is the first r bits of its evaluation. At a B-endpoint its suffix is the first $s - r$ evaluation bits and its bucket is slot data. Equality of the raw endpoint prefixes therefore gives the same full outward

label. Extra-dependent coefficients cause no exception because those extra values are part of their names.

Remove one edge at each end. By (4.2), the new endpoints agree to depth at least $h - s$, and the new threshold is $(p - 2)s/2 + M_k$. The prefixes are available at the new endpoints by the same propagation statement. The remaining subpath is still in the same band and consists of good vertices. If $p > 2$, this is a proper coherent return, a contradiction. If $p = 2$, the two edges entering its middle vertex have the same outward label there, so foldedness makes the original path an immediate backtrack. This too is impossible. $\square$

4.2 Reconstructing a counted object

The count separates choices from constraints. Once the labels and heights are fixed, two incident evaluations reconstruct the state at each internal vertex. A coherent path leaves only raw u-bits beyond the common endpoint prefix to be chosen; a theta has no free endpoint bits. A formal reconstruction need not lie in the prescribed state spaces. Its terminal coordinates must satisfy the active cutoffs, and at a theta's two branch vertices the third incident evaluation must agree with the first two. These are the constraints whose probabilities we estimate.

We record exactly what is fixed in the enumeration. A path pattern consists of its length p , initial side and height, band, edge labels, and the choices of next height allowed by those labels. There are at most three choices per edge if one does not use the height residues to determine the choice uniquely. A theta pattern consists of three positive branch lengths, the root position in that abstract graph, the initial side and height, edge labels, and allowed height changes. All slot names, coefficients, correction extras, and current cutoffs are fixed by this information. Patterns with inconsistent labels or height assignments are rejected before any random table is queried. We also reject equal coefficient names among the two incident slots at an internal vertex or the three slots at a branch vertex. Such a pair specifies the same incident slot, by the injectivity of the coefficient pools. At an actual vertex that slot supports just one edge, so its repetition would be an immediate backtrack or would identify two supposed branches. No immersed path or embedded theta is lost. At the coherent junction the corresponding exclusion is justified by Lemma 4.2.

There is a number L_s such that the number of choices of an oriented label together with its allowed next height is at most L_s , and, by increasing s after the finite type data,

$$L_s \leq 2^{1.001s}. \quad (4.5)$$

Indeed the additional factors here are at most a fixed multiple of the actual alphabet size. This deliberately loose bound is used only in counting, not in the later forest norm estimate.

For a coherent path we identify its two endpoints only through depth h . After that depth, choose all their remaining active *raw* u-bits freely and fix them as part of the pattern. Their inactive u-bits are fixed to zero. There are no free v-bits. For a theta there are no free endpoint bits.

Lemma 4.3 (Unique reconstruction and free-coordinate count). *Fix a coherent-path pattern with different junction coefficients, and fix its free endpoint u-bits. For any realization of the universal tables there is at most one assignment of vertex states realizing that pattern as a coherent path. If its endpoint scales are H_- and H_+ , the number of free bits is at most*

$$(H_- + H_+)/2 - h. \quad (4.6)$$

For a theta pattern, there is at most one assignment of states and there are no free bits. During either reconstruction all inputs and keys queried at a source block are determined before that block's tables are exposed.

Proof. Choose two incident slots at each theta branch vertex and use the two incident slots at every degree-two vertex. At the coherent junction use the two end slots through depth h . At a free end beyond h use its prescribed u-block and its single incident evaluation.

First recover every A-block A_0 . Its two evaluation heads are known from the labels. If their coefficients are $c_1 \neq c_2$, subtraction recovers the raw u-block, and then the raw v-block. At later blocks the known correction offsets can be moved to the right-hand sides, so the same two equations have the form

$$v + c_1 u = E_1, \quad v + c_2 u = E_2. \quad (4.7)$$

The difference $c_1 - c_2$ is a nonzero field element and is invertible on the full padded block. Thus these equations uniquely give

$$u = (c_1 - c_2)^{-1}(E_1 - E_2), \quad v = E_1 - c_1 u.$$

At a free endpoint, u is already prescribed and the one equation uniquely gives v .

The update of B-block j uses the two values of $P_{A,j-1}$ at the neighboring A-input blocks already recovered. The subsequent update of A-block j uses $P_{B,j}$ at the B-input blocks just recovered. Each key contains raw blocks strictly before its source block; current correction windows end before both coupled blocks. All these data are known before the current table is read. This proves the assertion inductively in the individual source-block order $A_0, B_1, A_1, B_2, A_2, \dots$

The induction is also an unambiguous *formal* reconstruction when a pattern is invalid: solve on full padded blocks, stop after all blocks that can meet the candidate cutoffs, and then test the required zeros, labels, and vertex conditions. An assignment violating an inactive-coordinate condition need not define a graph vertex, but it causes no ambiguity in earlier query data. Every actual realization is reproduced by the induction, which proves the claimed upper bounds on realizations. Ignoring some final validity conditions will consequently only increase our counts.

It remains to count the free endpoint u-bits. The active interleaving implies

$$r - s \leq \ell_A(i) - \ell_B(i) \leq r, \quad H_i = \ell_A(i) + \ell_B(i) + s - r.$$

Consequently

$$\ell_A(i) \leq H_i/2 + (2r - s)/2, \quad \ell_B(i) \leq H_i/2.$$

At aligned depth h an A-endpoint has specified $(h + 2r - s)/2$ u-bits, and a B-endpoint has specified $h/2$ u-bits. Adding the two residual lengths on their common side gives (4.6). A theta has two chosen evaluations at every vertex and hence has no freely prescribed coordinate. $\square$

4.3 The probability estimate at one source block

Let $\mathcal{F}_{<t}$ contain all universal tables at source blocks strictly preceding t , in every family, and all fixed pattern data. The last assertion of Lemma 4.3 says that keys, input arguments, gates, coefficient choices, and the widths of current tests are $\mathcal{F}_{<t}$ -measurable. In particular, no v-coordinate solved from a current evaluation is used in a current raw-u key.

By Proposition 3.4, an event B_t involving k_t whole current permutation tables satisfies

$$\mathbb{P}(B_t \mid \mathcal{F}_{<t}) \leq \exp(\epsilon_s k_t) \mathbb{P}_{\text{prod}}(B_t \mid \mathcal{F}_{<t}), \quad \epsilon_s \longrightarrow 0. \quad (4.8)$$

The right-hand law is the product of the indicated current uniform permutation laws, with earlier tables held fixed. It is legitimate to apply this bound to a union of output prescriptions, hence to a linear compatibility condition. Untested entries of a table in B_t are integrated out. The estimate is not an assertion about the last unknown entry of a permutation after its other entries have been prescribed.

Lemma 4.4 (Predictable accumulated rank). *Suppose that, at each of finitely many source blocks, a test is selected from $\mathcal{F}_{<t}$, and its product-law probability is at most 2^{-R_t} . The nonnegative integers R_t and k_t are $\mathcal{F}_{<t}$ -measurable. If an event A implies that all tests pass, that $\sum_t R_t \geq R$, and that $\sum_t k_t \leq K$, then*

$$\mathbb{P}(A) \leq \exp(\epsilon_s K) 2^{-R}.$$

The event A may additionally require later minimality, missed-gate counts, and full consistency of the reconstructed vertex states.

Proof. After a failed test give the process value zero. Otherwise multiply its value at step t by $2^{R_t} \exp(-\epsilon_s k_t)$ when that test passes. Conditional expectation does not increase, by (4.8). The initial value is one, so the final nonnegative variable has expectation at most one. On A its value is at least $2^R \exp(-\epsilon_s K)$, proving the claim. Additional final conditions are used only in this last lower bound, not in any conditional probability. $\square$

4.4 Terminal dimensions and equality of keys

The two-evaluation reconstruction solves for a full padded block, but only its active coordinates may be nonzero. At a terminal block this restriction imposes a number of compatibility conditions determined by the excess of evaluation bits over input bits. We first count these conditions, then determine when different vertices test distinct tables.

For a vertex put

$$\lambda_A(i) = r + \ell_B(i) - \ell_A(i), \quad \lambda_B(i) = s - r + \ell_A(i) - \ell_B(i).$$

These are its surplus evaluation lengths $|v| - |u|$. They lie in $[0, s]$, and

$$\lambda_A(i) + \lambda_B(i) = s, \quad |\lambda_A(i) + \lambda_B(j) - s| \leq b \quad (|i - j| \leq 1). \quad (4.9)$$

The last inequality follows because a height step changes just one input length by b .

Lemma 4.5 (Terminal compatibility). *If both incident common cutoffs equal the receiving vertex's own cutoff, the terminal constraints from two independent incident evaluations have product-law probability exactly $2^{-\lambda(v)}$. If each incident common cutoff is at most b bits below that receiver cutoff, one can test an event on those two current tables with probability at most*

$$2^{-\max\{0, \lambda(v) - 4b\}}.$$

The keys must be distinct between the tested edge occurrences; the input arguments must be known before the current tables are exposed, and the required terminal gates must be open. Only the actual uniform active prefix of a partial permutation is used in this statement.

Proof. Apart from the initial short block, which is never terminal, there are two terminal forms. If the final active input has t bits in its s -bit field, then its evaluation block has all s bits. The map in (4.7) sends an s -bit v and a t -bit u -subspace injectively into two s -bit evaluations. Its image has dimension $s + t$, so the compatibility codimension is $s - t$. The nonzero coefficient difference proves injectivity on the full field and therefore on the active subspace.

On the opposite side, the input has ended and the last evaluation has t bits. Its two incoming values are independent uniform t -bit values from the two active-prefix permutations. Since $u = 0$, they must agree after their known offsets are removed. This has codimension t . These two surpluses add to s . The boundary cases $t = 0, s$ simply move the one nonzero surplus to the other side. This proves the equal-cutoff statement.

For adjacent incidences the correction extras are fixed label data, and the lagged gates are determined by earlier raw blocks. For incidence a write its known correction as $o_a = C_a^v + K_a C_a^u$ and move the whole vector o_a to the other side of the equation. Its support may fill the output field; translation by it costs no random dimension. On a common active random prefix, with coordinate projection π and width w' , the map

$$(u, v) \mapsto (\pi(v + K_1 u - o_1), \pi(v + K_2 u - o_2))$$

has affine image of dimension at most $w' + t$ in $2w'$ bits. Thus its codimension is at least $\max\{0, w' - t\}$. This uses no invertibility of a truncated multiplier. For the terminal form with no input, replace t in this dimension count by zero.

Here are the active widths, including the boundary between blocks. Write $t \in \{b, 2b, \dots, s\}$ for the width of the last active block in the interleaved raw input stream at the receiver's height; each incident common cutoff is the receiver cutoff or is b bits earlier. If $t > b$, both evaluations of the partial-input side occur in the preceding source block and have the full s random bits. Their codimension is at least $s - t$. On the evaluation-only side the common random prefix has width at least $t - b$, giving that many compatibility bits. If $t = b$, the cutoff can cross a block boundary. On the partial-input side the preceding source block still has full random width: it is either ordinary or the full active prefix of a marked block. It therefore still contributes $s - b$ compatibility bits. On the opposite side a lower incidence can already be in its deterministic post-mark block. Its surplus is only b , so we use the certain event and charge zero, as permitted by $\max\{0, b - 4b\} = 0$. The same reasoning applies with A and B interchanged; $t = s$ includes the boundary represented by $t = 0$ on the other side. All retained positive-width outputs are actual independent uniform current-table prefixes. These bounds are stronger than the stated conservative loss of $4b$, and prove the probability assertion. $\square$

Lemma 4.6 (A repeated terminal key identifies receiver prefixes). *Suppose two distinct tested terminal edge occurrences at the same source block use the same table key. Their receiving vertices have the same side and agree in raw u and v through available aligned depth at least*

$$H_* - 8s,$$

where H_* is the minimum scale among the vertices of the counted path or theta. At any nonterminal update, equality of keys likewise identifies both receiver prefixes through the preceding complete receiver block.

Proof. At a B-block j update the queried table is $P_{A,j-1}$. Equality of its key identifies the family, full label, A raw inputs through block $j - 2$, and B raw inputs through block $j - 1$. In particular the B u-prefixes through $j - 1$ agree. To compute their B v-prefixes through $j - 1$, the equations use A inputs through $j - 2$, the same earlier keys, and the same coefficient and label offsets. Correction gates use still earlier raw pairs. Thus the evaluation prefixes agree, and $v = Q - Ku$ gives agreement of the v prefixes. At an A-block j update the queried table $P_{B,j}$ identifies A raw inputs through $j - 1$ and B raw inputs through $j - 1$. The A evaluation through $j - 1$ uses only this B data and earlier keys, giving the identical conclusion.

More explicitly, if the current table has source index k , its preceding complete receiver prefix has aligned depth exactly ks : for $P_{A,j-1}$ this is $2(j - 1)s$, and for $P_{B,j}$ it is $(2j - 1)s$. Write κ for the final common source index of an incident edge and $t \in \{b, 2b, \dots, s\}$ for its active length. Its common scale is $H_c = \kappa s + t$. The two terminal forms test source index $\kappa - 1$ or κ , respectively. Thus $ks \geq H_c - 2s$. The common scale is at least the receiving vertex scale minus b , and hence $ks \geq H_* - 2s - b$. If two adjacent incidences straddle a block boundary, restricting both to their

earlier common receiver prefix loses at most one more aligned step, of length $2s$. This is still greater than $H_* - 8s$ for $b < s$. Every prefix just used ends before the first partial receiving block and is therefore active at both receivers. Repeated keys at the same receiving vertex cannot occur among its two chosen slots: their full label names are different. $\square$

4.5 A deterministic end guard for minimal returns

Terminal compatibility probabilities multiply when the tested tables are distinct. For a minimal coherent return, a repeated key would identify a shorter coherent subpath unless both receivers lie near opposite ends. We omit terminal tests in fixed end intervals to remove this exception without conditioning the law on minimality.

Fix a coherent pattern of length p , initial scale H , and margin $M = M_k$. Let H_* be the minimum of its vertex scales. The height assignment is fixed and

$$H - bp \leq H_* \leq H_v \leq H + bp \quad \text{at every candidate vertex.} \quad (4.10)$$

Set

$$E_p = 100(s + bp), \quad J_p = \lceil 4E_p/s \rceil + 4.$$

The end guard consists of all receiver positions at distance at most J_p from either end of the abstract path. It depends only on the fixed pattern data, not on the tables.

Lemma 4.7 (Key separation outside the end guard). *In every minimal coherent return, all terminal keys tested outside the end guard are distinct at each source block.*

Proof. By the available endpoint depth,

$$p \leq 2(H_* - M + E_p)/s.$$

Suppose two keys at terminal receiver positions $a < b$ coincide. Lemma 4.6 gives agreement through depth at least $H_* - E_p$. The receivers have the same side. Their contiguous path segment is nonbacktracking and lies in the same band. If

$$(b - a)s/2 + M \leq H_* - E_p,$$

it is a proper coherent return, contrary to minimality, unless it is the entire original path. The latter case already has both positions in the guards. Otherwise

$$b - a > 2(H_* - M - E_p)/s \geq p - 4E_p/s.$$

It follows that $a < 4E_p/s$ and $p - b < 4E_p/s$. Both positions again lie in the guards. This proves the claim using only contiguous subpaths of the original return; the prefix identification does not close a new path. $\square$

In a counting experiment we reject a pattern at an update if its selected terminal keys outside the guard coincide, or if a required terminal gate is closed. These are predictable rejections because keys and gates are known in $\mathcal{F}_{<t}$. Lemma 4.7 says that no actual minimal return in X^{good} is lost by this rule. We do not condition on the return being minimal.

Choose one terminal compatibility test at each unguarded internal vertex with positive surplus. At a given update these tests use distinct tables and their product-law probabilities multiply. Pair successive internal vertices along the path in (4.9). Omitting the two endpoints and the guards

loses at most $(2J_p + 4)s$ surplus bits; the adjacent cutoff allowance loses at most $4bp$. Therefore the predictable terminal tests accumulated by an actual minimal return have total rank at least

$$R_{\text{ret}} \geq ps/2 - 10^4(s + bp). \quad (4.11)$$

The large absolute constant merely dominates the displayed guard, pairing, and $4b$ losses. If the right-hand side is negative the assertion follows from nonnegativity of rank. At most $2p$ whole tables are charged. Untested current outputs still participate in later formal reconstruction, but are integrated out in the current estimate.

Lemma 4.8 (Expected coherent-return count). *For sufficiently large s/b and then H_0/s , the expected number of minimal coherent returns based on either side at height i , counted over all bands containing them, is at most*

$$2^{.975H_i}.$$

This counts oriented rooted paths, including repetitions of vertices and edges.

Proof. Put $H = H_i$. The first endpoint has scale H , so (4.4) gives $p \leq 2(H - M)/s \leq 2H/s$. There are a fixed number of eligible bands and at most L_s^p label/height patterns of a given length. By Lemma 4.3, their free endpoint coordinates have at most

$$2^{H-h+bp}$$

choices. Each pattern and coordinate choice has at most one actual realization. Apply Lemma 4.4 using (4.11); the event being bounded is that this realization is a minimal return in the good graph. Its probability is at most

$$\exp(2\epsilon_s p) 2^{-ps/2 + 10^4(s+bp)}.$$

Thus its contribution, summed over label patterns and free bits, is at most

$$2^{H + .001sp - M + 10^5(s+bp)}.$$

Here we have used $h \geq ps/2 + M$, and the larger error constant also absorbs the exponential inflation; for large s , $\epsilon_s \leq 1$, and $b \geq 1$.

Since $M \geq .03H$ and $sp \leq 2H$, the exponent is at most

$$.972H + 10^5(s + 2bH/s).$$

Choose, for example, $s/b \geq 10^{12}$ and $H_0/s \geq 10^{12}$, in addition to the earlier requirements. The error here is less than $.001H$. The finite sum over lengths and bands is bounded by a fixed polynomial in H , which is absorbed by another $.002H$ on increasing H_0 if necessary. This proves the stated bound. These numerical choices are only convenient sufficient choices; no degree grows with tower height. $\square$

Delete every vertex on every minimal coherent return in every band, and denote the resulting induced graph by $X^{(1)}$.

Lemma 4.9 (The first deletion gives base girth). *No nonempty cyclically reduced closed path based at height i in $X^{(1)}$ has length at most $.93m_i$.*

Proof. Such a path lies in one of the interior bands chosen above. Its endpoints are the same vertex, and its available prefix depth is at least $H_i - 2s$. Its required depth is at most

$$\left(\frac{.93}{2(.50005)} + .05 \right) H_i < .98H_i.$$

The least aligned depth above this number adds less than $2s$. For sufficiently large H_0/s it is still available. Hence this path is a coherent return for that band's margin and contains a minimal return. At least one of its vertices, in fact all vertices of that minimal return, was deleted. This is a contradiction. $\square$

4.6 Theta constraints

Fix a bicycle of length $e < 1.52m_i$ with a root at height i in $X^{(1)}$, and put $H = H_i$. Along it $|H_v - H| \leq be$. A figure-eight or barbell contains two edge-disjoint cycles. By Lemma 4.9, their total length is greater than

$$1.86(1 - 1.52a)m_i > 1.52m_i, \quad a = b/\log_2 q,$$

for our choice of a . Thus only a theta is possible. Its shortest branch has length at most $e/3 < .51m_i$, and any two vertices are joined by a path of length at most $e/2 < .76m_i$: two edge-disjoint connecting paths have combined length at most e . Every theta vertex lies on a cycle of length at most e , so we may also restrict the count to $e > .9m_i$.

For a theta, the terminal tests will impose approximately $se/2$ compatibility bits. Against approximately se label bits, this alone does not put the count below the layer size 2^H throughout the required range of e . The additional saving comes from the third evaluation at each of the two branch vertices. We estimate those relations first and then arrange their tests at different source updates from the terminal tests.

Lemma 4.10 (Terminal keys in a surviving theta). *Let a theta in $X^{(1)}$ have $e < 1.52m_i$ edges and a root at height i , and set $H = H_i$. At every source-block update, the terminal keys used at its distinct vertices are distinct. If its two branch vertices have the same side, any key shared by their branch triples at a depth beyond $.65H$ is impossible, apart from a fixed initial/final block allowance that is omitted below.*

Proof. A repeated terminal key gives receiver-prefix agreement through $H - be - 8s$ by Lemma 4.6. Join the receivers by a shortest theta path. It has length below $.76m_i$ and lies in an interior band containing the theta. Its coherent threshold, including its margin, is at most

$$\left(\frac{.76}{2(.50005)} + .05 \right) H < .81H.$$

This is smaller than the available agreement depth for sufficiently small b/s and large H_0/s . The path is a coherent return, contradicting its survival in $X^{(1)}$.

For a shared key at the branch vertices, the last sentence of Lemma 4.6 identifies their preceding complete receiver prefixes. A source update at aligned depth at least $.65H$ identifies them to depth at least $.65H - 4s$. Their shortest branch has length below $.51m_i$, whose coherent threshold is less than

$$\left(\frac{.51}{2(.50005)} + .05 \right) H < .56H.$$

Again the available margin is much larger than the block and height errors. This gives the same contradiction. Equality of unordered triples in particular would give a shared key. $\square$

At a full ordinary receiving block, the three evaluations at a branch vertex satisfy

$$(c_2 - c_3)E_1 + (c_3 - c_1)E_2 + (c_1 - c_2)E_3 = 0. \quad (4.12)$$

The three coefficients are nonzero. Corrections have been incorporated into the known offsets of the E_j .

Lemma 4.11 (Ranks at the two branch vertices). *One branch-vertex relation in an ordinary full block with three open gates has product-law probability 2^{-s} . If two same-side branch vertices have different unordered key triples, their two relations have joint probability 2^{-2s} . If their triples are the same, it is always safe to charge just one relation.*

Proof. The three slots at one branch vertex have different labels and hence different keys. At their already known input arguments, the three fresh table values are independent uniform field elements. Equation (4.12) can be solved for any one of them and has probability 2^{-s} .

Two distinct three-element key sets each contain a key absent from the other set. Condition on all the *other whole tables*, leaving these two exclusive tables unexposed. Each exclusive table is used at one known argument in its branch-vertex equation. Its value is uniform and has a nonzero coefficient; it does not enter the other branch equation. The two equations therefore have probability 2^{-2s} . If other formal recovery queries use those same tables, their results have not been prescribed and are integrated out. With identical triples retain only one of the two necessary equations, giving the last assertion. $\square$

We now separate branch tests from terminal tests explicitly. All terminal updates of the fixed theta occur in the source-index interval

$$\left[\left\lfloor \frac{H - be}{s} \right\rfloor - 4, \left\lceil \frac{H + be}{s} \right\rceil + 2 \right].$$

Indeed its common incidence scales lie between $H - be - b$ and $H + be$, and the calculation in Lemma 4.6 places every terminal test at $\kappa - 1$ or κ when $H_c = \kappa s + t$. The displayed interval contains at most $9 + 2be/s$ integers. Enlarge this interval by ten indices at either end and omit it from branch charges. Also omit the first ten source indices and every marked split block or deterministic post-mark identity block at a branch-vertex slot. These omissions are fixed by the pattern and the universal mark locations. In particular, a current terminal test and a current branch test never use the same source-block update.

The mark spacing has been chosen so that, over the finite union of the six branch-vertex slot sequences and separately for both source parities, the deterministic omitted blocks cost less than $.001H + O(s + be)$ bits. The gate deletion is likewise normalized separately on A-source and B-source blocks of each slot. A vertex is good only when the proportion of closed ordinary gates in each such sequence is below .005. Thus the union of bad gates at the three slots of each of the two branch vertices costs at most $.015H + O(s + be)$ bits. This denominator convention is needed: a bound normalized on all interleaved indices without separating parity would permit twice this loss.

Before depth $.65H$, charge only the first chosen branch vertex when the two branch vertices have opposite sides. When they have the same side and their triples coincide, again charge one; when their triples differ we may charge either one or both. After that depth charge both branch vertices when their gates are open, rejecting any forbidden shared key predictably. All selection decisions are made from $\mathcal{F}_{< t}$. Each branch vertex has $H/(2s) + O(1 + be/s)$ full updates before the terminal band. Charging one before $.65H$ and two afterward gives at least

$$(.65/2 + .35)H - O(s + be) = .675H - O(s + be)$$

bits. Lemmas 4.10 and 4.11, followed by the two omitted-block budgets, consequently give

$$R_{\text{branch}} \geq .675H - .015H - .001H - 10^4(s + be) > .64H. \quad (4.13)$$

The last inequality follows from the same sufficient choices $s/b \geq 10^{12}$ and $H_0/s \geq 10^{12}$, enlarged if earlier finite constants require it.

For terminal tests, choose two slots at every theta vertex. Lemma 4.10 gives distinct keys, and the last-three-update gate requirement makes the necessary gates open. Pair adjacent internal vertices on each of its three branches in (4.9); there are at most six unpaired or branch vertices. The theta has $e - 1$ vertices. Lemma 4.5 therefore gives

$$R_{\text{term}} \geq se/2 - 100(s + be). \quad (4.14)$$

Every terminal update was omitted from the branch charges. The two rank contributions therefore compose chronologically under Lemma 4.4. Exclusivity among branch triples was used only to compute the branch rank at a single update; it does not assert independence from terminal tests.

Lemma 4.12 (Expected theta count). *For sufficiently large allowed parameters, the expected number of rooted embedded bicycles in $X^{(1)}$ of length below $1.52m_i$, based on either side at height i , is at most*

$$2^{.89H_i}.$$

Proof. Only theta patterns need be considered. For a fixed total length e , choose its three ordered positive branch lengths, the position and orientation of its root, and its bipartite side. The number of these choices is at most $100(e + 1)^4$. For example there are fewer than $(e + 1)^2$ length triples, at most $2e$ oriented root positions, two side choices, and a fixed finite number of choices of the first branch. Give its edges orientations for enumeration and choose at most L_s^e label/height patterns. Assigning heights along a spanning tree determines the remaining heights; any inconsistency on a remaining edge causes rejection.

Lemma 4.3 has no free coordinates for a theta. For each fixed pattern use the predictable terminal and branch tests above. Every actual theta in $X^{(1)}$ passes their key and gate rejections and has the stated accumulated ranks. The current-table count is at most $2(e - 1)$ for the terminal tests plus six per branch update. Since $e > .9m_i$, the latter count is $O(H_i/s) = O(e)$; a bound of $20e$ suffices after omitting the fixed initial updates. Lemma 4.4 bounds its probability by

$$\exp(20\epsilon_s e) 2^{-se/2 - .64H_i + 100(s + be)}.$$

After summing patterns, the exponent is at most

$$(1.001 - .5)se - .64H_i + 10^5(s + be).$$

For $e < 1.52H_i/ (.50005s)$ its main coefficient of H_i is

$$\frac{.501(1.52)}{.50005} - .64 = .88288771122 \dots < .883.$$

The error and the polynomial sum over e fit below $.007H_i$ by the allowed parameter choices. This proves the bound. Embeddedness and survival in $X^{(1)}$ are final predicates on the reconstructed assignment; they were not used to condition any current table law. $\square$

Delete every vertex on each bicycle with length less than $1.52m_i$ for at least one of its vertices of height i . Let X_{old} be the resulting induced graph.

4.7 From bicycle pruning to Heisenberg nonlaws

The counting estimates for both deletions are now established. Before choosing a realization with small losses in every layer, we derive the deterministic consequence of bicycle exclusion that the voltage choice requires.

We use the convention $[a, b] = aba^{-1}b^{-1}$ and write

$$\mathbf{H} = \langle x, y, z \mid [x, y] = z, [x, z] = [y, z] = 1 \rangle.$$

In particular, x and z have infinite order. This also follows directly from the realization of $\mathbf{H}$ by upper triangular integral 3×3 matrices with diagonal entries equal to one: take $x = I + E_{12}$, $y = I + E_{23}$, and $z = I + E_{13}$.

Choose an orientation of each unoriented edge of a graph X , and associate a distinct variable t_e to each such edge. An edge path determines a word by reading t_e on a positive traversal and t_e^{-1} on a negative traversal. Its word is an $\mathbf{H}$ -law if it evaluates to the identity for every independent assignment $t_e \in \mathbf{H}$. Assign each unoriented edge a positive length $\ell(e)$, and count every traversal with this length. For a subgraph S , set $\ell(S) = \sum_{e \in E(S)} \ell(e)$. Our application has unit edge lengths. The weighted formulation also allows lengths to record subdivision; alternatively one can apply the Lemma directly to the subdivided graph with a separate variable on each unit edge.

The edge-neutral analogue, in which each edge is traversed equally often in both directions, is the bicycle characterization of abelian girth in Friedman–Izsak–Silberman [14, Lemma 2.6]. The two-variable Heisenberg substitutions below give the stronger conclusion that every edge of a suitable bicycle is traversed at least four times.

Lemma 4.13 (Heisenberg laws contain a bicycle used four times). *Let W be a nonempty closed edge path in X , reduced as a based edge path: no two consecutive traversals in its linear ordering are mutually inverse. Suppose the word of W is a $\mathbf{H}$ -law. Then its support contains a bicycle B such that every edge of B is traversed at least four times by W . Consequently,*

$$\ell(W) \geq 4\ell(B).$$

In particular, if every bicycle in the support of W has length at least T , then $\ell(W) \geq 4T$.

Proof. We first use one-variable substitutions to balance every edge. We then use two-variable substitutions to isolate a balanced closed path supported on edges traversed at least four times. Its support will contain the required bicycle.

For an oriented edge e , let $n_e(V)$ denote the number of its positive traversals minus the number of its negative traversals in an edge word V . Assigning $t_e = x$ and all other variables the identity gives $x^{n_e(W)} = 1$, so

$$n_e(W) = 0 \quad \text{for every edge } e. \tag{4.15}$$

Thus every used edge has positive even traversal multiplicity. Call an edge *paired* if its multiplicity in W is exactly two, and call it *heavy* if its multiplicity is at least four.

We first examine a paired edge e . Its two signs are opposite by (4.15). Starting the cyclic word at its first occurrence, write

$$W = t_e^\varepsilon U t_e^{-\varepsilon} V, \quad \varepsilon \in \{1, -1\}.$$

Cyclically rotating a law preserves lawhood, since a cyclic rotation of a word is a conjugate of it under every assignment. Fix an edge $f \neq e$ and assign $t_e = x$, $t_f = y$, and all remaining variables the identity. Put $n = n_f(U)$. Equation (4.15) gives $n_f(V) = -n$, and the assigned word becomes

$$x^\varepsilon y^n x^{-\varepsilon} y^{-n} = z^{\varepsilon n}.$$

The infinite order of z implies $n_f(U) = 0$. The edge e does not occur inside U , so we have proved the following interval property:

$$n_f(U) = 0 \quad \text{for every edge } f. \quad (4.16)$$

Thus the open interval between the occurrences of a paired edge is balanced on every edge. The two-variable substitution gives the required pairwise area condition directly.

Fix the original linear ordering of the occurrences in W , and join the two positions of each paired edge by an interval. These intervals cannot cross. Indeed, if the intervals for e and f crossed, the open interval for e would contain exactly one occurrence of f , contradicting (4.16). If paired edges exist, choose an inclusion-minimal paired interval, and let U be its interior edge path. Every edge occurring in U is heavy: if a paired edge occurred there, the interval property would force both of its occurrences to lie there, giving a smaller paired interval.

Moreover, U is a closed edge path, because the inside endpoints of the two opposite traversals of the same edge coincide. It is nonempty, since an empty interior would give an immediate reversal in W . It is reduced as a based path, being a contiguous subpath of W , and it is balanced by (4.16). If there are no paired edges, take $U = W$; these same properties follow from the hypotheses and (4.15), and every edge of U is again heavy in W .

We claim that the connected support S of U has first Betti number at least two. Here it matters that U need not be cyclically reduced at its basepoint. Repeatedly remove its first and last traversals whenever they are mutually inverse. This expresses U as PCP^{-1} , where C is a cyclically reduced closed path. The path C is nonempty: if this process reached the empty path, its last removal would have removed two adjacent inverse traversals from the original reduced path. Each removal subtracts opposite traversals of the same edge, so C is still balanced.

Deleting a vertex of valence one and its incident edge from a finite graph cannot affect a cyclically reduced closed path: a visit to that vertex would require an immediate reversal. Such deletions also preserve first Betti number as long as an edge remains. If S had first Betti number zero, this deletion process would leave no edges, contradicting the existence of C . If S had first Betti number one, the remaining nonempty core would be a single cycle. To see the last assertion, the core is connected with minimum valence two and satisfies $|E| = |V|$; hence all its valences are exactly two. A nonempty cyclically reduced path on a cycle traverses it a nonzero number of times in one fixed direction. Its exponent sum on each cycle edge is therefore nonzero, contradicting the balance of C . This proves the claim, including when the basepoint of U lies on a path leading to its cyclic core.

Finally, choose a spanning tree of S and two distinct edges outside that tree. Let T' be the smallest subtree containing the endpoints of those two edges, and let B be their union with T' . The subgraph B is connected and has first Betti number two. Every leaf of T' is an endpoint of one of the added edges, so B has no vertices of valence one; the case where T' is a single vertex gives the same conclusion directly. Thus B is a bicycle contained in S . Each of its edges is heavy in W , and positivity of the edge lengths gives

$$\ell(W) = \sum_{e \in \text{supp}(W)} (\text{traversal multiplicity of } e) \ell(e) \geq 4 \sum_{e \in E(S)} \ell(e) \geq 4\ell(B).$$

The asserted lower bound follows. □

At a fixed comparison scale $m > 0$, exclusion of bicycles of length strictly below $1.52m$ therefore gives

$$\ell(W) \geq 4(1.52)m = 6.08m$$

for any H-law W whose support lies in the region to which that exclusion applies. In particular, every nonempty cyclically reduced closed path there of length less than $6.03m$ is a nonlaw. The final

proposition uses the gap between 6.08 and 6.04 to allow scale variation along a path; the smaller cutoff 6.03 leaves room for the integer rounding in the voltage selection.

The factor four in the lemma is sharp, and the lower bound is non-strict. Indeed, on a bouquet with two edges a, b , the cyclically reduced word

$$aba^{-1}b^{-1}a^{-1}bab^{-1} = [a, b][a^{-1}, b]$$

is a H-law and has length $4\ell(a) + 4\ell(b)$, attaining equality for its bicycle support. The same equality holds after subdividing the two cycles, including subdivisions with even cycle lengths.

4.8 Simultaneous small deletion in every layer

We now combine the two expectation bounds with the deterministic word lemma. The decay in H_i makes the total expected deleted fraction summable over all heights, so one table realization has every required property at once.

Proposition 4.14 (Pruned old graph). *Given any $\varepsilon > 0$, the allowed parameters and the universal tables can be chosen so that the additional two deletions in this Section remove less than an ε fraction of each $\mathcal{A}_i$ and $\mathcal{B}_i$. In the surviving induced old graph X_{old} :*

1. *the folded labeling, padded-prefix recovery of Lemma 2.4, and global word uniqueness of Proposition 2.6 remain valid;*
2. *a nonempty cyclically reduced closed path based at height i has length greater than $.93m_i$;*
3. *every bicycle containing a vertex of height i has length at least $1.52m_i$;*
4. *every nonempty cyclically reduced closed path based at height i with length at most $6.04m_i$ is not a law in the independent edge variables of H .*

The expansion conclusions for the undeleted incidences hold simultaneously, and the degrees remain fixed along the tower.

Proof. For a realization, let Z_i^A, Z_i^B be the additional deleted fractions. Bounds on rooted objects imply bounds on all their vertices, with only polynomial and small scale factors, as follows. An object of length ℓ rooted at height j can meet layer i only if $|i - j| \leq \ell$. Every counted object satisfies $\ell \leq 4H_j/s$. From $H_j = H_i + b(j - i)$ and $s \geq 20b$ we obtain

$$|i - j| \leq 5H_i/s, \quad |H_j - H_i| \leq 5bH_i/s.$$

Its vertex count is at most $\ell + 1 \leq 5H_i/s + 1$. There are at most $10H_i/s + 1$ possible root layers and at most a linear number of lengths. Lemmas 4.8 and 4.12 therefore imply, after division by $|\mathcal{B}_i| = 2^{H_i}$ or $|\mathcal{A}_i| = 2^{H_i+2r-s}$,

$$\mathbb{E}(Z_i^A + Z_i^B) \leq 2^{-.01H_i} \tag{4.17}$$

for sufficiently large s/b and H_0/s . More explicitly, the larger rooted bound contributes $2^{-.025H_i + .975(5bH_i/s)}$ times a fixed polynomial in H_i . The coefficient remains below $-.02$ for small b/s , and that polynomial is absorbed by another $.01H_i$ after increasing H_0 . This proves (4.17), including the cost of deleting vertices of an object whose qualifying root is in a different layer.

Since $H_i = H_0 + bi$,

$$\sum_{i \geq 0} \mathbb{E}(Z_i^A + Z_i^B) \leq \frac{2^{-.01H_0}}{1 - 2^{-.01b}}.$$

Choose H_0 still larger so the last expression is less than $\varepsilon/2$. Markov's inequality then gives positive probability that $\sum_i (Z_i^A + Z_i^B) < \varepsilon$. On that event each individual deleted fraction is below ε . Every table realization in the sequential law already has the undeleted expansion properties, so choose one in this positive-probability event. The earlier good-vertex deletion budget is separately supplied by Proposition 2.6 and can be made as small as required there.

Passing to an induced subgraph preserves foldedness and uniqueness of a word that occurred at most once before deletion. Lemma 4.9 proves the second assertion, and the definition of the second deletion proves the third. It remains to deduce the nonlaw assertion, allowing for the variation of scale along the closed path.

Let W be a nonempty cyclically reduced closed path of length $L \leq 6.04m_i$ based at height i in X_{old} . Every vertex v of its support has scale $m(v) \geq m_i - aL \geq (1 - 6.04a)m_i$. If W were a H-law, Lemma 4.13 would give a bicycle B in its support with $L \geq 4|E(B)|$. The third assertion, already proved by the deletion, gives

$$|E(B)| \geq 1.52(1 - 6.04a)m_i.$$

Consequently

$$L \geq 6.08(1 - 6.04a)m_i > 6.04m_i,$$

where the last inequality holds for $a \leq 10^{-6}$. This contradicts the choice of W . Independent edge variables here include every underlying edge, regardless of any scalar weight assigned later; reverse traversals use the inverse of the same variable. $\square$

5 Diffuse Heisenberg symbols and voltage approximation

We assign a Heisenberg voltage and a bounded matrix weight to every surviving old edge. The weighted incidences will approximate prescribed compact Bott symbols in the *full* regular operator norm, while the voltages will separate all the short closed walks specified below. These requirements use different features of the same distribution: its weighted mean realizes the symbol, and its diffuse horizontal coordinates prevent trivial holonomy. The horizontal coordinates remain diffuse when the matrix weight is zero.

For the voltage estimates we fix a surviving tower furnished by Proposition 4.14, with any prescribed finite collection of incidence types. All subsequent edge draws are independent on this fixed graph; the sequential table law has already served to select its incidences. Before deletion, a type between an $\mathcal{A}_i$ -layer and a $\mathcal{B}_k$ -layer, $|i - k| \leq 1$, is biregular. Denote its degrees by $d_{\tau,ik}$ and $D_{\tau,ik}$, its normalizing factor by

$$q_{\tau,ik} = (d_{\tau,ik}D_{\tau,ik})^{1/2},$$

and its surviving normalized incidence by $U_{\tau,ik}$. All normalizations use the original degrees. For fixed types and fixed b , there are positive constants c_*, C_* , independent of height and of sufficiently large s , such that

$$c_*d \leq d_{\tau,ik} \leq C_*d, \quad c_*D \leq D_{\tau,ik} \leq C_*D, \quad c_*q \leq q_{\tau,ik} \leq C_*q. \quad (5.1)$$

In particular $\|U_{\tau,ik}\| \leq 1$, because it is a compression of a normalized biregular incidence. The full surviving base graph has maximum degree at most C_*D . We shall use its base-girth bound $> .93m_i$ on comparable layers, and its local absence of Heisenberg laws of length at most $6.04m_i$ based at height i . These are inputs from Proposition 4.14; its proof applies Lemma 4.13 with the appropriate local scale comparison. The base-girth bound will control the centered moments, and the nonlaw bound will control the probability of trivial holonomy.

For definiteness the formulas are written in the left regular representation λ of H . The inversion unitary $\mathcal{I}\delta_g = \delta_{g^{-1}}$ carries $\lambda(a)$ to the right regular representation $\rho(a)\delta_g = \delta_{ga^{-1}}$. Thus all statements, including the matrix-valued ones, hold in either convention. The voltage and incidence orientations are fixed as follows. A positive old edge with label σ runs from A_v to B_w , and its forward voltage map is $(v, f) \mapsto (w, fh_e)$. The block $U_{\tau, ik}$ maps B-coordinates to A-coordinates, so its deck entry acts by

$$\rho(h_e)\delta_f = \delta_{fh_e^{-1}}.$$

Thus the sampled convolution variable v_e below is assigned as $h_e = v_e$; the reverse block is its adjoint, with inverse voltage. Equation (8.3) later identifies this same deck action in the embedded Cayley graph.

5.1 Compact Bott representatives and fixed smooth families

We use a smooth compactly supported form of the classical Bott representative associated to the Hopf line bundle; compare [2, Section 2]. The following formula fixes its support and orientation conventions. Put

$$h(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \zeta(r) = \frac{h(r - 1/4)}{h(r - 1/4) + h(1 - r)}, \quad r \geq 0,$$

and set $\theta(r) = 2r + (\pi - 2r)\zeta(r)$. Thus $\theta(r) = 2r$ near zero and $\theta(r) = \pi$ for $r \geq 1$, with a smooth constant extension at the latter endpoint. For $r = (t^2 + u^2)^{1/2} > 0$, define

$$p(t, u) = \frac{1}{2} \begin{pmatrix} 1 + \cos \theta(r) & \frac{\sin \theta(r)}{r}(t - iu) \\ \frac{\sin \theta(r)}{r}(t + iu) & 1 - \cos \theta(r) \end{pmatrix}, \quad e_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.2)$$

At zero use the continuous extension. The functions $\sin(2r)/r$ and $\cos(2r)$ are analytic functions of r^2 near zero; hence this extension is smooth. At $r = 1$ the difference from the constant matrix is flat. Direct multiplication, or the identity $(n \cdot \sigma)^2 = 1$ for the unit vector

$$n = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

shows that $p = p^* = p^2$ and $\text{Tr } p = 1$. In particular $p - e_\infty$ is a smooth compactly supported matrix function.

The compactification of this projection has degree of absolute value one. Indeed, its local unit eigenvectors are

$$\begin{pmatrix} \cos(\theta/2) \\ e^{i\phi} \sin(\theta/2) \end{pmatrix} \quad \text{near the origin,} \quad \begin{pmatrix} e^{-i\phi} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} \quad \text{near the exterior constant region.}$$

On their overlap the second is $e^{-i\phi}$ times the first, so the clutching function has winding number -1 . Equivalently, with the corresponding choice of Chern-form sign,

$$\frac{1}{2\pi i} \int_{\mathbb{R}^2} \text{Tr}(p \, dp \wedge dp) = \frac{1}{2} \int_0^1 \theta'(r) \sin \theta(r) \, dr = 1.$$

The winding calculation is independent of this sign convention. Thus the rank-zero difference represents one of the two Bott generators.

Fix

$$\alpha = \frac{\sqrt{5} - 1}{2}, \quad N_i = 2^{200H_i}, \quad Z_i = 2^{500H_i}, \quad J_i = 2^{50H_i}.$$

The commuting pair (z, x) identifies its group algebra with the continuous functions of torus coordinates (ϑ, ξ) . In fixed coordinate charts around $(\alpha, 0)$, set

$$\mathbf{p}_i(\vartheta, \xi) = p(Z_i(\vartheta - \alpha), N_i\xi), \quad \beta_i = \mathbf{p}_i - e_\infty, \quad (5.3)$$

and extend $\mathbf{p}_i$ by e_∞ . The initial height is chosen so that every support lies inside the charts. A change of variables in the preceding integral, or the same clutching function, proves that every β_i is the rank-zero Bott generator up to the fixed orientation sign.

Adjacent representatives are joined by the explicit rescaling

$$p(Z_i^{1-v} Z_k^v(\vartheta - \alpha), N_i^{1-v} N_k^v \xi), \quad 0 \leq v \leq 1, \quad |i - k| = 1.$$

Since $H_k - H_i = \pm b$, the ratios of these scales are fixed. In rescaled coordinates this is a compact smooth family, with all derivatives bounded independently of height. The matrix rotations

$$R(v) = \begin{pmatrix} \cos(\pi v/2) & -\sin(\pi v/2) \\ \sin(\pi v/2) & \cos(\pi v/2) \end{pmatrix} \otimes I_2$$

act by conjugation on $\mathbf{p}_i \oplus e_\infty$, joining it to $e_\infty \oplus \mathbf{p}_i$. For sampling we subtract the constant projection $e_\infty \oplus e_\infty$ from this rotation family and e_∞ from the rescaling family. Entries of these compact differences, their rescalings, and smooth cutoffs equal to one on their supports belong to a fixed compact smooth profile family. One may choose all cutoffs inside rectangles larger by a constant depending only on b . In particular, the carrier at height $i + 1$ may contain the supports of the height- i profiles used in the adjacent homotopy. Throughout the final accuracy sequence we keep b fixed, so these profile bounds and moduli of continuity are uniform also in the construction index.

More generally we will use matrix-valued functions of the form

$$f_i(\vartheta, \xi) = F(Z_i(\vartheta - \alpha), N_i\xi), \quad (5.4)$$

where F ranges over a fixed compact family in $C_c^\infty(\mathbb{R}^2; M_\ell(\mathbb{C}))$, supported in one fixed rectangle. Zero functions and any prescribed height selections are allowed. The matrix dimension ℓ is fixed. Passing between adjacent reference heights merely enlarges this profile family by fixed rescalings.

5.2 Spreading on disjoint torus translates

We record explicitly the separation property used here. If p is the nearest integer to $n\alpha$, and $\alpha' = (-\sqrt{5} - 1)/2$, then

$$(n\alpha - p)(n\alpha' - p) = p^2 + np - n^2$$

is a nonzero integer. Moreover $|n\alpha' - p| \leq \sqrt{5}|n| + 1/2 \leq 3|n|$ for $n \neq 0$. Consequently

$$\|n\alpha\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{1}{3|n|}, \quad n \neq 0. \quad (5.5)$$

Choose a nonzero real function $a \in C_c^\infty((-1, 1))$, and put

$$a_{ij} = \frac{a(j/J_i)}{(\sum_{k \in \mathbb{Z}} |a(k/J_i)|^2)^{1/2}}, \quad T_i = \sum_{j \in \mathbb{Z}} a_{ij} \lambda(y)^j.$$

All sums defining T_i are finite and $\sum_j |a_{ij}|^2 = 1$. Riemann sums show that the denominator is bounded above and below by positive constant multiples of $J_i^{1/2}$, once the initial height is large.

Let E_i be the Borel spectral projection of an enlarged rectangle

$$|\vartheta - \alpha| \leq R/Z_i, \quad |\xi| \leq R/N_i,$$

large enough to contain all symbols and cutoffs incident to height i . These Borel projections are auxiliary support operators; they are not asserted to belong to the continuous group algebra.

Lemma 5.1 (Supported spreading). *For sufficiently large initial height, the family of projections*

$$\lambda(y)^j E_i \lambda(y)^{-j}, \quad |j| \leq J_i,$$

is pairwise orthogonal. Consequently

$$E_i T_i^* T_i E_i = E_i, \quad P_i = T_i E_i T_i^*$$

is a projection, and

$$\tau_H(P_i) = \tau_H(E_i) \leq C_R 2^{-700H_i}. \quad (5.6)$$

If f is supported in both enlarged rectangles at heights i and k , and g in both enlarged rectangles at heights k and j , then

$$(T_i f T_k^*)(T_k g T_j^*) = T_i (fg) T_j^*, \quad (T_i f T_k^*)^* = T_k f^* T_i^*, \quad \|T_i f T_k^*\| \leq \|f\|.$$

Proof. The relation $[x, y] = z$ gives $y^j x y^{-j} = x z^{-j}$, so conjugating the torus rectangle changes the ξ -coordinate by $-j\vartheta$. If $0 < |j - k| \leq 2J_i$ and $|\vartheta - \alpha| \leq R/Z_i$,

$$\|(j - k)\vartheta\|_{\mathbb{R}/\mathbb{Z}} \geq \frac{1}{6J_i} - 2RJ_i/Z_i.$$

This exceeds $2R/N_i$ at all sufficiently large heights because $N_i = J_i^4$ and $Z_i = J_i^{10}$. The translated supports are disjoint. Expansion of $E_i T_i^* T_i E_i$ now leaves only its diagonal terms, proving the first identity. All subsequent product and norm identities follow by inserting the auxiliary support projections. The trace on the torus algebra is Haar measure: it vanishes on every nonidentity character $\lambda(z)^c \lambda(x)^n$, which determines its integral on continuous functions and hence on Borel spectral projections. The trace property and $E_i T_i^* T_i E_i = E_i$ give $\tau(P_i) = \tau(E_i)$, and the rectangle has area at most $4R^2/(Z_i N_i)$. In particular the number of translates does not multiply the trace of this isometric image. $\square$

5.3 The simultaneous approximation statement

The spread symbols now have the support and product identities needed for the Bott construction. We next realize any finite prescribed family of them by weighted graph incidences, while choosing the same voltages to eliminate short closed lifts.

Proposition 5.2 (Simultaneously sampled symbols). *Fix b , a finite matrix dimension, a positive accuracy η , and a finite list of compact smooth symbol profiles and height selections as above. The list may include cutoffs, both parities, the initial-height selection, and any finite mesh of the adjacent rescaling and rotation families. Allocate a prescribed incidence type for every required same or adjacent layer symbol. For sufficiently large s and then sufficiently large initial height, the surviving base graph admits a voltage on every edge and finite-net matrix weights such that the following hold simultaneously.*

1. The voltage cover has no nontrivial cyclically reduced closed walk of length at most $6.03m_i$ based at an old vertex of height i .
2. All weighted blocks may use the common denominator q . If $\mathcal{V}_{\tau,ik}$ is that normalized weighted convolution block and the prescribed symbol is $S_{\tau,ik} = T_i f_{\tau,ik} T_k^*$, then

$$\|\mathcal{V}_{\tau,ik} - U_{\tau,ik} \otimes S_{\tau,ik}\| \leq \eta$$

in the full regular operator norm, at every height. The tensor notation here means that each scalar incidence entry multiplies the same matrix-valued deck convolution symbol.

3. Weights are read from one finite set, independent of height. Appending their net indices to the existing edge labels preserves the labelled-graph uniqueness properties and gives

$$\log_2 |\mathcal{A}| \leq 1.00005s + O(\log s) + O_{b,\eta,\text{profiles}}(1) < \log_2(q^2)$$

for sufficiently large s . In particular $\sqrt{|\mathcal{A}|}/q$ tends to zero as s increases after the fixed data. Zero-weight edges are retained in the voltage cover.

The proof is completed in Section 5.8. It has three parts. A Fourier coefficient envelope will produce bounded matrix weights with the desired mean and independent diffuse horizontal coordinates. A polynomial zero estimate will then control the holonomy of each short nonlaw. Finally, a finite regular-norm test and a centered moment count will control the fluctuations of every weighted incidence block. Their failure probabilities are summable over the height, so one choice satisfies all the requirements.

5.4 A product envelope for the Fourier coefficients

To sample the two horizontal coordinates independently with small atoms, we need a coefficient bound that separates their contributions. The envelope below is uniform over the $O(J_i)$ possible y-powers; the sum of its x-frequency majorant is $O(J_i^{-1})$. These two factors compensate, giving both a uniform total coefficient mass and the product law used in Lemma 5.4. Control of the central Fourier tails will make every edge distribution finite.

Every element of $\mathbf{H}$ has a unique form $x^n y^l z^c$. For a matrix Fourier series, write

$$A = \sum_{n,l,c} A_{n,l,c} \lambda(x^n y^l z^c), \quad \|A\|_1 = \sum_{n,l,c} \|A_{n,l,c}\|.$$

This one-norm dominates the regular operator norm by the triangle inequality.

We first state an elementary rescaling estimate. If g is a smooth compactly supported matrix function on $\mathbb{R}$, periodized as $g(Z(\vartheta - \alpha))$ inside one chart, its central Fourier coefficients are

$$c_k = Z^{-1} e^{-2\pi i k \alpha} \hat{g}(k/Z).$$

For every nonnegative integer A , integration by parts $A + 2$ times, together with the trivial integral bound for $|k/Z| \leq 1$, gives

$$\sum_k (1 + |k|/Z)^A \|c_k\| \leq C_A \sum_{j=0}^{A+2} \|g^{(j)}\|_{L^1(\mathbb{R})}. \quad (5.7)$$

Indeed the pointwise bound after those integrations is a constant times $(1 + |k/Z|)^{-A-2}$ times the displayed derivative sum, and $Z^{-1} \sum_k (1 + |k/Z|)^{-2}$ is bounded uniformly for $Z \geq 1$. The same argument uses operator norms and therefore applies to a fixed matrix amplification without any scalar-entry assumption.

Lemma 5.3 (Fourier envelope). *Let $|i - k| \leq 1$, and let f belong to the profile family (5.4), expressed at reference height i . Write*

$$T_i f T_k^* = \sum_{n,l,c} A_{n,l,c} \lambda(x^n y^l z^c), \quad N = N_i, \quad Z = Z_i, \quad J = J_i.$$

For every pair of nonnegative integers A, P , with $P \geq 3$,

$$\sum_c (1 + |c|/Z)^A \|A_{n,l,c}\| \leq C_{A,P} N^{-1} (1 + |n|/N)^{-P} (1 + J \|n\alpha\|_{\mathbb{R}/\mathbb{Z}})^{-3}, \quad |l| \leq C_b J, \quad (5.8)$$

and the coefficients vanish for larger $|l|$. The constants are uniform in height and in the compact profile family. In particular these symbols have a uniformly bounded Fourier one-norm. For each $\eta > 0$ they can be truncated, with one-norm error at most η , to

$$|n| \leq L_\eta N, \quad |l| \leq C_b J, \quad |c| \leq L_\eta Z, \quad (5.9)$$

where L_η is independent of height.

Proof. We first bound the central coefficients at fixed n and l , and then sum the resulting horizontal majorant. Fourier transform only in ξ . With $t = Z(\vartheta - \alpha)$, its n -th coefficient is

$$f_n(\alpha + t/Z) = N^{-1} \Phi(t, n/N), \quad \Phi(t, v) = \int_{\mathbb{R}} F(t, u) e^{-2\pi i v u} du.$$

The function Φ is supported in one fixed compact t -interval and is Schwartz in v , uniformly with all t -derivatives and all profile parameters. For example integration by parts in u bounds $\|\partial_t^a \Phi(t, v)\|$ by $C_{a,M} (1 + |v|)^{-M}$ for every fixed a, M .

The Heisenberg commutation formula gives the coefficient at y -power l , before central Fourier transformation, as

$$f_n(\vartheta) S_{n,l}(\vartheta), \quad S_{n,l}(\vartheta) = \sum_j b_{j,l} e^{-2\pi i n j \vartheta}, \quad b_{j,l} = a_{ij} \overline{a_{k,j-l}}.$$

The ratios J_k/J_i are fixed. Thus $b_{j,l}$ is supported on $O_b(J)$ indices, is zero unless $|l| \leq C_b J$, and its r -th discrete differences satisfy

$$\sum_j |\Delta^r b_{j,l}| \leq C_r J^{-r}.$$

These statements follow by applying the fundamental theorem of calculus r times to the two fixed bump factors, whose normalization is comparable to J .

It is essential to retain the central residual phase. Treat $e^{-2\pi i n j t/Z}$ as part of the amplitude and sum by parts against $e^{-2\pi i n j \alpha}$. For all fixed nonnegative integers r, a , the discrete product rule and $|j| \leq C_b J$ give

$$\sum_j \left| \Delta^r \partial_t^a (b_{j,l} e^{-2\pi i n j t/Z}) \right| \leq C_{r,a} J^{-r} (1 + |n| J/Z)^{r+a}.$$

The trivial sum bound and r summations by parts therefore imply

$$|\partial_t^a S_{n,l}(\alpha + t/Z)| \leq C_{r,a} (1 + |n| J/Z)^{r+a} (1 + J \|n\alpha\|_{\mathbb{R}/\mathbb{Z}})^{-r}. \quad (5.10)$$

This also covers $n = 0$ by the trivial bound. To see the factor $1 + J \|n\alpha\|$ precisely, summation by parts divides by $|1 - e^{2\pi i n \alpha}|^r$, which is bounded below by a constant times $\|n\alpha\|^r$; taking the minimum with the trivial bound gives the stated expression.

Apply (5.7) to $g(t) = N^{-1}\Phi(t, n/N)S_{n,l}(\alpha + t/Z)$, using $r=3$ in (5.10). Since

$$NJ/Z = 2^{-250H_i} \leq 1, \quad 1 + |n|J/Z \leq 1 + |n|/N,$$

the polynomial factors introduced by every fixed number of central derivatives are absorbed by choosing the Schwartz exponent of Φ larger. This proves (5.8), for arbitrarily large n as well as for the principal x -frequency range.

It remains to obtain the compensating factor J^{-1} when summing over n . On any interval of M consecutive integers the points $n\alpha \bmod 1$ are separated by at least $1/(3M)$, by (5.5). Hence at most $1 + 6Mu$ lie at distance at most u from any fixed point of the circle. Integrating the identity

$$(1 + Jv)^{-3} = \int_v^\infty 3J(1 + Ju)^{-4} du$$

against this counting bound gives

$$\sum_{n \in I} (1 + J \|n\alpha\|)^{-3} \leq C(1 + M/J). \quad (5.11)$$

For u beyond half the circle the same upper bound still bounds the count, so the integral is valid as written. Split the integers into the central interval $|n| \leq N$ and dyadic annuli of lengths comparable to $2^v N$. Since $N \geq J$, (5.11) yields

$$\sum_n N^{-1}(1 + |n|/N)^{-3}(1 + J \|n\alpha\|)^{-3} \leq C/J. \quad (5.12)$$

Summation over the $O_b(J)$ possible l proves the one-norm assertion. Using a larger exponent P gives arbitrarily rapid polynomial tails when $|n| > LN$; using a larger A gives the same statement for $|c| > LZ$. Their sum proves (5.9). $\square$

5.5 Diffuse product sampling with bounded matrix weights

We turn the envelope into a probability distribution. Its independent horizontal marginals carry enough mass to dominate every Fourier coefficient; the remaining coefficient is placed in a uniformly bounded matrix weight. Unused mass is retained in the horizontal law even where the weight is zero.

Lemma 5.4 (Finite product sampler). *For every fixed $\eta > 0$, each symbol in Lemma 5.3 has a finite distribution of pairs (W, v) , with $W \in M_\ell(\mathbb{C})$ and $v = x^n y^l z^c \in \mathbf{H}$, satisfying*

$$\|W\| \leq M_\eta, \quad \|\mathbb{E}[W\lambda(v)] - T_i f T_k^*\|_1 \leq \eta.$$

The horizontal coordinates n and l are independent. Their maximal atoms satisfy

$$\sup_n \Pr(n) = O_b(J_i/N_i), \quad \sup_l \Pr(l) = O_b(J_i^{-1}). \quad (5.13)$$

All possible group coordinates, including those of inverses, are bounded by $2^{a_0 H_i}$ for a fixed exponent a_0 , for example $a_0 = 502$, after increasing the initial height. The weights can additionally be restricted to one finite adjoint-closed net, at the expense of any prescribed further one-norm error. All constants and the net are independent of height.

Proof. Truncate the Fourier series to (5.9), with error at most $\eta/2$, and denote its matrix coefficients by $A_{n,l,c}$. Set

$$w_n = N_i^{-1}(1 + |n|/N_i)^{-3}(1 + J_i \|n\alpha\|)^{-3}.$$

By Lemma 5.3, $R_{n,l} := \sum_c \|A_{n,l,c}\| \leq C_1 w_n$, while $\sum_n w_n \leq C_0/J_i$. On the finite allowed n -interval begin a probability measure with masses $J_i w_n/(2C_0)$, whose total is at most one half. Distribute the remaining mass uniformly on $\{-N_i, \dots, N_i\}$. Call the result μ_i . Let ν_i be uniform on the integer interval containing all allowed l , of length comparable to J_i . Then (5.13) holds and

$$R_{n,l} \leq M_\eta \mu_i(n) \nu_i(l)$$

for a height-independent constant. The finite cutoff L and fixed neighboring ratios are harmless in this inequality.

Draw n and l independently from μ_i and ν_i . If $R_{n,l} > 0$, draw c conditionally with probabilities $\|A_{n,l,c}\|/R_{n,l}$ on the nonzero coefficients, and set

$$W = \frac{R_{n,l}}{\mu_i(n) \nu_i(l)} \frac{A_{n,l,c}}{\|A_{n,l,c}\|}.$$

If $R_{n,l} = 0$, set $W = 0$ and choose $c = 0$. Thus zero-weight draws still have exactly the same diffuse horizontal product law. Directly summing the expectation recovers every truncated coefficient, and $\|W\| \leq M_\eta$.

Choose a finite net in the closed matrix ball of radius M_η , include zero, close it under adjoint, and round W within $\eta/2$, keeping zero fixed. The resulting expectation changes in one-norm by at most $\mathbb{E} \|W - W_{\text{rounded}}\| \leq \eta/2$. For a reversed edge use the adjoint weight and inverse voltage rather than an independent draw. The net cardinality depends on the accuracy and the fixed matrix dimension, not on height.

The coordinate bounds follow from (5.9). In the chosen normal form,

$$(n, l, c)(n', l', c') = (n + n', l + l', c + c' - ln'), \quad (n, l, c)^{-1} = (-n, -l, -c - ln).$$

The product $|nl|$ is bounded by a constant times 2^{250H_i} , much smaller than 2^{500H_i} . Every fixed prefactor is absorbed by raising the initial height, giving the stated exponent for both voltages and inverses. $\square$

The zero symbol is included in this construction. In particular a type may have mean zero on any prescribed set of heights, such as all but the initial height or one parity. This requires no new height labels: the finite weight net already contains zero. The underlying voltage edge is retained.

Lemma 5.5 (Polynomial avoidance). *Use independent edge draws as in Lemma 5.4. For a fixed base walk, reuse its one voltage variable on every occurrence of an unoriented edge and use its inverse in the reverse direction. If the associated word is not a law in H , the probability that its holonomy is the identity is at most*

$$C_b 2^{-50H_{\min}},$$

where $H_{\min}$ is the smallest reference height of an edge in the walk. The estimate is unaffected by matrix weights, including zero weights.

Proof. The polynomial estimate is the maximal-atom version of the usual polynomial zero bound [34, Lemma 1]; we include its short induction. If independent real-valued variables have atoms at most κ , a nonzero polynomial of total degree d vanishes with probability at most $d\kappa$. Induct on the number of variables. If its degree in the last variable is s , its leading coefficient is a nonzero

polynomial in the preceding variables of degree at most $d - s$. By induction that coefficient vanishes with probability at most $(d - s)\kappa$; otherwise the one-variable polynomial has at most s roots and costs at most $s\kappa$. Variables absent from the polynomial are discarded. A nonzero constant has zero vanishing probability, starting the induction.

By the normal-form multiplication law just displayed, the horizontal coordinates of any word are integral linear combinations of the edge coordinates n_e, l_e , with coefficients equal to signed occurrence counts. If any such count is nonzero, one horizontal polynomial is nonzero. If all counts vanish, every central coordinate c_e cancels, and the central coordinate of the word is a polynomial of degree at most two in the n_e, l_e . Were that polynomial identically zero, the word would vanish for every assignment in $\mathbf{H}$, contrary to the hypothesis. Thus a nonzero polynomial of degree at most two is tested on independent horizontal variables. Their maximal atoms are bounded by $C_b 2^{-50H_{\min}}$, since $J_i/N_i = 2^{-150H_i}$. Apply the polynomial bound. No conditioning on the central coordinate or on the matrix weight is performed. This proves the assertion also for an edge whose realized weight is zero. $\square$

5.6 Finite tests for the full regular norm

A sampled block acts on a finite base-layer space tensored with $\ell^2(\mathbf{H})$. To control its full operator norm, we first test it in a finite Heisenberg box, with exponentially small norm error and dimension at most exponential in the height scale. The subsequent moment order, proportional to $H_i/\log_2 d$, will overcome this dimension factor.

Lemma 5.6 (Heisenberg boxes and regular norm). *Suppose a finite set of group elements and their inverses have coordinates bounded by $2^{a_0 H}$, with $H \geq 1$. For every fixed B there is a finite box $\mathcal{B}_H \subset \mathbf{H}$, of cardinality at most $2^{K_0 H}$, such that*

$$\frac{|\mathcal{B}_H \triangle v \mathcal{B}_H|}{|\mathcal{B}_H|} \leq C 2^{-BH}$$

for all these elements. The exponent K_0 depends on a_0, B , but not on the number of elements.

Let A be a selfadjoint finite matrix of regular convolution sums whose convolution support is contained in the finite set just specified, with any fixed matrix amplification. Let R bound its absolute coefficient row sums; by selfadjointness it also bounds the column sums. If Q_H retains the deck coordinates in $\mathcal{B}_H$, then

$$\|Q_H A Q_H\| \leq \|A\| \leq \|Q_H A Q_H\| + C 2^{-BH} R. \quad (5.14)$$

The same statement holds for rectangular matrices after selfadjoint off-diagonal doubling, using the larger row or column coefficient bound.

Proof. For an integer T let

$$\mathcal{B}(T) = \{x^n y^l z^c : |n|, |l| \leq T, |c| \leq T^2\}.$$

If v has coordinates bounded by R_0 and $T \geq R_0$, left or right multiplication by v changes a horizontal coordinate by at most R_0 and the central coordinate by at most $R_0 + R_0 T$. This follows directly from the normal-form multiplication law. Points outside the corresponding boundary strips remain in the box. The total strip fraction is at most $C R_0 / T$. Multiplication is a bijection, so the same bound, with a factor two, controls the symmetric difference. Take $T = \lceil 2^{(a_0 + B + 2)H} \rceil$. The box has $(2T + 1)^2 (2T^2 + 1) = 2^{O(H)}$ elements, proving the first assertion uniformly over all possible translations.

For the norm estimate, let Q_t retain the deck coordinates in the right translate $\mathcal{B}_H t$. Right translation commutes with left convolution, so every $Q_t A Q_t$, acting on its range, has the same norm as $Q_H A Q_H$. For a finitely supported vector ξ ,

$$\sum_t \|Q_t \xi\|^2 = |\mathcal{B}_H| \|\xi\|^2.$$

Average $\langle A Q_t \xi, Q_t \xi \rangle$ over t and divide by $|\mathcal{B}_H|$. In this average the coefficient of $\lambda(v)$ is multiplied by $|\mathcal{B}_H \cap v \mathcal{B}_H| / |\mathcal{B}_H|$, which differs from one by at most $C 2^{-BH}$. The averaged quadratic form has absolute value at most $\|Q_H A Q_H\| \|\xi\|^2$. The difference from A has norm at most $C 2^{-BH} R$: to verify this last assertion, apply Cauchy–Schwarz to a matrix of nonnegative entry-norm bounds whose row and column sums are at most that number. Density and the selfadjoint quadratic form formula prove (5.14). None of these estimates separates coordinates of the matrix amplification. Doubling a rectangular matrix gives the final statement. $\square$

5.7 The centered moment count uses base girth

We use the classical centered-trace method: independence eliminates walks containing a singleton edge, and the remaining closed walks bound a high moment of the norm; compare [15, Sections 3.1–3.3]. Here base girth controls the cycle rank of their supports, while the preceding finite compression permits operator-valued entries. The next two lemmas give the support-rank bound and the walk count, including supports with cycles.

Lemma 5.7 (Length versus cycle rank). *Let S be a finite connected metric graph of total length at most Kg and girth greater than g . Then*

$$\dim H_1(S; \mathbb{Q}) \leq \max\{1, \lceil 20K \rceil\}^2.$$

Proof. Set $a = g/10$, and choose a maximal a -separated set P in the intrinsic metric. It is an a -net. If it has at least two points, the open balls of radius $a/2$ around them are disjoint and each contains a path interval of length $a/2$ pointing toward another net point. Hence $|P| a/2 \leq Kg$, and $|P| \leq \max\{1, \lceil 20K \rceil\}$, also covering a singleton net. Join each pair of distinct net points at distance at most $3a$ by one edge of an abstract graph L , mapped to a chosen shortest path in S . This map is surjective on first homology. To check this, subdivide a path of S into pieces of length at most a , choose net points within a of the subdivision points, and replace each piece and its two endpoint connectors by the chosen edge path. The replacement bounds a closed path of length at most $6a < g$, which is null-homotopic: cancellation of immediate reversals in a nontrivial closed graph path leaves a path containing a simple cycle no longer than itself. Connectors cancel consecutively around a loop. Thus L is connected and its mapped loops span first homology. Finally $\dim H_1(S; \mathbb{Q}) \leq \dim H_1(L; \mathbb{Q}) \leq |P|^2$. $\square$

Lemma 5.8 (Repeated-edge walk count). *Let a graph have maximum degree $\Delta \geq 1$. Consider length- $2h$ closed walks from a fixed vertex, with no unoriented edge occurring exactly once. Multiple edges are treated as distinct edges. If every support under consideration has cycle rank at most R , their number is at most*

$$[72(2R + 2)^2 \Delta]^h.$$

Proof. A support has at most h edges. Prune leaves other than the starting vertex, retaining its rooted core K . A tree leaves just that vertex. Otherwise K has no degree-one vertex except possibly the root and has the same cycle rank as the support. The identity

$$\sum_{v \in K} (\deg_K(v) - 2) = 2 \dim H_1(K; \mathbb{Q}) - 2$$

gives $\deg_K(v) \leq 2R + 2$. The removed pieces are trees attached at single core vertices, since neither a cycle nor a path between two core vertices could be removed by this pruning.

There are at most $4^e \Delta^e$ possible rooted connected subgraphs K with e edges. For clarity this includes cyclic K . Fix orders on ambient incident edges and explore K by *marking edges*. When an unmarked K -edge is available, mark and traverse it, push it onto a stack, and recursively explore at its endpoint, even if that vertex was already visited. When the recursive call has exhausted its unmarked edges, return along the pushed edge and pop it. Every edge is pushed and popped exactly once. The balanced push/pop word has at most 4^e possibilities, and at each push an ambient edge has at most Δ choices. These data recover the traversal and its edge set.

For fixed K , classify each walk step as a core step, a tree descent away from the core, or a forced tree return. There are at most 3^{2h} type words. Each core step has at most $2R + 2$ choices. If t denotes the number of descents and c the number of core steps, closure gives $2t + c = 2h$. Every core edge is traversed at least twice, so $c \geq 2e$ and $t \leq h - e$. Descents have at most Δ choices and returns are determined by their stack. Thus the count is bounded by

$$\sum_{e=0}^h 4^e \Delta^e 3^{2h} (2R + 2)^{2h} \Delta^{h-e} \leq [72(2R + 2)^2 \Delta]^h,$$

using $h + 1 \leq 2^h$. Invalid formal encodings only increase this upper bound. No assertion that a short backtracking walk is a simple path is used. $\square$

Lemma 5.9 (Centered block concentration). *Fix a surviving incidence type and one of its same or adjacent layer blocks. Assign independent samples from Lemma 5.4 to its unoriented edges. Let X_{ik} be its unnormalized centered convolution matrix. For every $\eta > 0$, the choices can be arranged in the order described below so that*

$$\Pr\{\|X_{ik}\| > \eta q_{\tau,ik}\} \leq 2^{-6H_i}, \quad (5.15)$$

uniformly in height. All estimates are under the independent sampling law, without conditioning on voltage girth.

Proof. An edge entry is $W_e \lambda(v_e) - \mathbb{E}[W_e \lambda(v_e)]$ and has norm at most $2M_\eta$. Double the block off the diagonal to make it selfadjoint. Its absolute row and column coefficient sums are at most $C_* M_\eta D$.

We first make precise why a fixed finite-test exponent is available before choosing s . Require $H_0 \geq s$, so

$$D \leq 2^{.60005H_i}, \quad |\mathcal{A}_i| = 2^{H_i + .2s} \leq 2^{1.2H_i}.$$

After raising H_0 to absorb fixed constants, the row bound is at most 2^{2H_i} , and the number of base and matrix indices is at most 2^{3H_i} . The coordinates of every possible voltage and inverse have the fixed exponent of Lemma 5.4. Apply Lemma 5.6 with a boundary exponent larger than four. Its box error tends to zero exponentially even after the row loss, and its total finite compression dimension is at most $2^{K_1 H_i}$ for a fixed K_1 . Enlarge H_0 further so the box error is at most $\eta q_{\tau,ik}/2$. These enlargements do not change K_1 .

Now fix $C > 2(K_1 + 6)$, increasing it strictly if needed, and put

$$h = \left\lceil \frac{CH_i}{\log_2 d} \right\rceil.$$

Every no-singleton support in the $2h$ -moment has at most h base edges. It lies in the two fixed layers of this block. Their base girth is at least $.9H_i/\log_2 q$ after increasing H_0 , since adjacent H -values differ by only b . Because $\log_2 q/\log_2 d < 1.25$, the ratio of support length to this girth is

at most $2C$, again increasing H_0 for the ceiling. Lemma 5.7 gives a rank bound depending on C alone. This is a use of *base* girth, not the as-yet unproved voltage girth.

Let $\hat{X}$ denote the finite selfadjoint compression. Expand $\mathbb{E} \text{Tr}(\hat{X}^{2h})$ along base vertices and individual edges. Any summand with a singleton unoriented edge has expectation zero: after conditioning on the other edge draws it is a fixed left factor, one centered compressed entry or its adjoint, and a fixed right factor. Compression does not change this independence. For every other summand its absolute trace on the deck and matrix indices is at most their dimension times $(2M_\eta)^{2h}$. Lemma 5.8, with $\Delta \leq C_*D$, proves

$$\mathbb{E} \text{Tr}(\hat{X}^{2h}) \leq 2^{K_1 H_i} (C_1 D)^h,$$

where C_1 is fixed after C and the coefficient data, but before s . Since $\hat{X}$ is selfadjoint, its largest absolute eigenvalue to the power $2h$ is at most this nonnegative trace. Markov's inequality and (5.1) give

$$\Pr\{\|X_{ik}\| > \eta q_{\tau,ik}\} \leq 2^{K_1 H_i} \left(\frac{C_2}{\eta^2 d} \right)^h.$$

Choose s large enough that $C_2/\eta^2 \leq d^{1/2}$. The last expression is at most $2^{(K_1 - C/2)H_i} \leq 2^{-6H_i}$, by the choice of C . The potentially large dependence of C_2 on C is not circular: s is chosen after both constants. This proves the claim. $\square$

5.8 Simultaneous selection over all heights

Proof of Proposition 5.2. For a block put $a_{\tau,ik} = q/q_{\tau,ik}$. This scalar lies in a fixed compact subinterval of $(0, \infty)$. Apply Lemma 5.4, with smaller tolerances, to the symbol $a_{\tau,ik} S_{\tau,ik}$. Multiplication of the profile family by these bounded scalars preserves every uniform estimate. Round these already rescaled weights to one finite net, and normalize the weighted incidence by the common denominator q . Within a block use the same law at every edge, independently over all underlying unoriented edges. Its expectation is

$$\frac{q_{\tau,ik}}{q} U_{\tau,ik} \otimes \mathbb{E}[W_e \rho(h_e)],$$

whose distance from $U_{\tau,ik} \otimes S_{\tau,ik}$ is bounded by a fixed constant times the assigned one-norm error. Lemma 5.9, with its tolerance reduced using $q_{\tau,ik}/q \leq C_*$, supplies the remaining error. This common- q normalization avoids any need to recover an unbounded height from a label.

We verify the voltage-girth probability for the same law. A forbidden closed lift projects to a cyclically nonbacktracking closed base walk. Test all such base walks of length at most $\lceil 6.03m_i \rceil$ from height i . For large m_i , this length is below $6.04m_i$; hence the local bound in Proposition 4.14 says that each is a non-law. This also removes any rounding issue at exactly $6.03m_i$. Along such a walk,

$$H_{\min} \geq H_i - b \lceil 6.03H_i / \log_2 q \rceil \geq (1 - 7b / \log_2 q) H_i$$

after raising the initial height. Taking $b / \log_2 q \leq 10^{-6}$, Lemma 5.5 bounds its identity probability by 2^{-40H_i} , with all fixed factors absorbed by the initial height.

The number of possible rooted walk instructions is bounded by

$$(|\mathcal{A}_i| + |\mathcal{B}_i|) \sum_{a \leq \lceil 6.03m_i \rceil} (C_* D)^a \leq 2^{9H_i}$$

for sufficiently large s and then H_0/s . Indeed its leading exponent divided by H_i is

$$1 + 6.03 \frac{.60005}{.50005} < 8.24;$$

the finite type factor, the side-size offset $.2s$, and rounding fit the remaining margin. The union bound for height i is therefore at most 2^{-31H_i} . It counts every edge, even those with realized weight zero. Triviality of holonomy is independent of the chosen point in the initial deck fiber, so no additional union over that infinite fiber is needed.

There are finitely many block types at each height. Since $H_i = H_0 + bi$, the sum over all heights of their norm-failure bounds and the girth-failure bounds can be made less than one by raising H_0 . No conditioning on the girth events has entered the centered moment calculation. For completeness, one can obtain a simultaneous choice without invoking an infinite-product measure. Enumerate all the required block inequalities and walk tests. Any finite list involves finitely many edge variables and, by the same summed probability bound, has a satisfying assignment. Extend those assignments arbitrarily to all other edges. Each edge has finitely many possible voltage/net-weight pairs, so successive subsequences stabilize the choice at the first edge, the second edge, and so on. The diagonal limit satisfies every condition, because each condition depends on finitely many edge choices. Fix such an assignment.

Finally append the finite matrix-weight index to each old label, and use the inverse label with adjoint weight on its reverse. Each refined positive label now specifies its matrix weight, and every type uses the common normalizer q^{-1} . These are the coefficients of the finite Cayley polynomial in Equation (8.15). Its comparison with the sampled blocks on retained sheet domains, and the estimate on the remaining edges, are proved in Section 8. Equality of refined words implies equality of the original words, so all original uniqueness conclusions persist. The original leading label count is $d2^r = 2^{1.00005s}$; prescribed types, the weight net and extras contribute height-independent factors, and the residue flags contribute $2^{O(\log s)}$. Large voltage coordinates are separate edge data and do not enter this alphabet. Since $\log_2(q^2) = 1.0001s$, the displayed sharp margin follows. The ratio is more precisely $2^{-.000025s + O(\log s) + O_{b,\eta,\text{profiles}}(1)}$, not a claimed constant-prefactor exact decay. This proves all assertions. $\square$

5.9 Choice order and the interface with constant extraction

To apply Proposition 5.2 along a sequence whose errors tend to zero, fix b large enough for the uniform gap $\rho < 1$ of the unweighted incidences, and keep b fixed throughout. Also fix α , the Bott profile, enlarged support cutoffs, and the matrix sizes. These choices give uniform profile bounds and homotopy moduli throughout the accuracy sequence.

At accuracy ϵ , choose a finite extraction length k sufficiently large for the gap term in Equation (8.13). Choose a finite mesh of the two adjacent-coordinate homotopies using their uniform profile modulus. Collect the cutoffs, mesh symbols, parity selections and initial selection into the finite list in Proposition 5.2. Set their individual tolerances small enough compared with ϵ/k , and choose the finite Fourier boxes and weight net. Then fix the finite-test exponent K_1 , the moment constant C , choose s sufficiently large for all concentration, label and upstream graph requirements, and finally choose H_0 sufficiently large. In abbreviated form the order is

$$b, \text{ profiles and matrix size}; \quad \epsilon, k, \text{ mesh, types, net}; \quad K_1, C, s, H_0.$$

The extraction length and finite type count may grow along the accuracy sequence; they are fixed inside each tower before s is chosen.

For the assembly interface, let χ_i be one of the larger cutoffs equal to one on every required incident symbol support. The preceding proposition supplies, in particular, the full-norm approximations to

$$U_{ii} \otimes T_i \chi_i T_i^*, \quad U_{ik} \otimes T_i f_{ik} T_k^*, \quad |i - k| \leq 1.$$

The constant-extraction argument in Section 8 uses finite products of these same-layer blocks and their adjoints, followed by a desired cross block. The cutoffs leave the target compact symbols unchanged by Lemma 5.1. Proposition 5.2 controls each factor on the whole regular space, including the complementary modes. For a product of r bounded factors, replacing each factor by an approximant of error δ costs at most $r\delta$ times the appropriate fixed product norm bound, by telescoping.

Because b is fixed, finite meshes approximate the model homotopies uniformly in both height and construction index. Passing from these abstract block estimates to operators in the physical reduced algebra requires the domain and residual-forest conclusions of Proposition 7.4 and the weighted forest estimate in the proof of Lemma 8.4. Once the finite mesh operators have been realized there, linear interpolation in each coordinate reduced algebra has error tending to zero uniformly in the path parameter. These uniform approximations give the continuous paths in the norm quotient established in Lemma 8.5.

6 The graphical group and its embedded sheets

We now realize the sampled voltage cover inside a Cayley graph. A private completion first makes the cover connected. We then impose its closed-walk words as group relators and prove that the cover embeds in the resulting Cayley graph and its Heisenberg deck group embeds in the group. The translated copies of the cover are the sheets used in the rest of the proof. Our second goal is to control polygons through distinct sheets while keeping their contact vertices fixed; this will give the intersection bounds needed in Section 7.

The argument uses graphical small-cancellation methods developed by Gromov, Ollivier, and Gruber [16, §2.3][31, §§2–3] [19, Definition 1.3 and Lemma 4.1]. In particular, the reductions below compare lifted paths modulo label-preserving deck transformations and then apply a finite-diagram curvature argument. We give the required form with its varying scale, bridges, and prescribed boundary contacts. The graphs in this section may be infinite; every path, diagram, and word used in a proof is finite.

6.1 Conventions and the input from the preceding construction

An edge of a labeled graph has a specified positive orientation and one formal letter. Traversing it backwards reads the inverse formal letter. A labeling is *folded* if, at each vertex, each signed formal letter is read on at most one outgoing edge incidence. In particular, a path and its initial vertex are determined by its word whenever that path exists. Word products are read in traversal order. Thus the formal right Cayley graph of a group with alphabet S has one positive edge

$$(g, s): g \longrightarrow gs \quad (g \in \Gamma, s \in S).$$

These are distinct cells indexed by (g, s) , even if their unordered endpoints coincide with those of another cell. We never replace this graph by an undirected simple graph. Left translation preserves all its labeled, oriented cells. The corresponding right regular representation is $R(a)\delta_g = \delta_{ga^{-1}}$; the unitary $I\delta_g = \delta_{g^{-1}}$ satisfies $IR(a)I = \lambda(a)$.

Write X_o for the surviving old graph, with its old alphabet S_o and chosen voltages $h_e \in \mathbf{H}$ on its positive edges; put $h_{\bar{e}} = h_e^{-1}$. Here and throughout this Section

$$a = \frac{b}{\log_2 q} \leq 10^{-6}, \quad L = 1.003, \quad m_{\min} \geq 10^6.$$

The use of a in this Section is solely for the slope of the scale function. At an old vertex of level i put $m(v) = m_i$. The inputs furnished by Propositions 2.6, 4.14, and 5.2, in the form used here, are:

1. X_o is countable, folded, and locally finite, with maximum degree at most $K_{b,\epsilon}D$. Its edges join the same or consecutive levels, so $|m(v) - m(w)| \leq a$ on every old edge.
2. If an immersed old path starts at v and its word has two different projected occurrences in X_o , its length is less than $Lm(v)$. The assertion applies to either occurrence and to the reversed path.
3. In the old voltage graph, every simple cycle C satisfies $|C| > 6.03m(v)$ at every vertex v of C .

For the second assertion, the recovery length from the preceding construction is $\lceil 1.002m(v) \rceil$; the bound displayed here follows because $\lceil 1.002m(v) \rceil < 1.003m(v)$ when $m(v) \geq 10^6$. An immersed path means one with no immediate edge reversal. Foldedness identifies this condition with free reduction of its word. These three explicit assertions, rather than a small-cancellation theorem, are the inputs to the proofs below.

6.2 A private completion with a controlled scale

The following construction uses group-valued voltage covers in the sense of Gross and Gross–Tucker [17, 18]; we specify the multiplication convention to fix the later operator formulas. The added paths connect the old components, and two added loops make the cover connected. Letters used on single added base edges will also identify the full label-preserving automorphism group with the deck group. Their lengths are chosen to preserve the local girth and recovery bounds.

Lemma 6.1 (Private completion). *There is a connected, countable, folded labeled graph X containing X_o as its old subgraph, and an extension of the voltages, with the following properties.*

1. *Each added edge has its own new positive letter, used on no other base edge. Every new vertex has degree two and has only private incident edges. The maximum degree is at most $K_{b,\epsilon}D + 6$.*
2. *The function m extends to the new vertices, remains at least $m_{\min}$, and satisfies $|m(v) - m(w)| \leq a$ on every edge.*

3. *The voltage cover $\tilde{X}$, with edges*

$$(v, h) \longrightarrow (w, hh_e) \tag{6.1}$$

over $e: v \rightarrow w$, is connected. Its deck action is the left action $f(v, h) = (v, fh)$ of H .

4. *Every simple cycle C of $\tilde{X}$ satisfies $|C| > 6.03m(v)$ at every vertex v of C .*
5. *If two paths in $\tilde{X}$ have the same word and their projected occurrences in X differ, neither contains a private edge. If they are immersed and start at v, w , their common length is less than both $Lm(v)$ and $Lm(w)$.*

The scale on the cover is the pullback of the scale on X .

Proof. Choose an old vertex in each component of X_o and enumerate the components. Join the chosen vertices in a chain by new subdivided paths; if there are finitely many components use a finite chain. The chain has no new branching vertices. For each joining path with endpoints v, w , choose its integer length n to satisfy

$$n \geq \max \left\{ 20 \max\{m(v), m(w)\}, \frac{|m(v) - m(w)|}{a} \right\}. \tag{6.2}$$

Interpolate m linearly along the path. All of its voltages may be the identity. At one old basepoint v_* attach two further subdivided based loops, each of length at least $20m(v_*)$, with new interior vertices. Keep $m = m(v_*)$ on these loops. Assign their voltages so that their ordered products are respectively x and y ; for example use identity voltages except on one positively traversed edge of each loop. Give every added positive edge a distinct new letter. This preserves foldedness. The chain adds at most two incidences at any old vertex and the two loops add four at v_* . Their lengths are greater than two, so all new vertices have degree two. The interpolation proves the second assertion.

The completed base graph is connected. A path from v_* to a given base vertex v has some voltage h_0 . By first traversing based words in the two added loops, one can obtain any prescribed voltage in H , since x, y generate H . Following the chosen path thereafter reaches any specified lift (v, h) . Thus the cover in (6.1) is connected. Its displayed left action is free on vertices and transitive on each fiber.

Consider a simple cycle that uses a private edge. Since every private interior vertex has degree two, the cycle traverses an entire lift of a joining path or one of the two added subdivided loops. Let v be an old endpoint of that lifted path and let ℓ be the length of the cycle. Then $\ell \geq 20m(v)$. At any other vertex w of the cycle the Lipschitz bound gives $m(w) \leq m(v) + a\ell$. Consequently

$$6.03m(w) \leq 6.03(1/20 + a)\ell \leq 0.30150603\ell < \ell.$$

A cycle using only old edges satisfies the required inequality by the input hypothesis. This proves the fourth assertion, and in particular the cover has neither loops nor two-edge circuits.

A private letter in a common word identifies its unique oriented base edge. Foldedness then propagates equality of the two projected paths forwards and backwards from that edge, without any assumption that the paths are immersed. This proves the first statement of the last assertion. When different projected occurrences are immersed they therefore consist of old edges, and the old recovery bound applies. Reversing the paths gives the bounds at their other endpoints as well. $\square$

For future use, every nonempty cyclically immersed closed walk W based at v , including a nonsimple one, has

$$|W| > \frac{6.03}{1 + 6.03a} m(v). \quad (6.3)$$

Indeed it contains a simple cycle based at some vertex w reached within $|W|$ steps. Thus $|W| > 6.03m(w) \geq 6.03(m(v) - a|W|)$. Also, if consecutive immersed comparison paths, each bounded by Lm at its own initial point, number at most j , their total length is at most

$$L \sum_{t=0}^{j-1} (1 + La)^t m(v). \quad (6.4)$$

This follows by induction from the Lipschitz bound. In particular,

$$L \sum_{t=0}^2 (1 + La)^t < 3.009004 < 3.014, \quad (6.5)$$

$$L \sum_{t=0}^5 (1 + La)^t < 6.018016 < 6.029963 < \frac{6.03}{1 + 6.03a}. \quad (6.6)$$

6.3 Finite word diagrams and their perimeter budget

Put $S = S_o \sqcup S_p$, where S_p is the countable private alphabet. Write $\text{lab}(C)$ for the word read along an oriented path C , and define

$$\Gamma = \left\langle S \mid \text{lab}(C) = 1 \text{ for every finite closed walk } C \text{ in } \tilde{X} \right\rangle. \quad (6.7)$$

Each occurrence of a relator in a diagram will retain an actual closed walk of $\tilde{X}$ reading it. Two occurrences of the same free word need not be the same projected walk.

The relators make the group element represented by a path from a fixed source vertex depend only on its endpoint. To obtain an embedding, we must prove the converse: a source path with trivial label has equal endpoints. We will use diagrams to prove this closure property. Keeping the source walks lets us distinguish two genuinely different projected occurrences from two lifts of the same base path.

We use finite plane multigraph diagrams. Such a diagram is a finite connected graph embedded in the sphere, with one complementary face specified as the exterior. Each edge is oriented and labeled by a formal letter. Every other face has a specified source closed walk in $\tilde{X}$ reading its oriented facial contour. A facial contour is a walk and can repeat vertices and bridges. Perimeters count edge incidences, including both traversals of a bridge. The *filling cost* is the sum of the perimeters of the bounded faces. A one-vertex graph with no edges is allowed.

Lemma 6.2 (Elementary word-diagram construction). *If a finite word w is trivial in (6.7), there is a finite plane multigraph diagram with exterior contour reading the literal word w . More precisely, if in the free group on S*

$$w = \prod_{j=1}^n u_j r_j u_j^{-1},$$

where each r_j is read on a chosen closed source walk, a diagram may be chosen with filling cost at most $\sum_j |r_j|$. Conversely, the exterior word of any such diagram is a product of conjugates of its bounded-face words in the free group, using each bounded facial word once with the compatible orientation. Its labeled one-skeleton therefore has a well-defined vertex map to the formal Cayley graph after prescribing the image of one vertex.

Proof. Only the finitely many letters in the displayed expression matter. Let B be their finite bouquet of oriented circles. The elementary free-cancellation rule constructs a null homotopy in B for a freely trivial word: delete adjacent inverse traversals one pair at a time. Take an oriented sphere with one disk removed for each of the words w and $r_1, \dots, r_n$, reading w with the opposite boundary orientation. Choose a cutting tree joining these boundary circles, and prescribe the words u_j on its cuts. Cutting along the tree gives a disk whose boundary reads the freely trivial word supplied by the displayed identity. The cancellation homotopy extends its boundary map to B . Regluing the cuts gives a map from the punctured sphere to B with the specified word maps on all boundary circles.

Choose one interior point of each circle of B . A piecewise linear approximation relative to the boundary can be made transverse to these points as follows: triangulate the source sufficiently finely so each triangle maps into an interval of a circle or into a small tree around the bouquet vertex; choose the interior points outside the images of source vertices. Inside each triangle the inverse image of a selected point is a line segment or is empty. These segments fit to disjoint arcs and circles. Their boundary endpoints are precisely the occurrences of the selected letters on the word boundaries, and each arc joins opposite signed occurrences of the same letter. Closed inverse-image circles may be ignored. Thicken each remaining arc to a narrow disjoint band, cap

the word boundaries by disks, and contract each cap to a vertex and each band to an edge. This is an embedded ribbon multigraph in the sphere; its cyclic order at a word vertex is exactly the cyclic order of that word's letters.

If w is nonempty, retain the component containing its word vertex and take the planar dual, specifying that vertex's dual face as exterior. Every bounded dual face corresponds to one of the retained r_j word vertices, with its original source walk, so the perimeter budget follows. This argument permits loops, multiple edges, and nonsimple dual face contours. If w is empty, the one-vertex empty diagram suffices. A freely trivial but nonempty w is covered by the same construction, even when there are no relator word vertices in its component.

For the converse, take a thin regular neighborhood of the embedded graph. Its inner boundary circles correspond to the bounded facial contours and its outer boundary to the exterior contour. Cut between these boundary components along a tree of arcs. The cut surface is a disk, and the neighborhood retracts to the labeled graph. Reading the disk boundary in B gives the exterior word as a product of conjugates of the bounded face words, each occurring once with its compatible orientation. The same conclusion holds for the contour of each simple cycle of the graph, by taking the portion on its bounded side. Consequently every closed walk in the graph is trivial in Γ : split a walk at repeated vertices until it consists of simple cycles and immediate edge reversals. Labels of paths from a chosen vertex to any other vertex are therefore independent of the chosen path in Γ . This is the asserted vertex map; each edge maps to its formal Cayley cell. $\square$

Two elementary facts about plane multigraphs will be used repeatedly. An edge has the same complementary face on both sides if and only if it is a bridge. Removing such an edge separates the graph into two components. Between its two traversals, the incident face follows the exterior contour of the appropriate component. These assertions follow by the simple-cycle criterion for a bridge and the Jordan separation property of a simple cycle; they remain valid when a contour repeats other bridges.

6.4 Null words close in the cover

Lemma 6.3 (Closure of a null-labeled source path). *A path in $\tilde{X}$ whose label is trivial in Γ has equal initial and terminal vertices.*

Proof. We argue by a minimal counterexample. Coincident projected edge occurrences will permit reductions that lower the total face perimeter. The remaining comparisons are short by recovery, so each bounded face needs at least seven arcs; the final Euler count will rule out such a diagram.

Suppose a counterexample exists. Among all such source paths and all finite diagrams for their words, minimize first the filling cost c , then the exterior path length, and finally the number of diagram edges. All three quantities are nonnegative integers. The exterior path has one marked end-to-start join. Its source endpoints are required to be distinct, but may change when considering a competing counterexample. This global choice is needed for the bridge argument below.

Each bounded source face may be assumed nonempty and cyclically immersed. To justify this without assuming anything about diagram regularity, use the free-group face expression from Lemma 6.2. Foldedness turns every free cancellation in a source walk into an actual edge reversal. A freely trivial closed face word can be omitted from that expression. Otherwise delete its backtracks and its cyclic conjugating tail, absorbing the tail into its conjugator in the expression. The remaining word is read on a closed source walk with smaller perimeter. The construction part of Lemma 6.2 gives a diagram for the unchanged exterior word with strictly smaller filling cost. Minimality excludes these possibilities. In the same way the exterior source path is linearly immersed: deleting any

backtrack preserves its two source endpoints, does not increase the least filling cost, and decreases its length.

We next compare the two source occurrences of each diagram edge incident to a bounded face, orienting them to read the same signed letter. There are three cases, according to the two face incidences. In each case, coincident projections can be eliminated; a bridge on one bounded face can be eliminated without even requiring coincident projections.

First, if two different bounded faces project their common edge to the same base edge of X , a unique deck transformation aligns the two lifted edges. Delete this diagram edge. It is not a bridge, so the remaining graph is connected and the faces merge. Splice their aligned closed source walks, omitting the two opposite edge traversals. The new source walk is closed, and its perimeter is the sum of the old two perimeters minus two. Other shared edges are retained and may occur twice on the new contour. The new diagram is permitted and has smaller filling cost, a contradiction.

Second, suppose an edge has the same bounded face F on both sides. It is a bridge. The exterior face of the original diagram, including its marked join, lies entirely on one side of that bridge. Let D_1 be the other component after deleting the bridge. In the source walk assigned to F , the excursion into D_1 is a subpath χ reading the bridge, the exterior contour of D_1 , and the reverse bridge. All bounded faces of D_1 are former bounded faces other than F . Lemma 6.2 shows that $\text{lab}(\chi) = 1$ in Γ and gives a filling of cost strictly less than c , because the positive perimeter of F is absent. If the source endpoints of χ differed, it would be a smaller counterexample to the globally chosen one. They therefore agree. Delete this source excursion, the bridge, and D_1 . The remaining walk assigned to F still closes, the original exterior source path is unchanged, and the filling cost has decreased by at least the two bridge incidences. This is again impossible. Notice that this argument did not assume embeddedness and did not need the two projected bridge occurrences to coincide.

Third, if a bounded face and the exterior path project their common edge to the same edge of X , align their lifts by a deck transformation. Replace this step of the exterior source path by the complementary path of the bounded face, oriented to have the same lifted endpoints. Delete the shared diagram edge, thereby absorbing that bounded face into the exterior. The source endpoints of the exterior path remain distinct, and its filling cost drops. Any subsequent source backtrack deletions preserve that failure. This also contradicts minimality.

These reductions leave only different projected occurrences along edges incident to bounded faces. We now collect them into arcs. Retain the marked join and suppress all other vertices of degree two. There is no unmarked vertex of degree one: it would force an immediate reversal in a cyclically immersed bounded face or within the linearly immersed exterior path. The marked vertex has degree at least one if any edge is present. Each resulting edge, called an *arc*, has constant face incidences and a constant exterior-path incidence along its interior. An arc incident to a bounded face reads an immersed word on both its occurrences. By the preceding reductions those projected occurrences differ. Lemma 6.1 therefore bounds its length by Lm at the initial point of either occurrence.

The length bounds force every bounded face to have at least seven arc incidences. Otherwise apply (6.4) to its at most six arcs. Its nonempty cyclically immersed closed source walk would violate (6.3) and (6.6).

If there are no bounded faces, the diagram is a tree. A nonempty finite tree has at least two leaves, whereas only the marked vertex could be a leaf. The empty diagram has empty word and cannot represent a counterexample. Thus at least one bounded face is present. Suppression is also defined when the graph is a lone cycle, because its marked join is retained. Write V, E, F for the compressed vertex, edge, and total face counts. Euler's formula and the vertex degrees give

$$6F - 2E = 12 + 2 \sum_v (\deg(v) - 3) \geq 8. \quad (6.8)$$

On the other hand this expression equals $\sum_f (6 - \deg(f))$. Each bounded-face term is at most -1 , and the one exterior term is at most 6 . Their sum is less than 8 , a contradiction. The retained exceptional join is the reason for using 8 , rather than assuming the stronger bound 12 . $\square$

The closure lemma supplies the injectivity needed to pass from the cover to its Cayley image. The private letters then determine which translations preserve that image.

Proposition 6.4 (Graphical realization). *There is an injective map of vertices and formal oriented edges*

$$j: \tilde{X} \longrightarrow \text{Cay}(\Gamma, S)$$

preserving labels and sending $(v_, 1)$ to the identity. The subgroup of vertices over v_* defines an injective homomorphism $\phi: \mathbf{H} \hookrightarrow \Gamma$. If $t_v = j(v, 1)$, then*

$$j(v, h) = \phi(h)t_v. \quad (6.9)$$

The full group of label-preserving automorphisms of $\tilde{X}$ is its deck group. Consequently the distinct translated images, called sheets, are indexed by $\Gamma/\phi(\mathbf{H})$, and the stabilizer of $gj(\tilde{X})$ under left translation is $g\phi(\mathbf{H})g^{-1}$.

Proof. Assign to a source vertex the element represented by any path from $(v_*, 1)$ to that vertex. Two choices differ by a closed source walk, whose word is a relator, so the assignment is well defined. If two vertices have the same image, a source path between them has null label; Lemma 6.3 identifies its endpoints. This proves vertex injectivity. Edges with the same formal Cayley image have the same initial image and the same signed label. Their initial vertices agree by injectivity and their source edges agree by foldedness. Hence formal edges also embed.

For $f \in \mathbf{H}$, let $\phi(f) = j(v_*, f)$. A source path from $(v_*, 1)$ to (v_*, f_1) , followed by the left deck translate by f_1 of a path to (v_*, f_2) , ends at $(v_*, f_1 f_2)$ and reads the product of their words. Thus $\phi(f_1 f_2) = \phi(f_1)\phi(f_2)$. It is injective by vertex injectivity. Following a path to (v_*, h) and then the left deck translate of a path to $(v, 1)$ proves (6.9). More generally, connectedness and label preservation show

$$j(fu) = \phi(f)j(u) \quad (u \in \tilde{X}).$$

Fix one of the private positive base edges. Its letter appears on no other base edge. A label-preserving source automorphism must send a chosen lift of this edge to a lift of the same base edge. Compose the automorphism with a deck transformation to fix that edge's initial vertex. An automorphism of a connected folded labeled graph fixing a vertex fixes every path from that vertex, hence every vertex and edge. It is the identity. Thus every source automorphism is a deck transformation.

If a left translation by g stabilizes $j(\tilde{X})$, its restriction, conjugated by j , is a label-preserving source automorphism. It is deck multiplication by some f . Evaluating at $(v_*, 1)$ gives $g = \phi(f)$. Conversely (6.9) shows that every $\phi(f)$ stabilizes the image. The remaining stabilizer and coset assertions follow by conjugation and translation. $\square$

We identify $\mathbf{H}$ with its image from now on. Each sheet is equipped with its source vertices, edges, and scale m_Q . These do not depend on the representative g of its coset: changing g by an element of $\mathbf{H}$ changes the source parametrization by a deck transformation, which preserves the base vertex and scale. There is a useful rigidity observation. If two source-parametrized sheet maps agree at one source vertex after deck alignment, they agree everywhere by connectedness and label propagation. Thus two different sheets meeting at a physical vertex cannot have the same projected base vertex there. In particular, a common path in two different sheets always has different projected occurrences. This observation concerns nonconstant paths; it does not exclude isolated contacts at private vertices.

6.5 Polygons with fixed sheet contacts

We next control polygons whose vertices are prescribed intersections of sheets. The sides may change during diagram reduction, but their contact vertices must stay fixed. This is the form needed to compare the intrinsic distances between contacts in Section 7.

Lemma 6.5 (The exterior-arc bound). *Let $Q_1, \dots, Q_k$ be distinct sheets, $k \geq 2$, and let $g_i \in Q_i \cap Q_{i+1}$, with indices modulo k . Contacts may coincide. Choose a source path in Q_i from g_{i-1} to g_i for each i . There is a finite diagram and a choice of these paths, with every prescribed endpoint g_i fixed, such that:*

1. *Each side is linearly immersed. After retaining all junction vertices and suppressing other degree-two vertices, every arc has two different projected occurrences in X . Each occurrence is immersed, contains no private edge, and has length less than $Lm_Q(v)$ at its initial point. Each entire nonconstant side is the concatenation of its exterior arc occurrences.*
2. *Every bounded face has at least seven arc incidences.*
3. *If B counts exterior arc incidences, counting an exterior bridge twice, then*

$$B \leq 4k - 6. \tag{6.10}$$

A diagram with no edges has $B = 0$ and all its sides constant. In particular, a nonconstant side cannot have a private vertex as an endpoint. A side whose intrinsic endpoint distance is greater than $3.014m_Q(v)$ at its initial endpoint has at least four arc incidences.

Proof. The concatenation of any chosen sheet paths is a closed walk in the Cayley graph and has null word. Lemma 6.2 supplies a finite diagram. Over all choices with the prescribed contacts fixed, minimize lexicographically the filling cost, the total side length, and the number of edges. Retain every boundary vertex carrying a junction mark; there are at most k such vertices, even if several junctions coincide.

The source-path and bounded-face reductions at the beginning of Lemma 6.3 apply unchanged: bounded source faces are nonempty and cyclically immersed, while each side is linearly immersed. Coincident projected edges between two bounded faces can be merged after deck alignment, decreasing filling cost. A bridge with both incidences on one bounded face can be removed as in that proof. Here there is no preliminary injectivity issue: its source excursion has null label by the cut-off component, and Lemma 6.3 makes it close. The component containing the exterior retains all marked contacts.

If a bounded face and a boundary side have coincident projected occurrences along an edge, align the closed face lift to that side and replace the one side step by the complementary face path. The two source endpoints of the replaced step agree with the old ones, and therefore both prescribed endpoints of the full side remain fixed. Deleting the separating edge absorbs the face and decreases filling cost. This is impossible by minimality.

It remains to analyze edges whose two sides are exterior. Such an edge is a bridge. If its two traversals occur in the same side, the interval between them within that side is an excursion with no junction in its interior. Its endpoints have the same Cayley vertex, and hence the same source vertex by Proposition 6.4. Delete the excursion from that side and remove the bridge and its cut-off component. No prescribed junction is lost. Filling cost does not increase and total side length decreases, again contradicting minimality. If instead the two traversals belong to different sides, their sheets are distinct. Coincident projected occurrences would be deck-aligned and agree at a

physical vertex; the rigidity observation after Proposition 6.4 would identify the two sheets. Thus these projected occurrences differ too.

Suppress every unmarked degree-two vertex. An unmarked leaf would force a backtrack in one side or one bounded face, so all unmarked vertices have degree at least three. Marked vertices have degree at least one when edges occur. The face and side incidences of each resulting arc are constant along its interior: a side change occurs at a retained mark, and a face change requires branching. Both of its occurrences therefore lie within immersed source paths. The preceding reductions show they have different projections. Apply Lemma 6.1 to obtain all the assertions about arc lengths and private edges. The proof using (6.3) and (6.6) again gives at least seven arcs on every bounded face.

For completeness the Euler count uses the exterior degree itself, rather than the coarser estimate by all edges. Let f be the number of bounded faces and let V, E count the compressed vertices and edges. When edges occur,

$$E = V + f - 1, \quad 2E \geq 3V - 2k, \quad E \leq 3f + 2k - 3.$$

Every edge has two face incidences, so

$$2E - B = \sum_{\text{bounded } F} \deg(F) \geq 6f.$$

Consequently $B \leq 2E - 6f \leq 4k - 6$. If the diagram has no edges, all of its marked vertices coincide and every side is constant, giving the stated separate case.

Every edge incident to a private source vertex is private. If a side with such an endpoint were nonconstant, its initial or terminal arc would contain a private edge, contradicting the arc conclusion already proved. Finally, (6.5) bounds a side of at most three arcs by $3.014m_Q(v)$ at its initial endpoint. The contraposition proves the last assertion. $\square$

7 Disjoint domains, cone chains, and an integral detector

The fixed-contact polygon bound will supply both disjoint analytic domains and the integral detector. We first show that every finite family can be peeled: one sheet meets the remaining sheets only inside a vertex set whose incident edges form a forest. Peeling gives a unique decomposition of every finite Cayley cycle into sheet cycles. From these cycle coordinates the argument branches. A compactness argument produces disjoint domains and a residual forest for the analytic realization. Independently, coning the full sheets gives an exact cellular complex, which proves torsion-freeness and extends an integral detector from the Heisenberg torus to the ambient group.

Write $\mathcal{C}$ for the formal right Cayley graph of Γ supplied by Proposition 6.4. A positive edge is a separate cell (g, σ) , directed from g to $g\sigma$; cells with the same unordered endpoints are not identified. Let $\mathcal{Q}$ be the family of distinct translated images of the connected cover $\tilde{X}$. Each $Q \in \mathcal{Q}$ retains its intrinsic graph metric and the type of each of its vertices in X . In particular, “old” and “private” refer to the vertex in that sheet, not just to its physical image in $\mathcal{C}$.

We use the following conclusions of Lemma 6.1, Proposition 6.4, and Lemma 6.5. The sheets are embedded, their stabilizers are conjugates of H , and every private vertex has only private incident edges. Enlarging the fixed degree constant absorbs the six added incidences, so the maximum intrinsic degree is at most $\Delta \leq K_{b,\epsilon}D$. On each sheet the scale function m_Q satisfies

$$|m_Q(u) - m_Q(v)| \leq a d_Q(u, v), \quad a = \frac{b}{\log_2 q} \leq 10^{-6}, \quad m_Q \geq 10^6. \quad (7.1)$$

At an old vertex in layer i , its value is $m_i = H_i / \log_2 q$. Every simple circuit in Q , based at v , has length greater than $6.03m_Q(v)$. Here a comparison arc is an arc in the reduced fixed-contact filling

of Lemma 6.5. Every such arc has length less than $Lm_Q(v)$ when read from either endpoint v , where $L = 1.003$, and contains no private edge. Finally, a polygon through $k \geq 2$ distinct sheets, with its contact vertices fixed, has a reduced filling whose exterior boundary has at most $4k - 6$ comparison-arc incidences, counting both traversals of an exterior bridge. Each entire nonconstant side is a concatenation of those arcs. The last fixed-endpoint assertion will be used even when a contact vertex is private.

7.1 Finite peeling, including private contacts

Put $\kappa = 3.014$. If a path is a concatenation of at most three comparison arcs and starts at v , the successive initial scales are at most $m_Q(v)$, $(1 + La)m_Q(v)$, and $(1 + La)^2 m_Q(v)$. Consequently its length is less than

$$L(1 + (1 + La) + (1 + La)^2)m_Q(v) < 3.010m_Q(v) < \kappa m_Q(v). \quad (7.2)$$

Indeed the coefficient on the left is $3.009 + 3.018027a + 1.009027027a^2$, which is less than 3.010 for $a \leq 10^{-6}$.

For a vertex set $B \subseteq V(Q)$, let $\text{Star}_Q(B)$ denote the subgraph consisting of all edges of Q with at least one endpoint in B , together with their endpoints. This is an edge-incidence neighborhood, rather than the induced graph on B .

Lemma 7.1 (Allowed removed sets). *For an old vertex $v \in Q$, set*

$$B_Q(v) = \{w \in V(Q) : d_Q(v, w) \leq \lfloor \kappa m_Q(v) \rfloor\}.$$

The graphs $\text{Star}_Q(B_Q(v))$ and $\text{Star}_Q(\{w\})$, for a private vertex w , are forests. The empty set also has this property.

After increasing s as permitted by the parameter order, if $B_Q(v)$ contains an old vertex of layer i , then

$$|B_Q(v)| \leq 2^{10H_i}. \quad (7.3)$$

A private singleton contains no old vertex.

Proof. Let $m = m_Q(v)$ and $R = \lfloor \kappa m \rfloor$. All vertices of the induced ball $B_Q(v)$ can be joined to v by paths inside that ball; choosing a predecessor at distance one less gives a shortest-path tree. If the ball had a non-tree edge, that edge and its tree path would contain a simple circuit of length at most $2R + 1$. If an outside vertex had two edge incidences into the ball, these incidences and a path in the tree would contain a simple circuit of length at most $2R + 2$. This includes the possibility of parallel edges. Either circuit has a vertex u in the ball, at which $m_Q(u) \geq m - aR \geq (1 - \kappa a)m$. Both possibilities contradict the local girth bound, because

$$6.03(1 - \kappa a)m > 2\kappa m + 2 \geq 2R + 2. \quad (7.4)$$

For the declared parameters, the difference between the first two expressions is at least $(0.002 - 6.03\kappa \cdot 10^{-6})10^6 - 2 > 0$. Thus the ball is a tree and each outside vertex has at most one edge incidence into it. Its incidence neighborhood is a tree with possible additional leaves, hence a forest. The incidence neighborhood of a singleton is a star: loops or repeated neighbors would give a circuit of length one or two, which the same local girth bound excludes.

The constants in $K_{b,\epsilon}$ are fixed before s . Since $\log_2 D = .60005s$, we may arrange $\log_2(\Delta + 1) \leq .61s$. The ball has at most $(\Delta + 1)^{R+1}$ vertices. If an old vertex of layer i lies in it, Equation (7.1) gives $m \leq m_i/(1 - \kappa a)$. Therefore

$$\log_2 |B_Q(v)| \leq \left(\frac{\kappa m_i}{1 - \kappa a} + 1 \right) .61s = H_i \left(\frac{.61\kappa}{.50005(1 - \kappa a)} + \frac{.61}{.50005m_i} \right) < 4H_i < 10H_i.$$

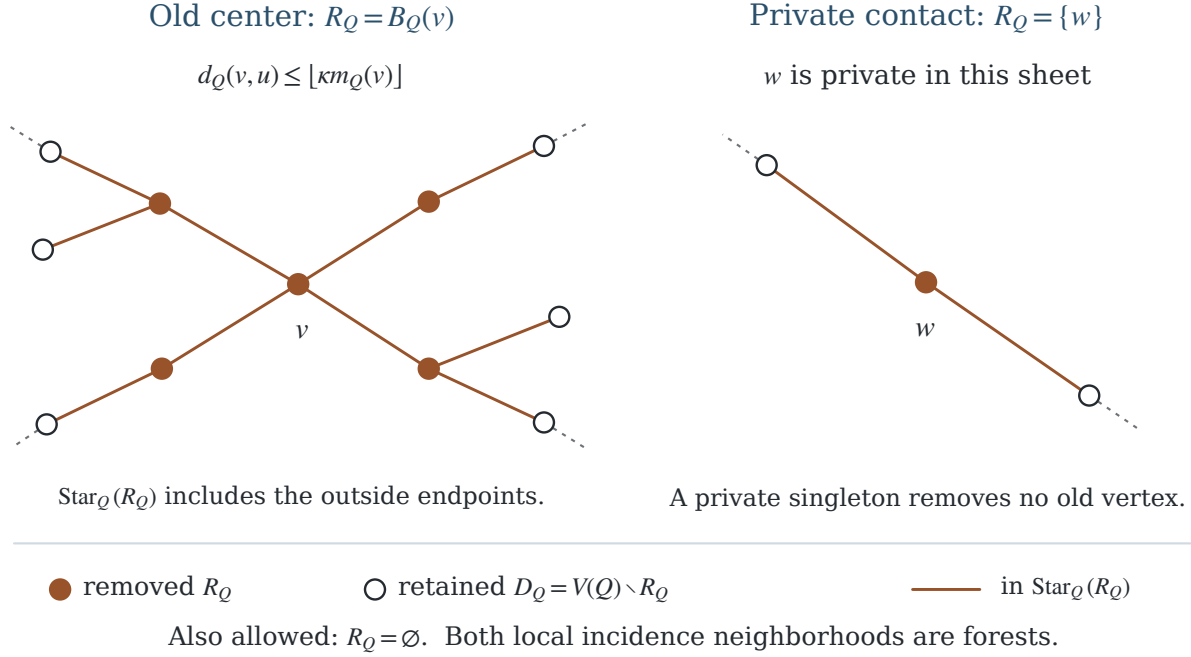


Figure 3: Schematic incidence neighborhoods of allowed removed sets: an old-centered ball or a private singleton of degree two. Filled vertices lie in R_Q ; hollow vertices remain in D_Q . Colored edges have an endpoint in R_Q , so their outside endpoints also belong to $\text{Star}_Q(R_Q)$. These neighborhoods are forests by Lemma 7.1. Dashed continuations represent unspecified sheet paths. Graph-metric lengths and cardinalities are not drawn to scale.

The displayed strict estimate follows from $a \leq 10^{-6}$ and $m_i \geq 10^6$. This proves Equation (7.3). $\square$

For an allowed removed set R_Q , write $D_Q = V(Q) \setminus R_Q$. Figure 3 illustrates the two nonempty choices and their incidence neighborhoods.

For a finite family $\mathcal{F} \subseteq \mathcal{Q}$, the contact set of $Q \in \mathcal{F}$ is

$$I_{\mathcal{F}}(Q) = V(Q) \cap \bigcup_{R \in \mathcal{F} \setminus \{Q\}} V(R).$$

These sets need not be finite. In particular, the fact that comparison arcs have no private edges does not exclude a singleton contact at a private vertex. The next lemma includes that case explicitly.

Lemma 7.2 (Finite peeling). *Every nonempty finite family $\mathcal{F}$ has a sheet Q for which $I_{\mathcal{F}}(Q)$ is either empty, contained in $B_Q(v)$ for some old contact v , or equal to a singleton private contact. Consequently the family can be ordered $Q_1, \dots, Q_n$ with allowed removed sets R_j of those three forms such that*

$$V(Q_j) \cap \bigcup_{\ell > j} V(Q_\ell) \subseteq R_j. \quad (7.5)$$

Proof. Suppose that no sheet has one of the three properties. Every sheet then has a contact. We will construct a polygon in distinct sheets with all but one side forced to use either a private edge or at least four comparison arcs. The exterior-arc bound will exclude this polygon.

Enter a sheet at a contact u . If u is old, choose a contact $w \notin B_Q(u)$; its integer distance from u is strictly greater than $\kappa m_Q(u)$. If u is private, choose any contact $w \neq u$, which is possible because

the contact set is not that private singleton. Transfer at w to another sheet containing it and repeat. Since the family is finite and consecutive sheets are different, a sheet eventually repeats. At the first repetition, retain the intervening $k \geq 2$ distinct sheets and close the path inside the repeated sheet. The resulting polygon has $k - 1$ chosen nonconstant sides; only the closing side is unrestricted.

Apply Lemma 6.5 with all these contact vertices fixed. If a chosen side has a private endpoint in its sheet, any nonconstant path with those endpoints starts or ends with a private edge. Some private edge is therefore unavoidable in every replacement path with the same distinct endpoints. Yet every arc of the reduced side is a comparison arc, none of which contains a private edge. This is impossible. Otherwise all $k - 1$ chosen sides have old endpoints and have intrinsic endpoint distance greater than $\kappa m_Q(u)$ at their incoming contacts. By Equation (7.2), each such side requires at least four exterior arc incidences. There are therefore at least $4(k - 1) > 4k - 6$ exterior incidences, again a contradiction. The argument does not require the closing side to be nonconstant or long.

Remove the sheet just obtained and apply the same assertion to the remaining finite family. At each removal take its whole contact ball, its private singleton, or the empty set, respectively. This gives Equation (7.5). $\square$

7.2 Finite cycle coordinates

An oriented edge chain over a commutative ring R always means a finite-support chain. The cycle module of a graph Y is $Z_1(Y; R) = \ker(\partial : C_1(Y; R) \rightarrow C_0(Y; R))$. A forest has no nonzero such cycle: the support of a proposed cycle is a finite forest, and its coefficient at an edge incident to a leaf must vanish; repeated leaf removal removes the whole support.

Lemma 7.3 (Direct cycle coordinates). *For every commutative coefficient ring R , the sum of the inclusions is an isomorphism*

$$\Phi_R : \bigoplus_{Q \in \mathcal{Q}} Z_1(Q; R) \longrightarrow Z_1(\mathcal{C}; R), \quad (c_Q)_Q \longmapsto \sum_Q c_Q. \quad (7.6)$$

Both the direct sum and every chain appearing here have finite support.

Proof. First, every finite cycle chain in a graph is a finite linear combination of circuit chains. To see this, take a spanning forest of its finite support graph. For each edge outside that forest, subtract its coefficient times the circuit formed with the forest path between its endpoints. This removes all edges outside the forest. The remainder is a cycle supported on a forest and is zero. Loops and pairs of parallel edges are included in this argument.

Let a circuit in $\mathcal{C}$ start at g and read the word w . The definition of Γ means that, in the free group on the formal labels, w is a finite product of conjugates of words read on closed walks of $\tilde{X}$, and their inverses. Follow that product from g in $\mathcal{C}$. Each conjugating stem is traversed once in each direction, so its oriented chain cancels. Each relator walk lies in a translated sheet. Free cancellation of adjacent inverse letters also cancels the corresponding oriented edge chains in the formal Cayley graph. Thus the circuit chain is a finite sum of sheet cycle chains. The preceding circuit decomposition proves surjectivity of Φ_R .

For injectivity, suppose that $\sum_{j=1}^n c_{Q_j} = 0$. Order the finitely many sheets in this relation by Lemma 7.2. An edge of Q_1 with a nonzero coefficient in c_{Q_1} must occur in another nonzero coordinate, since the total coefficient is zero. Both endpoints of that edge are contacts with a later sheet and belong to R_1 . Hence c_{Q_1} is supported on $\text{Star}_{Q_1}(R_1)$, a forest by Lemma 7.1, and is zero. Remove this coordinate and repeat. Every coordinate is zero. $\square$

7.3 Simultaneous domains for the analytic realization

We now choose removed sets on all sheets at once, so that the retained domains are disjoint and every Cayley circuit contains an edge internal to one of them. The cycle coordinates just proved provide finite witnesses for this second requirement, allowing both requirements to pass to a limit.

Proposition 7.4 (Simultaneous disjoint domains). *There are sets $R_Q \subseteq V(Q)$, one for each sheet, such that each R_Q is empty, an old-centered ball $B_Q(v)$, or a private singleton, and the sets $D_Q = V(Q) \setminus R_Q$ have the following properties.*

1. *The physical vertex sets D_Q are pairwise disjoint.*

2. *Put*

$$E_{\text{int}} = \bigcup_{Q \in \mathcal{Q}} \{e \in E(Q) : \text{both endpoints of } e \text{ lie in } D_Q\}.$$

Every edge in this union has a unique such sheet, and $(V(\mathcal{C}), E(\mathcal{C}) \setminus E_{\text{int}})$ is a forest.

3. *$\text{Star}_Q(R_Q)$ is a forest. If R_Q contains an old vertex of layer i , then $|R_Q| \leq 2^{10H_i}$. A private singleton removes no old coordinate.*

No equivariance of the choices R_Q is asserted or needed.

Proof. We first describe the possible limits of allowed removed sets. We then express disjointness and the residual-forest requirement by conditions on finitely many membership coordinates, verify them on finite families by peeling, and take a diagonal limit.

For a fixed sheet Q , let $\mathcal{K}_Q$ be the collection of all allowed removed sets in the statement, represented by their characteristic functions in $\{0, 1\}^{V(Q)}$. Every fixed vertex p belongs to only finitely many allowed old-centered balls. Indeed, if $p \in B_Q(v)$, then

$$d_Q(p, v) \leq \kappa m_Q(v) \leq \kappa(m_Q(p) + ad_Q(p, v)), \quad d_Q(p, v) \leq \frac{\kappa m_Q(p)}{1 - \kappa a}.$$

The last radius is finite, and Q is locally finite. Also p belongs to at most one allowed private singleton. It follows that $\mathcal{K}_Q$ is closed in the characteristic-function topology. For completeness, if a pointwise limit is nonempty, choose a vertex p in it. All sufficiently late removed sets contain p , so they belong to a fixed finite collection. A subsequence has one constant value, which must equal the pointwise limit. If the limit is empty, it is an allowed value already. This also explains why the empty choice must be retained.

All vertices, sheets, finite circuits, and their incidences are countable. Choose an orientation for each finite circuit C of $\mathcal{C}$ and let c_C be its integral circuit chain. Using $\Phi_{\mathbb{Z}}$ from Lemma 7.3, fix its finite decomposition

$$c_C = \sum_{Q \in \mathcal{F}_C} c_{C,Q}, \quad |\mathcal{F}_C| < \infty. \quad (7.7)$$

We impose the following countable collection of finite-coordinate conditions on the removed sets:

1. For each physical vertex x and distinct sheets Q, R containing it, x is not retained in both D_Q and D_R .
2. For each circuit C , there are $Q \in \mathcal{F}_C$ and $e \in E(C) \cap E(Q)$ such that both endpoints of e lie in D_Q .

Each condition is a finite Boolean condition, hence is preserved by coordinatewise limits. In the second condition both the list of edges and the list of candidate sheets are finite. Replacing that list by all sheets would not give the same closedness argument.

Any finite collection of these conditions can be satisfied. Take the finite union of all sheets appearing in the selected first-type conditions and of all sets $\mathcal{F}_C$ for the selected second-type conditions. Peel this finite family by Lemma 7.2 and use its allowed removed sets. Assign the empty removed set to every sheet outside the family. The selected first-type conditions hold by Equation (7.5).

Suppose that a selected circuit condition failed. In its decomposition (7.7), take the earliest peeled sheet with a nonzero coordinate, after discarding earlier zero coordinates. An edge of this coordinate whose two endpoints lie outside its removed set cannot occur in any later coordinate: if it did, its endpoints would be contacts in the removed set. If this edge belonged to C , it and the present sheet would witness the selected condition, contrary to its failure. Hence it lies outside C , and its coefficient in c_C , and therefore in the present coordinate, is zero. Thus the coordinate is supported on the incidence neighborhood of its removed set and is a cycle in a forest. It is zero. Induction removes all its coordinates, contradicting $c_C \neq 0$. This proves finite satisfiability.

Here one can finish by an explicit diagonal argument. Enumerate all the conditions, and for each n choose a full assignment satisfying the first n . Enumerate the countably many membership coordinates (Q, v) . Pass successively to an infinite subsequence on which the first coordinate is constant, one on which the second is constant, and so on, and take the diagonal subsequence. It converges pointwise. The preceding closedness argument shows that each sheet's limiting removed set belongs to $\mathcal{K}_Q$. Every one of the finitely supported conditions holds in the limit.

The first conditions give disjoint domains. An edge internal to two domains would have an endpoint in their intersection, so its sheet is unique. A circuit in the residual graph would violate its own second-type condition. The residual graph is therefore a forest. The last assertions follow from Lemma 7.1. $\square$

Lemma 7.5 (Incidence coordinates). *The map sending a sheet incidence to its physical vertex and its base type is a bijection*

$$\{(Q, u) : Q \in \mathcal{Q}, u \in V(Q)\} \longrightarrow \Gamma \times V(X). \quad (7.8)$$

In particular, a physical vertex occurs in at most $n_i^{\text{full}} = |\mathcal{A}_i| + |\mathcal{B}_i|$ old layer- i incidences, and

$$n_i^{\text{full}} = (1 + 2^{2r-s})2^{H_i} \leq C_s 2^{H_i}. \quad (7.9)$$

Proof. Normalize the embedded reference sheet as $j(v, f) = ft_v$, with $f \in \mathbf{H}$ and $v \in V(X)$. The deck group is identified with its embedded image in Γ . A sheet is hQ_0 for a coset $h\mathbf{H}$, because its stabilizer is $\mathbf{H}$. At a physical vertex u and specified base type v , the equality $u = hft_v$ forces and is achieved by the unique coset $h\mathbf{H} = ut_v^{-1}\mathbf{H}$. Embeddedness makes its source vertex unique. This proves the bijection. The displayed cardinality follows from the definitions of the two old layer state sets. It counts only old types; private singleton removals add no old incidence or active old coordinate. $\square$

7.4 The cone complex and exact chains

The cone-chain and prime-order obstruction method follows [32, Section 7.6]. We prove its exact-chain interfaces for the present sheets below. The integral detector extension and Bott realization are separate arguments of this manuscript.

We return to the cycle coordinates to obtain the topological consequences. This argument uses the full sheets and is independent of the domain choices. Start with $\mathcal{C}$, add one vertex c_Q for every $Q \in \mathcal{Q}$, and add a spoke $\sigma_{Q,u}$ directed from c_Q to each $u \in V(Q)$. For every positive edge $e : u \rightarrow v$ of Q , attach a separate triangular cell $\tau_{Q,e}$ with cellular boundary

$$\partial\tau_{Q,e} = e + \sigma_{Q,u} - \sigma_{Q,v}. \quad (7.10)$$

The resulting two-dimensional CW complex is denoted $\mathcal{Z}$. Different sheet occurrences give different triangles and spokes, even when their images in $\mathcal{C}$ agree. This cell description also defines the complex for infinite sheets without using any infinite chain sums.

Lemma 7.6 (Exact cone chains). *For every commutative coefficient ring R , the augmented cellular complex*

$$0 \rightarrow C_2(\mathcal{Z}; R) \xrightarrow{\partial_2} C_1(\mathcal{Z}; R) \xrightarrow{\partial_1} C_0(\mathcal{Z}; R) \xrightarrow{\varepsilon} R \rightarrow 0 \quad (7.11)$$

is exact. This assertion depends on finite cycle coordinates, not on the compactness step in Proposition 7.4.

Proof. For an edge chain b in a sheet Q , let $c_Q * b$ denote the same linear combination of its triangles, and let $\sigma_Q(t)$ denote the spoke chain with coefficients given by a vertex chain t in that sheet. Equation (7.10) reads

$$\partial(c_Q * b) = b - \sigma_Q(\partial b).$$

Any finite 2-chain determines a finite family of sheet edge chains b_Q . If its boundary vanishes, its components on the distinct spoke sets give $\partial b_Q = 0$ for every Q . Its original-edge component is then $\sum_Q b_Q = 0$. Injectivity of Φ_R in Lemma 7.3 forces every $b_Q = 0$, proving injectivity at C_2 .

Let $b + \sum_Q \sigma_Q(t_Q)$ be a finite 1-cycle, with b on $\mathcal{C}$. Its coefficient at c_Q says $\varepsilon(t_Q) = 0$. Since Q is connected, t_Q is the boundary of a finite edge chain a_Q : join the finitely many supporting vertices to one chosen vertex by finite paths and use their coefficients. Adding $\sum_Q \partial(c_Q * a_Q)$ removes all spoke terms. The remaining 1-cycle lies on $\mathcal{C}$, and surjectivity of Φ_R writes it as a finite sum of sheet cycles. Each such cycle is the boundary of the identical combination of its cone triangles. Hence the original 1-cycle was a boundary.

Finally $\mathcal{C}$ is connected and every cone is attached to it. A finite vertex chain of total coefficient zero is the boundary of paths to a fixed vertex. This proves exactness at C_0 and at the augmentation. There are no cells of dimension greater than two. Every argument involved finite-support chains only. $\square$

Left translation acts cellularly on $\mathcal{Z}$ and preserves all specified positive orientations. Write $A = \mathbb{Z}\Gamma$ and $B = \mathbb{Z}H$. Its cellular modules satisfy

$$C_1(\mathcal{Z}; \mathbb{Z}), C_2(\mathcal{Z}; \mathbb{Z}) \text{ are free } A\text{-modules}, \quad C_0(\mathcal{Z}; \mathbb{Z}) \cong A \oplus A \otimes_B \mathbb{Z}. \quad (7.12)$$

Indeed stabilizing a formal generator edge fixes its initial group vertex; stabilizing a spoke fixes its group-vertex endpoint; and stabilizing a triangle fixes its specified positive base edge. All three stabilizers are trivial. The group vertices form one free orbit, and the cone vertices form the orbit Γ/H . These observations prove Equation (7.12) without assuming torsion-freeness of Γ and without making the domain choices equivariant.

7.5 The prime-order obstruction

We record the elementary resolution comparison used both here and in the detector argument below. Modules and resolutions in this paragraph may have arbitrary free ranks.

Lemma 7.7 (Comparison of free resolutions). *Let $P_\bullet \rightarrow M$ and $Q_\bullet \rightarrow N$ be exact augmented complexes of free modules over a ring A . Every homomorphism $M \rightarrow N$ lifts to a chain map $P_\bullet \rightarrow Q_\bullet$. Any two such lifts are chain homotopic. In particular, two free resolutions of one module are chain homotopy equivalent, and remain so after tensoring with a right A -module or applying $\text{Hom}_A(-, L)$.*

Proof. Choose the image of each free basis vector of P_0 to lift its prescribed image under augmentation. Having defined the map through degree $n - 1$, the desired image of the boundary of a basis vector of P_n is a cycle in Q_{n-1} . Exactness supplies a preimage in Q_n . Choose it and extend linearly. This constructs the chain map in every degree.

For two lifts, let f be their difference. Its augmentation is zero. Inductively suppose that h_{n-1} has been chosen with the earlier homotopy identities. The map $f_n - h_{n-1}\partial_n$ takes values in the cycles of Q_n ; this follows on applying the boundary and using the chain-map identity and the preceding homotopy identity. Choose preimages on the free basis of P_n to obtain h_n . Then $f_n = \partial_{n+1}h_n + h_{n-1}\partial_n$. For $n = 0$ the same argument uses the zero augmentation. Thus the lifts are homotopic. Lifts of the identity in both directions give the stated chain homotopy equivalence. Tensor and Hom preserve the displayed homotopy equations. No step requires a finite basis. $\square$

Lemma 7.8 (No prime-order subgroup). *The group Γ is torsion-free.*

Proof. The integral Heisenberg group is torsion-free. For example, in matrix coordinates $(\alpha, \beta, \gamma) \in \mathbb{Z}^3$, the normal-form element $x^n y^l z^c$ from Section 5 has coordinates $(\alpha, \beta, \gamma) = (n, l, c + nl)$. Its law is

$$(\alpha, \beta, \gamma)(\alpha', \beta', \gamma') = (\alpha + \alpha', \beta + \beta', \gamma + \gamma' + \alpha\beta'). \quad (7.13)$$

If a positive power is the identity, its first two coordinates force $\alpha = \beta = 0$, and its third then forces $\gamma = 0$.

Suppose $P \leq \Gamma$ has prime order p . The stabilizer of a cone vertex is a conjugate of H , so it intersects P trivially. The other cell stabilizers are already trivial by Equation (7.12). Thus P acts freely on the set of cells of every dimension. This is freeness of setwise cell stabilizers, not just absence of pointwise-fixed edges. In particular, no involution can invert a formal positive Cayley edge. Choosing orbit representatives gives free modules over $A_p = \mathbb{F}_p[P]$. Lemma 7.6, with coefficients $\mathbb{F}_p$, would give a free resolution of its trivial module of length two.

Let $t = u - 1$ for a generator u of P . Then $A_p \cong \mathbb{F}_p[t]/(t^p)$, and the complex

$$\cdots \xrightarrow{t} A_p \xrightarrow{t^{p-1}} A_p \xrightarrow{t} A_p \xrightarrow{t^{p-1}} A_p \xrightarrow{t} A_p \longrightarrow \mathbb{F}_p \longrightarrow 0 \quad (7.14)$$

is exact. In the basis $1, t, \dots, t^{p-1}$, the kernels of multiplication by t and t^{p-1} are respectively (t^{p-1}) and (t) . Tensoring this resolution with the trivial right module $\mathbb{F}_p$ makes every differential zero; its homology is $\mathbb{F}_p$ in every nonnegative degree. Tensoring the alleged length-two resolution gives zero homology in every degree above two. This contradicts Lemma 7.7. Hence Γ has no subgroup of prime order. A nonidentity element of finite order n would have a power of order p for any prime divisor p of n , which proves the assertion. $\square$

7.6 The primitive class on the Heisenberg torus

We first construct an integral class on $\mathbf{H}$ that evaluates to -1 on the ordered torus (z, x) , and then prove that it extends to Γ . We use an explicit bar model both to compute this evaluation and to identify the extension map with pullback on classifying spaces. For a group Λ , let $\mathcal{B}_n(\Lambda)$ be the free abelian group on ordered tuples $(g_0, \dots, g_n) \in \Lambda^{n+1}$, with diagonal left action, alternating deletion boundary, and augmentation one on every vertex. It is free over $\mathbb{Z}\Lambda$: each orbit has a unique representative with $g_0 = 1$. Prepending 1 gives an abelian-group contracting homotopy, since direct cancellation in the deletion formula gives $\partial s + s\partial = \text{id}$, including the augmentation. Thus $\mathcal{B}_\bullet(\Lambda)$ is a free resolution of the trivial module.

The realization of the semi-simplicial set of these ordered tuples is a contractible free Λ -CW complex. On the simplex $(g_0, \dots, g_n)$ with barycentric coordinates $(t_0, \dots, t_n)$, the contraction has value in $(1, g_0, \dots, g_n)$ with coordinates $(u, (1-u)t_0, \dots, (1-u)t_n)$, for $0 \leq u \leq 1$. These formulas agree under deletion of zero-coordinate faces. The quotient has one vertex, loops $[g]$, and triangular relations $[g][h] = [gh]$, so its fundamental group is Λ . Lifting a simplex from an initial vertex g_0 produces the tuple $(g_0, g_0g_1, g_0g_1g_2, \dots, g_0g_1 \cdots g_n)$; its face rules are precisely the preceding deletion rules. Thus the contractible tuple complex is its universal cover, and the quotient is a model for $B\Lambda$. Its integral cellular cochain complex is $\text{Hom}_{\mathbb{Z}\Lambda}(\mathcal{B}_\bullet(\Lambda), \mathbb{Z})$. Consequently its cohomology is $H^*(B\Lambda; \mathbb{Z})$, or equivalently $\text{Ext}_{\mathbb{Z}\Lambda}^*(\mathbb{Z}, \mathbb{Z})$ computed with this resolution. A subgroup inclusion induces the inclusion of the tuple complexes, so restriction of these cochains is precisely pullback on the indicated classifying spaces.

We write $[g_1 \mid \cdots \mid g_n]$ for the orbit representative $(1, g_1, g_1g_2, \dots, g_1 \cdots g_n)$. In particular,

$$\partial[g \mid h] = g[h] - [gh] + [g], \quad \partial[g \mid h \mid k] = g[h \mid k] - [gh \mid k] + [g \mid hk] - [g \mid h].$$

With trivial coefficients these are the inhomogeneous group cochain formulas used below.

Lemma 7.9 (A primitive Heisenberg detector). *In the coordinates of Equation (7.13), the integer function*

$$\omega((\alpha, \beta, \gamma), (\alpha', \beta', \gamma')) = \alpha\gamma' + \binom{\alpha}{2}\beta' \quad (7.15)$$

is a normalized group 2-cocycle. Its class $c_{\mathbf{H}}$ evaluates to -1 on the torus with ordered generators (z, x) .

Proof. The binomial expression $\binom{\alpha}{2} = \alpha(\alpha-1)/2$ is an integer also for negative α . Normalization follows immediately from Equation (7.15). Using Equation (7.13), the identity

$$\binom{\alpha + \alpha'}{2} = \binom{\alpha}{2} + \binom{\alpha'}{2} + \alpha\alpha'$$

gives, for all $g, h, k \in \mathbf{H}$,

$$\omega(g, h) + \omega(gh, k) = \omega(h, k) + \omega(g, hk).$$

Explicitly, the only mixed term added to the central coordinate of hk is $\alpha'\beta''$; multiplying it by α agrees with the mixed term $\alpha\alpha'\beta''$ in the binomial identity. All other terms on the two sides coincide separately. This proves the cocycle equation in the inhomogeneous bar complex. Equivalently, the multiplication

$$(t, g)(t', h) = (t + t' + \omega(g, h), gh)$$

on $\mathbb{Z} \times \mathbf{H}$ is associative, with central kernel $\mathbb{Z} \times \{1\}$, and defines the corresponding integral central extension. For the canonical lifts $X = (0, x)$, $Y = (0, y)$, $Z = (0, z)$ and the positive central kernel

generator $T = (1, 1)$, one has $[X, Y] = Z$, $[X, Z] = T$, and Y, Z, T commute. This is the unipotent shear extension that lengthens the Heisenberg block.

The subgroup generated by $z = (0, 0, 1)$ and $x = (1, 0, 0)$ is freely abelian of rank two: $z^u x^v = (v, 0, u)$ in these coordinates. For its oriented torus, the bar 2-chain

$$[z \mid x] - [x \mid z]$$

represents the oriented fundamental class. To check the sign and the normalization directly, subdivide the fundamental square with successive vertices $1, z, zx, x$ by the diagonal from 1 to zx . The two oriented triangles give $(1, z, zx) - (1, x, zx)$, which are the displayed inhomogeneous bar chains. Sending grid vertices (u, v) to $z^u x^v$ supplies the equivariant map from the triangulated universal cover of the torus into the bar model. Evaluation is therefore

$$\omega(z, x) - \omega(x, z) = 0 - 1 = -1.$$

□

7.7 Extending integral degree-two classes

The class c_H now gives the detector on the embedded torus. It remains to extend it from H to Γ . The exact cone chains prove the stronger assertion that every integral degree-two class on H extends.

Lemma 7.10 (Integral restriction is onto in degree two). *The inclusion of the deck subgroup induces a surjection*

$$H^2(B\Gamma; \mathbb{Z}) \longrightarrow H^2(BH; \mathbb{Z}). \quad (7.16)$$

Proof. Put $A = \mathbb{Z}\Gamma$, $B = \mathbb{Z}H$, $M = \mathbb{Z}$, and $C_0 = C_0(\mathcal{Z}; \mathbb{Z})$. By Lemma 7.6 and Equation (7.12), the module $K = \ker(C_0 \rightarrow M)$ has a length-one free resolution

$$0 \longrightarrow P_1 = C_2(\mathcal{Z}; \mathbb{Z}) \xrightarrow{d_P} P_0 = C_1(\mathcal{Z}; \mathbb{Z}) \longrightarrow K \longrightarrow 0.$$

This is the vanishing of $\text{Ext}_A^2(K, M)$ that makes the degree-two restriction map surjective. The proof has two parts: we construct the resulting surjection in Ext using a free resolution, and then identify it with restriction by comparing the induced Heisenberg resolution with the bar resolution of Γ .

As a right B -module, A is a direct sum of copies of B , using representatives of the cosets Γ/H . Therefore

$$I_\bullet = A \otimes_B \mathcal{B}_\bullet(H)$$

is an exact free A -resolution of $A \otimes_B M$. Let $Q_\bullet$ resolve $C_0 = A \oplus A \otimes_B M$ by taking $Q_0 = A \oplus I_0$, $Q_n = I_n$ for $n \geq 1$, with the extra A summand concentrated in degree zero. Lift $K \hookrightarrow C_0$ to a chain map $f : P_\bullet \rightarrow Q_\bullet$ as follows. Choose f_0 on a free basis to lift that inclusion. The elements $f_0 d_P(P_1)$ have zero augmentation in C_0 , so exactness of $Q_\bullet$ permits a lift $f_1 : P_1 \rightarrow Q_1$ with $d_Q f_1 = f_0 d_P$. There are no higher P terms.

Define a free augmented complex $T_\bullet \rightarrow M$ by

$$T_0 = Q_0, \quad T_n = Q_n \oplus P_{n-1} \quad (n \geq 1),$$

where $P_j = 0$ for $j \geq 2$, with boundary

$$d_T(q, p) = (d_Q q + f_{n-1} p, -d_P p) \quad (n \geq 2), \quad d_T(q, p) = d_Q q + f_0 p \quad (n = 1).$$

The augmentation is $Q_0 \rightarrow C_0 \rightarrow M$. The identities defining f give $d_T^2 = 0$. This complex is exact. In degree two, a cycle (q, p) has $d_P p = 0$, hence $p = 0$, and then q bounds in $Q_\bullet$. The same is immediate in degrees at least three. In degree one, if $d_Q q + f_0 p = 0$, augmentation in C_0 implies that $p = d_P p_1$ for some p_1 . Then $q + f_1 p_1$ is a Q -cycle and equals $d_Q q_2$; thus $(q, p) = d_T(q_2, -p_1)$. In degree zero, an element mapping to zero in M has augmentation in K ; subtract a suitable $f_0 p$ and then lift the remainder through d_Q . Surjectivity of the augmentation is immediate. Hence $T_\bullet$ is a free resolution of M .

The inclusion $Q_\bullet \rightarrow T_\bullet$ lifts $C_0 \rightarrow M$. Any degree-two cocycle $\phi : Q_2 \rightarrow M$ extends to $\psi : T_2 = Q_2 \oplus P_1 \rightarrow M$ by $\psi(q, p) = \phi(q)$. It is still a cocycle because $T_3 = Q_3$ and its boundary has zero P_1 component. Therefore the induced map

$$\mathrm{Ext}_A^2(M, M) \longrightarrow \mathrm{Ext}_A^2(C_0, M) \quad (7.17)$$

is surjective. This is explicitly the degree-two portion of the Ext sequence for $0 \rightarrow K \rightarrow C_0 \rightarrow M \rightarrow 0$. There is a cochain isomorphism

$$\mathrm{Hom}_A(A \otimes_B \mathcal{B}_n(\mathbf{H}), M) \longrightarrow \mathrm{Hom}_B(\mathcal{B}_n(\mathbf{H}), M), \quad \phi \longmapsto (b \mapsto \phi(1 \otimes b)). \quad (7.18)$$

Its inverse sends η to $a \otimes b \mapsto a\eta(b)$; this is well-defined by B -linearity and is A -linear. The maps commute with the boundaries. The extra free A summand in Q_0 contributes no cohomology in degree two, so Equation (7.18) identifies the target of Equation (7.17) with $H^2(B\mathbf{H}; \mathbb{Z})$. This proves the needed instance of Shapiro's isomorphism at the cochain level.

It remains to identify the map, rather than just its two groups. Let $\mathcal{B}_\bullet(\Gamma)$ be the preceding bar resolution. Choose a comparison map $T_\bullet \rightarrow \mathcal{B}_\bullet(\Gamma)$ lifting the identity of M . The composite

$$I_\bullet \longrightarrow Q_\bullet \longrightarrow T_\bullet \longrightarrow \mathcal{B}_\bullet(\Gamma)$$

lifts the augmentation $A \otimes_B M \rightarrow M$. Another such lift is the explicit map

$$\theta(a \otimes (h_0, \dots, h_n)) = a(h_0, \dots, h_n),$$

where the tuple on the right is viewed in Γ . Lemma 7.7 makes these lifts chain homotopic. After Equation (7.18), precomposition with θ restricts a Γ -cochain to tuples from $\mathbf{H}$. As observed in the bar model, this is exactly pullback along $B\mathbf{H} \rightarrow B\Gamma$. Thus the surjection in Equation (7.17) is the map in Equation (7.16). $\square$

Proposition 7.11 (Torsion-freeness and the integral detector). *The graphical group Γ is torsion-free. The map $i_\Gamma : \mathbb{Z}^2 \rightarrow \Gamma$ sending the ordered basis to (z, x) is injective, and there is a class $c_\Gamma \in H^2(B\Gamma; \mathbb{Z})$ such that*

$$\langle (Bi_\Gamma)^* c_\Gamma, [T^2] \rangle = -1. \quad (7.19)$$

Proof. Lemma 7.8 proves torsion-freeness. The deck embedding of Proposition 6.4 and the coordinate calculation in Lemma 7.9 prove injectivity of i_Γ . Choose c_Γ mapping to $c_\mathbf{H}$ under the surjection of Lemma 7.10. Its map was identified there with actual restriction in integral cohomology. Equation (7.19) now follows from the explicit evaluation in Lemma 7.9. $\square$

8 Realization of the Bott cancellation in the reduced algebra

We now prove that the rank-zero Bott class of the embedded torus $\langle z, x \rangle$ vanishes in the reduced group algebras along a tail of the compatible accuracy sequence constructed below. The localized

Bott profiles from Section 5 can be moved between neighboring heights. We realize two families of these moves by finite Cayley formulas, using the disjoint domains of Proposition 7.4. As the errors tend to zero, the formulas give exact projection homotopies in a norm sequence quotient. Their cancellation removes the initial-height profile. We then identify that profile with the physical torus Bott class and recover an exact finite witness in a single reduced group algebra, before passing to a finitely generated subgroup.

We use the right Cayley graph and the conventions of Proposition 6.4. Thus a positive formal edge is $g \rightarrow g\sigma$, the deck group acts on the left, and, after choosing one lift of each base vertex v , the embedded cover has coordinates

$$j(v, f) = ft_v \quad (f \in H). \quad (8.1)$$

We identify H with its embedded image in Γ . Its distinct sheets are the left translates $h\tilde{X}$, indexed by the cosets hH . On $\ell^2(\Gamma)$ put

$$R(g)\delta_h = \delta_{hg^{-1}}.$$

This is the right regular representation with the convention that $R(g)R(k) = R(gk)$. The unitary $I\delta_h = \delta_{h^{-1}}$ satisfies

$$IR(g)I = \lambda(g). \quad (8.2)$$

Consequently the norm closure of the right regular group algebra is a copy of $C_r^*(\Gamma)$. We use it throughout the estimates and apply (8.2) at the end. This equivalence sends the represented group element g to the same element g , so it does not introduce an antihomomorphism on group K -theory. We orient each positive old incidence edge from its $\mathcal{A}$ endpoint v to its $\mathcal{B}$ endpoint w . If its label is σ and its forward voltage is h_e , then

$$t_v\sigma = h_et_w, \quad R(\sigma)\delta_{ft_w} = \delta_{fh_e^{-1}t_v}. \quad (8.3)$$

Thus the $\mathcal{B}$ -to- $\mathcal{A}$ incidence block of $R(\sigma)$ is the deck operator $R_H(h_e)$. For our chosen direction a sampled term $R_H(v_e)$ is implemented by the forward voltage $h_e = v_e$. The opposite, forward transition operator has the adjoint convention.

We will take a sequence of constructions, indexed by ν . The integer b is fixed once, including across this sequence. The parameters $s = s_\nu$ and $H_0 = H_{0,\nu}$ may increase; the degree is fixed at all heights of any one construction. All matrix sizes in the Bott representatives below are fixed. The number of incidence types, the finite weight net, and the lengths of operator products may depend on ν ; these choices precede the choice of s_ν .

8.1 Compact Bott profiles and common carriers

Use throughout the compact Bott projection p in Equation (5.2), with $m = 2$, $e = e_\infty = \text{diag}(0, 1)$, and $a = p - e$. The explicit eigenvectors and winding calculation in Section 5 prove that $[p] - [e]$ is a compactly supported Bott generator. This projection and its signed class are fixed for the entire accuracy sequence.

In angular torus coordinates $(\zeta, \xi) \in (\mathbb{R}/\mathbb{Z})^2$ corresponding to (z, x) , put

$$Z_i = 2^{500H_i}, \quad N_i^{\text{freq}} = 2^{200H_i}, \quad J_i = 2^{50H_i}, \quad a_i(\zeta, \xi) = a(Z_i(\zeta - \alpha), N_i^{\text{freq}}\xi).$$

The functions are defined in a small coordinate chart and extended by zero; their supports lie strictly inside that chart. Thus $e + a_i$ is a smooth projection on the torus. It has the same first Chern number as p , so

$$[e + a_i] - [e] = \beta \quad \text{in } K_0(C(\mathbb{T}^2)), \quad (8.4)$$

where β is the torus realization of this fixed rank-zero Bott generator. The positive rescalings preserve its sign.

The rescaling path

$$a_{i,t}(\zeta, \xi) = a(2^{500bt} Z_i(\zeta - \alpha), 2^{200bt} N_i^{\text{freq}} \xi), \quad 0 \leq t \leq 1, \quad (8.5)$$

joins a_i to a_{i+1} . Each $e + a_{i,t}$ is a projection. In the rescaled coordinates all these functions form one compact smooth family. In particular there is a common modulus of norm continuity in t , independent of i and ν , because b is fixed.

Choose real continuous smooth cutoffs χ_i , with $0 \leq \chi_i \leq 1$, equal to one on the supports of all profiles used at level i , including the profiles $a_{i-1,t}$ transported from the preceding level. Their supports are contained in fixed enlargements, depending only on b , of the rectangle at level i . Choose an additional slightly larger Borel rectangle projection e_i^{car} containing $\text{supp } \chi_i$. We use the same choices on both sides of a layer. They may be chosen so that every function used in a block (i, k) , $|i - k| \leq 1$, is supported where both χ_i and χ_k equal one. Indeed the largest required support at level i is contained in the rectangle of level $i - 1$, and the ratios between successive widths are the fixed numbers 2^{500b} and 2^{200b} . At $i = 0$ the notation $i - 1$ merely prescribes a fixed enlargement, not an additional graph layer.

Apply Proposition 5.2 with these fixed enlargements. The spread operators from Lemma 5.1, written in the right regular representation, have the form

$$T_i = \sum_j a_{ij} R(y)^j, \quad \sum_j |a_{ij}|^2 = 1, \quad e_i^{\text{car}} T_i^* T_i e_i^{\text{car}} = e_i^{\text{car}}.$$

Their range carriers

$$P_i = T_i e_i^{\text{car}} T_i^* \quad (8.6)$$

are projections in the containing von Neumann algebra and satisfy

$$\tau_{\mathbf{H}}(P_i) = \tau_{\mathbf{H}}(e_i^{\text{car}}) \leq C_{\nu} 2^{-700H_i}. \quad (8.7)$$

The constant is uniform in the height. Increasing the starting height ensures the disjointness of the translates even for the specified enlarged rectangles. These Borel carriers are used only for Hilbert-space estimates. Actual group-algebra formulas below use continuous functions and the finite sums T_i ; no Borel carrier is asserted to belong to a reduced C^* -algebra.

For one construction put $\mathcal{L} = \ell^2(\Gamma)$ and regard e_i^{car} as a projection on $\mathcal{L}$ through the subgroup's right regular operators. The model space is

$$\mathcal{K} = \bigoplus_{i \geq 0} e_i^{\text{car}} \mathcal{L}. \quad (8.8)$$

A band profile $f = (f_{ik})_{|i-k| \leq 1}$ from our fixed compact profile families acts on this space from the k th to the i th summand by its torus function f_{ik} . Support in both carriers makes each block well defined; the uniform block norm and width one give a bounded operator f^{mod} . All cosets of $\mathbf{H}$ carry the same profile.

The profiles we must realize have the four diagonal height patterns

	0	1	2	3	...
ev	a_0	0	a_2	0	...
od	0	a_1	0	a_3	...
ev, +	0	0	a_2	0	...
0	a_0	0	0	0	...

Each entry is a compactly supported difference from the common scalar projection e ; a zero entry retains that scalar part in the full projection. Rotating between the disjoint pairs $(0, 1), (2, 3), \dots$ and then applying (8.5) moves the even pattern to the odd pattern. The pairs $(1, 2), (3, 4), \dots$ similarly move the odd pattern to the positive-even pattern. The even pattern also splits into the initial-height and positive-even patterns. Our analytic task is to realize these four profiles and the two paths in one fixed matrix size. The estimates below first pass from this model to normalized constants on the sheets, and then from the sheets to finite Cayley formulas.

8.2 Domains, trimming, and bounded maps

For this subsection fix one construction. Let

$$\mathcal{H}_{\text{abs}} = \bigoplus_Q \ell^2(Q)$$

be the abstract orthogonal sum over its sheets. A fixed finite matrix factor can be tensored onto every space without changing the argument. On each old vertex of height i use the carrier P_i of (8.6). More precisely, using the regular deck coordinate on the lifted vertex set, let $E_{Q,i}$ be the sum of these carriers over the old base vertices on both sides of height i in sheet Q . Put

$$E_i = \bigoplus_Q E_{Q,i}, \quad E = \bigoplus_{i \geq 0} E_i.$$

All these projections are zero on private-vertex coordinates.

Let D_Q be the domains from Proposition 7.4, and let D be coordinate restriction to their abstract sum. Their physical images are pairwise vertex-disjoint. Thus coordinate inclusion defines a partial isometry

$$U: \mathcal{H}_{\text{abs}} \longrightarrow \ell^2(\Gamma), \quad U^*U = D,$$

which is an isometry on $D\mathcal{H}_{\text{abs}}$ and is zero on its orthogonal complement.

Lemma 8.1 (Uniform trimming estimate). *For a construction whose constants have already been fixed,*

$$\|(1 - D_Q)E_{Q,i}\|^2 \leq C_\nu 2^{-690H_i}, \quad \delta_\nu := \|(1 - D)E\| \leq C_\nu 2^{-345H_0}.$$

In particular $H_{0,\nu}$ can be chosen so that $\delta_\nu \rightarrow 0$. If $A = EAE$ and $B = EBE$, and

$$\Psi_\nu(A) = UDADU^*,$$

then

$$\|\Psi_\nu(A)\Psi_\nu(B) - \Psi_\nu(AB)\| \leq \delta_\nu^2 \|A\| \|B\|. \quad (8.9)$$

Proof. For a projection P and a finite coordinate set B_0 , the ordinary Hilbert–Schmidt norm gives

$$\|1_{B_0}P\|^2 \leq \|1_{B_0}P\|_{\text{HS}}^2 = \sum_{u \in B_0} \langle P\delta_u, \delta_u \rangle.$$

In a matrix amplification one sums over the fixed matrix basis as well. Every regular deck translate of a given old vertex has the same diagonal trace. If the removed set in Q is an old-centered ball, then Proposition 7.4 bounds its entire cardinality by 2^{10H_i} whenever it meets height i . Equation (8.7) therefore gives the first estimate, with all repeated deck lifts counted separately. An empty removed set costs nothing. A private singleton also costs nothing, since $E_{Q,i}$ has zero private coordinates.

This statement is about indexed sheet coordinates even if the same physical vertex is old in another sheet.

Both D and E preserve the height decomposition. The norm of their orthogonal direct sum is the supremum of the block norms, which gives the bound on δ_ν . Finally, using $U^*U = D$, the difference in (8.9) is $UDA(D-1)BDU^*$. Its norm is at most $\|A\|\|B\|\|E(1-D)E\|$, and $\|E(1-D)E\| = \|(1-D)E\|^2$. $\square$

Trimming gives the almost multiplicative map needed for all model profiles. To compare the initial-height profile with the physical torus class, we also need to sum its normalized constants over sheets. Let $\mathcal{J}$ initially sum finitely supported vectors from finitely many abstract sheets into their physical vertex coordinates. This notation does not presume that untrimmed summation is bounded.

Lemma 8.2 (Synthesis on the active carriers). *Let n_i be the number of surviving old base vertices on both sides of height i . Then*

$$\|\mathcal{J}(1-D)E_i\|^2 \leq n_i \sup_Q \|(1-D_Q)E_{Q,i}\|^2 \leq C_\nu 2^{-689H_i}. \quad (8.10)$$

Consequently $\mathcal{J}E$ extends to a bounded operator and

$$\|\mathcal{J}E - UDE\| \leq C_\nu \sum_{i \geq 0} 2^{-344.5H_i} = C_\nu \frac{2^{-344.5H_0}}{1 - 2^{-344.5b}}. \quad (8.11)$$

Proof. For a physical vertex u and a specified old base vertex v , the only sheet coset that can contain u at base type v is $ut_v^{-1}\mathbf{H}$, by (8.1). This also follows from the incidence bijection in Lemma 7.5. Thus at height i each physical vertex receives at most n_i indexed coordinates. Pointwise Cauchy–Schwarz gives the first inequality in (8.10), initially on the stated dense domain. The layer cardinalities give $n_i \leq C_s 2^{H_i}$; Lemma 8.1 gives the second inequality. The bounded extensions at individual heights may be summed in operator norm, because $H_i = H_0 + bi$ and the square-root estimates are summable. On their common dense domain $\mathcal{J}E = UDE + \mathcal{J}(1-D)E$, proving (8.11) and the claimed extension. $\square$

8.3 Finite Cayley formulas and extraction of constants

The incidences supplied by Propositions 2.6, 3.4, and 4.14 will be denoted by $U_{ik}^0: \ell^2(\mathcal{B}_k) \rightarrow \ell^2(\mathcal{A}_i)$ for a pre-deletion normalized incidence, $|i - k| \leq 1$, suppressing its finite type index. It sends the normalized constant vector to the normalized constant vector, its adjoint does the same, and its centered norm is at most the fixed $\rho < 1$. Let U_{ik} be its compression to the surviving vertex sets. The deletion fraction is at most ε_{del} in each side and layer. We always normalize constant vectors using the actual surviving vertex counts.

Lemma 8.3 (Deleting a small fraction of vertices). *There are contractions $\widehat{U}_{ik}$ on the surviving spaces that send normalized constants exactly to normalized constants in both directions, have centered norm at most ρ , and satisfy*

$$\|U_{ik} - \widehat{U}_{ik}\| \leq C\sqrt{\varepsilon_{\text{del}}}.$$

The constant is uniform in the height and the incidence type.

Proof. For one incidence let a, b be the old normalized constants and let P_A, P_B be the surviving coordinate projections. Their normalized restrictions a', b' differ from a, b , after extension by zero, by at most $C\sqrt{\varepsilon_{\text{del}}}$. Put $Q_A = 1 - |a'\rangle\langle a'|$ and $Q_B = 1 - |b'\rangle\langle b'|$ on the surviving spaces and define

$$\widehat{U} = |a'\rangle\langle b'| + Q_A U Q_B.$$

Write $U^0 = |a\rangle\langle b| + U_0$. Then $Q_A P_A a = 0$ and $Q_B P_B b = 0$, so $\|Q_A U Q_B\| \leq \|U_0\| \leq \rho$. The two displayed summands of $\widehat{U}$ have orthogonal source and range spaces, proving its asserted gap and norm. The estimates $\|U b' - a'\| + \|U^* a' - b'\| \leq C\sqrt{\varepsilon_{\text{del}}}$ bound the remaining three blocks in $U - \widehat{U}$. $\square$

We embed the model space (8.8) using normalized constants on the surviving $\mathcal{B}_i$ vertices. The subgroup operators preserve $\mathcal{L} = \bigoplus_{h\mathbf{H}} \ell^2(h\mathbf{H})$. For each coset and height, map its vector to the corresponding sheet's normalized constant array, applying T_i to its deck coordinate. This defines an isometry

$$\mathcal{I}: \mathcal{K} \longrightarrow \mathcal{H}_{\text{abs}}, \quad \mathcal{I}\mathcal{I}^* \leq E.$$

Changing the representative of $h\mathbf{H}$ changes both coordinates by a left deck translation. The right regular functions and T_i commute with that translation, so the map is well defined. We write $\mathcal{K}_\nu$ and $\mathcal{I}_\nu$ when the construction index is needed.

The $\mathcal{A}_i$ coordinates supply finite routes between these chosen $\mathcal{B}_i$ constant coordinates. On the full abstract sheet space consider the compactly supported mean blocks

$$\begin{aligned} V_i &= U_{ii} \otimes T_i \chi_i T_i^*, \\ W_{ik}(f) &= U_{ik} \otimes T_i f_{ik} T_k^*. \end{aligned}$$

Here V_i maps the $\mathcal{B}_i$ side to the $\mathcal{A}_i$ side, and f_{ik} is supported where $\chi_i = \chi_k = 1$. All functions can be matrix-valued in a fixed dimension. Take the sums over sheets and heights, writing $V = \bigoplus_i V_i$ and $W(f)$ for the band matrix with blocks $W_{ik}(f)$, $|i - k| \leq 1$. They are supported on E , and their norms are uniformly bounded by the ordinary block Schur estimate.

For $k \geq 1$ define the finite word in these operators

$$\mathcal{R}_k(f) = (V^* V)^k V^* W(f) (V^* V)^k. \quad (8.12)$$

On the block from the $\mathcal{B}_j$ side to the $\mathcal{B}_i$ side its incidence factor is

$$(U_{ii}^* U_{ii})^k U_{ii}^* U_{ij} (U_{jj}^* U_{jj})^k.$$

The function factor is exactly $T_i f_{ij} T_j^*$: every extra cutoff is one on $\text{supp } f_{ij}$. Replacing all incidences by Lemma 8.3's hatted incidences changes the product by at most $C_f k \sqrt{\varepsilon_{\text{del}}}$. On constants the hatted incidence product is exactly the map between normalized constants. To see the suppression on the complement, let Π_i be the projection onto normalized constants in the surviving $\mathcal{B}_i$ space. Then

$$\widehat{U}_{ii}^* \widehat{U}_{ii} = \Pi_i + C_i, \quad \Pi_i C_i = C_i \Pi_i = 0, \quad \|C_i\| \leq \rho^2.$$

Each flanking k th power therefore has centered norm at most ρ^{2k} . The intervening block preserves constants and their orthogonal complements, so the full product has centered norm at most $C\rho^{2k}$. A band matrix of width one has norm at most three times the supremum of its block norms. We have therefore proved

$$\|\mathcal{R}_k(f) - \mathcal{I} f^{\text{mod}} \mathcal{I}^*\| \leq C_f (\rho^{2k} + k \sqrt{\varepsilon_{\text{del}}}). \quad (8.13)$$

Thus the route extracts the prescribed profile on the model coordinates and suppresses the other incidence modes. The estimate is in the full operator norm, including all nonconstant coordinates.

Lemma 8.4 (Finite formulas for a prescribed profile family). *Fix a finite collection of band profiles $f = (f_{ik})_{|i-k| \leq 1}$ of the kind above, a fixed finite matrix size, and a target error $\eta > 0$. One can choose the construction parameters so that for each such profile there is a finite matrix group-algebra element b_f satisfying*

$$\|b_f - UDI f^{\text{mod}} \mathcal{I}^* DU^*\| < \eta. \quad (8.14)$$

The estimate holds simultaneously over all heights and sheets. Profiles may be set to zero on any prescribed parity pattern, on all positive heights, or on the initial height, without a height-dependent enlargement of the alphabet.

Proof. First choose k so that the gap term in (8.13) is sufficiently small. Include the finitely many cutoff and symbol blocks V and $W(f)$ as designated incidence types in Proposition 5.2. That Proposition supplies a finite weight net. We first compare each sampled factor with its compressed sheet mean, and then control the products in (8.12).

For a type τ define the actual finite polynomial

$$C_\tau = q^{-1} \sum_{\sigma \in \mathcal{S}_\tau} W_\sigma R(\sigma) \in M_m(R(\mathbb{C}[\Gamma])), \quad (8.15)$$

where the positive augmented label σ determines its matrix weight W_σ . These weights already incorporate the bounded factor $q/q_{\tau,ik}$ in Proposition 5.2, before finite-net rounding; the polynomial uses the single common normalizer q^{-1} . The set $\mathcal{S}_\tau$ is finite. On a sheet Q , sum the polynomial's contributions over the prescribed source edges of type τ in Q . Equation (8.3) identifies this source-edge sum, on each basis vector, with the sampled $\mathcal{B}$ -to- $\mathcal{A}$ convolution for that type, with forward voltage equal to the sampled group element. The sampled operator differs in full regular norm from its prescribed mean by an arbitrarily small uniform amount.

Compressing to the disjoint domains and taking the orthogonal sum does not increase the sampling error. It remains to estimate the edges outside that sum. The physical Cayley polynomial and this domain sum differ precisely on the remaining formal edges, with their specified weights. These include nonsource Cayley chords and unused weight-tag edges, even when both endpoints lie in one retained domain. An edge is removed from the remainder only when it is an edge of its retained source sheet. Disjointness of the domains prevents any ambiguity. The remaining edges form a forest by Proposition 7.4. A matrix adjacency on a forest of degree at most L and edge weight norm at most w has norm at most $2w\sqrt{L}$: root each component, take the operator sending parent coordinates to child coordinates, use that each child has a unique parent, and add its adjoint. For a rectangular or directed block, self-adjoint doubling has support in the bipartite double cover of this forest, which is again a forest. Consequently the residual error here is at most

$$C \frac{\sqrt{|\mathcal{S}_{\text{wt}}|}}{q},$$

where $\mathcal{S}_{\text{wt}}$ is the finite weighted old alphabet. All private letters have coefficient zero. Each normalized nonzero weight has norm at most C/q , with C fixed before s . The sharp alphabet bound is

$$\frac{\sqrt{|\mathcal{S}_{\text{wt}}|}}{q} \leq 2^{-.000025s + O(\log s) + O_{b,\eta}(1)},$$

which tends to zero after the finite type and net choices. Thus every factor of (8.12) has a finite Cayley formula close to its Ψ_ν image in full norm.

We now use those finite formulas in the same finite word (8.12). Equation (8.9) controls replacing the product of compressed means by the compression of their product. Ordinary telescoping controls

perturbing the factors: for a product of r factors of norm at most L , per-factor error θ costs at most $r\theta(L + \theta)^{r-1}$. The numbers r, L are fixed before the degree choice, so the sampling and forest errors can be chosen small enough. Choose s and then H_0 also to make the deletion term and the trimming term as small as required. Equation (8.13) now proves (8.14).

There is no physical projection selecting a height in this argument. Instead, each designated symbol has its prescribed value, including zero, at each block. On a zero block Proposition 5.2 uses zero scalar weights with diffuse voltages. The finite type and weight tags record everything needed for the finite Cayley polynomial. A parity pattern or an initial-height exception changes this finite prescription but introduces no family of new labels indexed by height. $\square$

8.4 An exact source algebra in a norm sequence quotient

Finite-profile realization gives approximations with errors tending to zero. We pass to norm sequence quotients to make these approximations exact. For construction ν let A_ν be the right regular copy of $C_r^*(\Gamma_\nu)$ on $\mathcal{L}_\nu = \ell^2(\Gamma_\nu)$. Define

$$\mathcal{Q}_r = \frac{\prod_\nu A_\nu}{\bigoplus_\nu A_\nu}, \quad \mathcal{Q}_{\text{op}} = \frac{\prod_\nu B(\mathcal{L}_\nu)}{\bigoplus_\nu B(\mathcal{L}_\nu)}.$$

Products mean bounded sequences and direct sums mean norm-null sequences. The faithful coordinate representations give an isometric inclusion $\mathcal{Q}_r \subset \mathcal{Q}_{\text{op}}$.

For the model spaces in (8.8), put

$$\mathcal{M} = \frac{\prod_\nu B(\mathcal{K}_\nu)}{\bigoplus_\nu B(\mathcal{K}_\nu)}.$$

Using the normalized-constant embeddings already constructed, set

$$V_\nu = U_\nu D_\nu \mathcal{I}_\nu.$$

Lemma 8.1 gives

$$\|V_\nu^* V_\nu - 1_{\mathcal{K}_\nu}\| \leq \delta_\nu^2. \quad (8.16)$$

It follows that

$$\Theta: \mathcal{M} \longrightarrow \mathcal{Q}_{\text{op}}, \quad [(b_\nu)] \longmapsto [(V_\nu b_\nu V_\nu^*)] \quad (8.17)$$

is a $*$ -homomorphism. Indeed it is norm bounded and respects adjoints; the multiplicative defect is at most $\delta_\nu^2 \|b_\nu\| \|c_\nu\|$. It sends norm-null sequences to norm-null sequences. This establishes the exact algebraic map without presuming that its entire image belongs to the reduced algebra quotient.

Let $A_{\text{ev}}, A_{\text{od}}, A_{\text{ev},+}, A_0$ denote the fixed-size matrix elements of $\mathcal{M}$ represented by the four diagonal profile families specified above. The scalar matrix e acts on the fixed Bott matrix factor, identically at all heights.

The two height shifts implement the Eilenberg-swindle pattern used in coarse operator K -theory [25, Section 10]. Here we construct bounded operator families and use their homotopies through finite identities in the K -theory of the algebra they generate. The two continuous projection paths are explicit. On a pair of consecutive summands $(i, i+1)$ first use the compact difference

$$\begin{pmatrix} \cos^2(t)a_i & \cos(t)\sin(t)a_i \\ \cos(t)\sin(t)a_i & \sin^2(t)a_i \end{pmatrix}, \quad 0 \leq t \leq \pi/2. \quad (8.18)$$

Adding $\text{diag}(e, e)$ gives a projection. This follows either by multiplication or from $ea_i + a_i e + a_i^2 = a_i$ and $\cos^2(t) + \sin^2(t) = 1$. The off-diagonal function is supported in the intersection of the two

carriers. This path moves a_i from summand i to summand $i + 1$. Then apply (8.5) on summand $i + 1$ to replace a_i by a_{i+1} . Apply this simultaneously to the disjoint pairs $(0, 1), (2, 3), \dots$ to obtain a path from the even to the odd family, and to $(1, 2), (3, 4), \dots$ to obtain a path from the odd to the positive-even family. Their norms and moduli of continuity are uniform in both i and ν : the pairs are orthogonal, and all rescaling ratios are fixed by b . The displayed two-by-two blocks use the two height summands of $\mathcal{K}_\nu$; the external Bott matrix size is still m .

Define the exact source algebra $\mathcal{B}$ to be the C^* -subalgebra of $\mathcal{M}$ generated by the scalar matrix entries of these four compact differences and of the two paths just defined. Thus this is a specified algebra generated by bounded operator families; it is not a formal infinite sum of K -classes. It suffices to use the path values at rational parameters, since the paths are norm continuous. Treat $\mathcal{B}$ as a possibly nonunital algebra and use its external unitization

$$\mathcal{B}^+ = \mathcal{B} \oplus \mathbb{C}, \quad (b, \lambda)(c, \mu) = (bc + \lambda c + \mu b, \lambda\mu).$$

All the displayed projections belong to a fixed matrix algebra over $\mathcal{B}^+$, with scalar part e (or its indicated block sum).

Lemma 8.5 (Reduced membership of the source algebra). *The constructions can be chosen so that $\Theta(\mathcal{B}) \subset \mathcal{Q}_r$.*

Proof. Take any sequence $\eta_\nu \downarrow 0$. At stage ν , first choose the extraction length from the fixed gap. Next choose finite meshes of the two uniform model paths, with mesh error at most η_ν . Collect their finitely many scalar matrix profiles, the four endpoint profiles, and the larger cutoffs. Apply Lemma 8.4 to that finite family, with errors sufficiently small compared with η_ν . Its finite type and weight choices precede s_ν ; choose s_ν and then $H_{0,\nu}$ to make all errors tend to zero. In particular require both δ_ν and the right-hand side of (8.11) to be at most η_ν . The value of b remains fixed.

Each endpoint family now has an actual bounded sequence in a fixed matrix algebra over A_ν with the same $\mathcal{Q}_{\text{op}}$ class as its Θ image. For an arbitrary path parameter, linearly interpolate the physical mesh values. The interpolated sequence differs from the Θ image of the model path by a norm-null sequence, uniformly in the parameter. The uniform model modulus proves that this is a continuous path in the quotient. Thus every generating path value has image in $\mathcal{Q}_r$. This latter algebra is a closed C^* -subalgebra of $\mathcal{Q}_{\text{op}}$, so it contains $\Theta(\mathcal{B})$.

Only a finite profile family was sampled at each stage. Uniform approximation of the specified compact parameter spaces supplies every path value in the quotient. No claim about arbitrary coordinatewise continuous paths is used. $\square$

The restricted homomorphism extends by

$$\Theta^+ : \mathcal{B}^+ \longrightarrow \mathcal{Q}_r, \quad \Theta^+(b, \lambda) = \Theta(b) + \lambda 1_{\mathcal{Q}_r}.$$

This is the only unit extension used. It does not claim that the identity operator on all active model carriers has been realized in the reduced algebra.

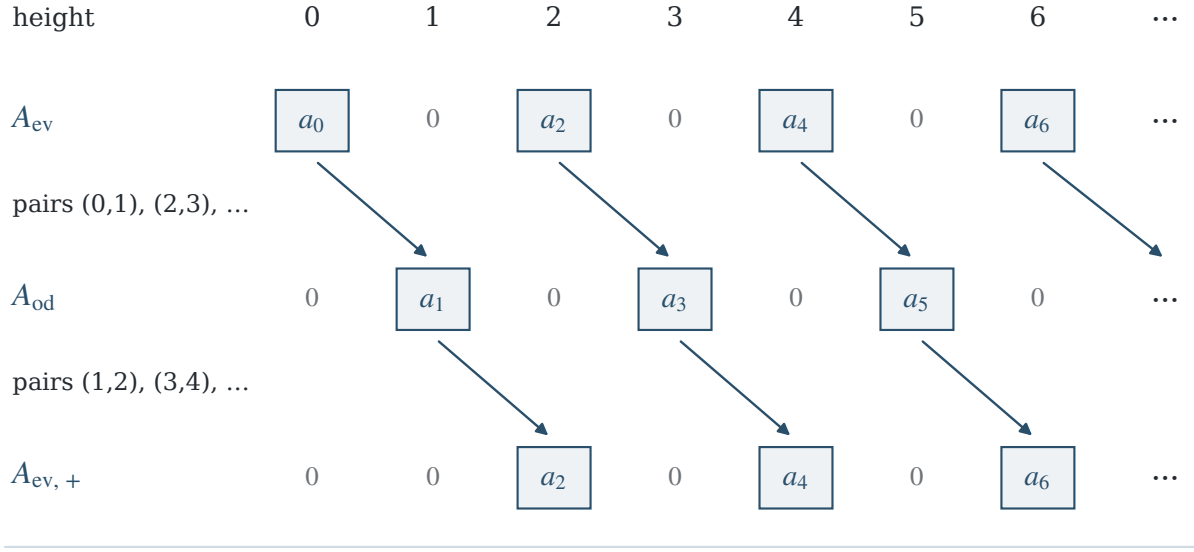
8.5 Finite parity identities in the source algebra

Write

$$b_I = [e + A_I] - [e] \in K_0(\mathcal{B}), \quad I \in \{\text{ev}, \text{od}, (\text{ev}, +), 0\}.$$

The common scalar rank makes each difference an element of the kernel of $K_0(\mathcal{B}^+) \rightarrow K_0(\mathbb{C})$, which is the definition of $K_0(\mathcal{B})$. The two explicit paths give

$$b_{\text{ev}} = b_{\text{od}} = b_{\text{ev},+}. \tag{8.19}$$



$$b_{\text{ev}} = b_{\text{od}} = b_{\text{ev},+} \quad b_{\text{ev}} = b_0 + b_{\text{ev},+} \quad \implies \quad b_0 = 0$$

$b_I = [e + A_I] - [e] \in K_0(\mathcal{B})$; finite additivity splits off height 0.

Figure 4: Height supports of three bounded operator families. Each arrow represents the rotation in (8.18) followed by rescaling from a_i to a_{i+1} . The disjoint pairs give the two homotopies. Finite additivity splits off the root family. Ellipses describe the supports of actual operators in $\mathcal{B}$; they do not denote an infinite sum of K -classes. A zero entry is a zero compact difference; the full projection retains its scalar part.

We give the finite-additivity step explicitly. If compact self-adjoint differences a', a'' satisfy $a'a'' = a''a' = 0$ and $e + a', e + a''$ are projections, then

$$\begin{pmatrix} e + a' + \cos^2(t)a'' & \cos(t)\sin(t)a'' \\ \cos(t)\sin(t)a'' & e + \sin^2(t)a'' \end{pmatrix}$$

is a projection path from $\text{diag}(e + a' + a'', e)$ to $\text{diag}(e + a', e + a'')$. Direct multiplication uses the two projection relations and the two zero products. Apply this to $a' = A_0$ and $a'' = A_{\text{ev},+}$, whose height supports are disjoint. It follows that

$$b_{\text{ev}} = b_0 + b_{\text{ev},+}.$$

Together with (8.19), this proves

$$b_0 = 0 \quad \text{in } K_0(\mathcal{B}). \tag{8.20}$$

There are only finitely many additions and subtractions of K -classes here. The all-height objects are actual elements of the specified source algebra, and Lemma 8.5 has proved their reduced quotient membership.

Figure 4 displays the three supports.

8.6 The physical root and a supported projection homotopy

Let $J_{0,\nu}: e_{\nu 0}^{\text{car}} \mathcal{L}_\nu \rightarrow \mathcal{K}_\nu$ be inclusion in its zeroth summand. If n_0^B is the number of surviving $\mathcal{B}_0$ vertices, define the finite operator

$$S_\nu = (n_0^B)^{-1/2} \sum_{v \in \mathcal{B}_0^{\text{surv}}} R(t_v^{-1}) T_0, \quad v_\nu = S_\nu \chi_0 \in A_\nu. \quad (8.21)$$

All sums are finite. The unrestricted norm of S_ν need not be uniformly bounded. For a source coset $h\mathbf{H}$, however, $R(t_v^{-1})\delta_{hf} = \delta_{hft_v}$, so (8.21) is exactly the normalized constant transport in (8.1), after spreading. Thus on the localized source space

$$v_\nu = \mathcal{J}\mathcal{I}_\nu J_{0,\nu} \chi_0.$$

Lemma 8.2, or just its single-height estimate, gives

$$\|v_\nu - V_\nu J_{0,\nu} \chi_0\| \leq C_\nu 2^{-344.5H_{0,\nu}} \rightarrow 0. \quad (8.22)$$

In particular the v_ν form a uniformly bounded sequence, because $\|V_\nu\| \leq 1$ and $\|\chi_0\| \leq 1$. The factor n_0^B has been accounted for by the multiplicity estimate, rather than suppressed by assuming that untrimmed sheet summation is an isometry.

Let $a_\nu^{\text{phys}} = a_0(R(z), R(x))$ and let $p_\nu^{\text{phys}} = e + a_\nu^{\text{phys}}$. These are fixed-size matrices and the latter are exact projections in $M_m(A_\nu)$. Write $a^{\text{phys}} = [(a_\nu^{\text{phys}})]$ and $v = [(v_\nu)]$ in the reduced quotient, tensoring v with I_m when it acts on Bott matrices. The cutoff is real, supported inside the controlled source carrier, and is one on $\text{supp } a_\nu^{\text{phys}}$. Equations (8.16) and (8.22) imply

$$v^* v a^{\text{phys}} = a^{\text{phys}} = a^{\text{phys}} v^* v, \quad \Theta(A_0) = v a^{\text{phys}} v^*. \quad (8.23)$$

For example $v_\nu^* v_\nu - \chi_0^2$ is norm null after restricting to the cutoff's carrier, and multiplication by a_ν^{phys} removes both cutoffs. The second equality follows by multiplying (8.22) on both sides of the compact difference. This argument uses a cutoff supported in the larger controlled carrier; merely asking that a cutoff equal one on the smaller Bott support would not by itself control its remaining norm.

Lemma 8.6 (Supported Bott transport). *Let B be a C^* -algebra, $e \in M_m(\mathbb{C})$ a projection, and $a = a^* \in M_m(B)$ such that $e + a$ is a projection in $M_m(B^+)$. Suppose $v \in B$ is bounded, acts as $v \otimes I_m$ on the matrix factor, and satisfies $v^* v a = a$. Then*

$$[e + a] - [e] = [e + v a v^*] - [e] \quad \text{in } K_0(B).$$

No contraction assumption on v or identity projection for a carrier is needed.

Proof. The adjoint relation is $av^*v = a$. Put $c = \cos t$, $s = \sin t$ and

$$P_t = \begin{pmatrix} e + c^2 a & cs a v^* \\ cs v a & e + s^2 v a v^* \end{pmatrix}, \quad 0 \leq t \leq \pi/2. \quad (8.24)$$

It is self-adjoint. With $W_t = (cI_m, sv)^T$ and $E_{\text{sc}} = \text{diag}(e, e)$, this is $E_{\text{sc}} + W_t a W_t^*$. The scalar-factor assumption gives $E_{\text{sc}} W_t = W_t e$. Moreover $a W_t^* W_t a = a^2$ and $ea + ae + a^2 = a$. Multiplication now gives $P_t^2 = P_t$. The endpoints are $\text{diag}(e + a, e)$ and $\text{diag}(e, e + v a v^*)$, proving the equality after one fixed stabilization. All nonscalar entries belong to $M_{2m}(B)$ and the path is norm continuous. Its modulus depends only on $\|a\|$ and $\|v\|$. In particular it is uniform for bounded coordinate sequences. $\square$

Apply Lemma 8.6 to (8.23). By (8.20) and the homomorphism Θ^+ ,

$$[(p_\nu^{\text{phys}})] - [e] = 0 \quad \text{in } K_0(\mathcal{Q}_r). \quad (8.25)$$

The size in (8.24) is $2m$, independently of the number of transports, graph layers, sheets, or construction index.

8.7 From the quotient to one finitely generated group

The remaining two steps recover finite witnesses. First, equality in the norm sequence quotient gives equality at every sufficiently late coordinate. Then the group elements needed to approximate one such witness generate the required subgroup. These are concrete forms of projection stability and continuity in C^* -algebra K -theory [9, Sections 4.5 and 5.2, Theorem 5.6.1]; we give the witness constructions to keep their finite matrix sizes explicit.

Lemma 8.7 (Eventual coordinate equality). *Let B_ν be unital C^* -algebras and let $P_\nu, Q_\nu \in M_m(B_\nu)$ be projections in a fixed size. If their classes agree in $K_0((\prod B_\nu)/(\oplus B_\nu))$, then $[P_\nu] = [Q_\nu]$ in $K_0(B_\nu)$ for all sufficiently large ν .*

Proof. The definition of K_0 supplies a projection R in a finite matrix algebra over the quotient and, after finite zero padding, a partial isometry w with

$$w^*w = P \oplus R =: E, \quad ww^* = Q \oplus R =: F.$$

Lift R to a bounded self-adjoint sequence r_ν . Its idempotence defect tends to zero. On a tail its spectrum avoids $1/2$, and spectral cutting gives projections R_ν with $\|R_\nu - r_\nu\| \rightarrow 0$. Form the corresponding E_ν, F_ν and lift w to a bounded sequence b_ν . Set $z_\nu = F_\nu b_\nu E_\nu$. Then

$$\|z_\nu^* z_\nu - E_\nu\| \rightarrow 0, \quad \|z_\nu z_\nu^* - F_\nu\| \rightarrow 0.$$

For large ν these products are invertible in the respective corners. The element $z_\nu(z_\nu^* z_\nu)^{-1/2}$ is a partial isometry with initial projection E_ν and final projection F_ν ; the latter assertion uses invertibility in the F_ν corner. Thus $[P_\nu] + [R_\nu] = [Q_\nu] + [R_\nu]$, and cancellation in $K_0(B_\nu)$ proves the result. Every stabilization used is finite and fixed after the quotient witness has been chosen. $\square$

Proposition 8.8 (Reduced Bott vanishing). *For all sufficiently large ν in the chosen compatible sequence, the image of the rank-zero Bott generator of $K_0(C_r^*(\langle z, x \rangle))$ is zero integrally in $K_0(C_r^*(\Gamma_\nu))$.*

Proof. Apply Lemma 8.7 to (8.25). At every coordinate (8.4) identifies the difference with the physical torus Bott generator. Equation (8.2) transfers the equality to the usual left reduced algebra without changing the subgroup elements or their functional-calculus projections. $\square$

Lemma 8.9 (A finite witness lies in a finitely generated subgroup). *Let $H \leq \Gamma$ be finitely generated, and let P, Q be fixed finite matrix projections over $C_r^*(H)$. If their images have equal K_0 classes over $C_r^*(\Gamma)$, then some finitely generated subgroup G_{inj} with $H \leq G_{\text{inj}} \leq \Gamma$ already has $[P] = [Q]$ in $K_0(C_r^*(G_{\text{inj}}))$.*

Proof. Choose one finite stabilization projection and a partial isometry witnessing the equality over $C_r^*(\Gamma)$. Approximate their finitely many entries by group polynomials, using self-adjoint approximations for the projection. Adjoin all group elements in these finite supports to generators of H ; they generate a finitely generated subgroup G_{inj} . The reduced inclusion $C_r^*(G_{\text{inj}}) \rightarrow C_r^*(\Gamma)$ is isometric, since restriction of the regular representation of Γ to G_{inj} is a direct sum of regular

representations of G_{inj} . Thus the required small projection and partial-isometry defects have the same bounds in $C_r^*(G_{\text{inj}})$. Spectral cutting gives the stabilizing projection in $C_r^*(G_{\text{inj}})$; compressing the approximate partial isometry to the two corners and polar-normalizing it gives an exact witness there, just as in Lemma 8.7. Functional calculus stays inside $C_r^*(G_{\text{inj}})$. The original projections P, Q were retained exactly throughout. $\square$

9 Compatible choices and the conclusion

We choose parameters satisfying all the preceding estimates, then combine the resulting integral detector and reduced Bott vanishing to prove Theorem 1.1. The two indices have different roles: $i \geq 0$ is a height in one infinite tower; $\nu \geq 1$ indexes a sequence of increasingly accurate constructions. Degrees are independent of i . No uniform degree bound in ν is required.

9.1 The order of parameters

First fix the integer b with $\sqrt{5 \cdot 2^{1-b}} < 3/5$. This gives the height-independent gap $\rho = 1 - .05^2/2$ of Proposition 3.4. Retain the explicit compact Bott projection in (5.2), its fixed signed class, and its matrix size. Fix enlarged support profiles for all neighboring scales; their ratios depend on b , which remains fixed for every ν . The rescaling and rotation paths therefore have one common modulus of continuity.

For each ν , take a target error $\epsilon_\nu \downarrow 0$. Perform the following finite choices in the indicated order.

1. Choose the constant-extraction length k_ν from the fixed gap, so that the term ρ^{2k_ν} in (8.13) is sufficiently small. Choose finite meshes of the two projection paths. The four endpoint families, those mesh profiles, and their cutoffs specify a finite collection of same-height and adjacent-height types. Choose all per-factor error tolerances small enough that the telescoping bound in Lemma 8.4 is below ϵ_ν .
2. Choose the finite Fourier truncations and finite matrix-weight net of Lemma 5.4. Its support exponents, the finite regular-norm test exponent, and the moment constant in Lemma 5.9 are now fixed. Also fix a desired deletion fraction small enough that $k_\nu \sqrt{\epsilon_{\text{del}}}$ has the required error bound.
3. Choose s_ν as a sufficiently large multiple of 20000 and $5b$. Besides the coefficient, probability, and moment requirements, impose

$$\frac{s_\nu}{b} \geq 10^{12}, \quad \frac{b}{.50005s_\nu} \leq 10^{-6}, \quad 2^{-.000025s_\nu + O(\log s_\nu) + O_\nu(1)} \ll \epsilon_\nu.$$

The last estimate uses the actual refined alphabet, after the finite type and weight choices. Its exponent has a strictly negative linear term, so it can be imposed along with every previous finite requirement. The uniform gate budget and the height-separation term depending only on s_ν, b can be made smaller than the allocated deletion budget at this step.

4. Choose $n_{0,\nu}$, hence $H_{0,\nu} = (2n_{0,\nu} + 1)s_\nu$, sufficiently large. Impose $H_{0,\nu}/s_\nu \geq 10^{12}$ and all initial-height requirements in Sections 2–8. In particular make the remaining separation budget small, make the sum of the expected pruning fractions smaller than its allocation, and make the sum of all voltage and norm failure probabilities less than one. Increase $H_{0,\nu}$

further to make both the domain-trimming error and the physical-synthesis error less than their allocations. Their bounds have the forms

$$C_\nu 2^{-345H_{0,\nu}}, \quad C_\nu \frac{2^{-344.5H_{0,\nu}}}{1 - 2^{-344.5b}},$$

where C_ν is already independent of height and of this final increase. Thus these requirements are compatible.

We explain the choices of random data after these parameters. Choose the coefficient pools once as in Lemma 2.1; the same finite pools are reused on all full blocks. The sequential law then gives expanding auxiliary incidences for every table realization in its support. The deletion estimates in Proposition 2.6 hold for every such realization. Proposition 4.14 selects a realization with all further layer deletions small. Only after fixing this old graph apply Proposition 5.2. Its one independent voltage-and-weight law controls both kinds of failure before their simultaneous avoidance. There is consequently no change of measure hidden in the centered moment estimate.

The finite label set is unchanged as heights increase. It is refined only by the chosen finite weight net. The private completion in Lemma 6.1 adds a countable alphabet, whose coefficients in every operator polynomial are zero. The resulting graphical group Γ_ν is countable; finite generation is obtained from a finite K -theory witness after the quotient argument.

This order also avoids a potential circularity in the norm sequence quotient. At stage ν only finitely many profiles are required. Their meshes become finer with ν , and their uniform modulus of continuity supplies the complete paths in the quotient. One does not choose a separate nonuniform approximation for each height or each path parameter.

9.2 Proof of the main result

Proof of Theorem 1.1. Make the sequence of compatible constructions just described. Proposition 6.4 embeds $\mathbf{H}$ in every Γ_ν . Proposition 7.11 proves torsion-freeness and provides an integral class whose evaluation on the ordered torus (z, x) is -1 . Proposition 8.8 proves that, for all sufficiently large ν , the rank-zero Bott class of this torus is zero in $K_0(C_r^*(\Gamma_\nu))$.

Fix one such ν . Apply Lemma 8.9 with $H = \mathbf{H}$ and the physical Bott projections to obtain a finitely generated subgroup $\mathbf{H} \leq G_{\text{inj}} \leq \Gamma_\nu$ in which that equality already holds. The subgroup G_{inj} is torsion-free. Restrict the negative of the detector class to BG_{inj} , and denote it by c . Let $i : \mathbb{Z}^2 \hookrightarrow G_{\text{inj}}$ be the inclusion with ordered generators (z, x) . The detector evaluation is unchanged by restriction, so

$$\langle (Bi)^*c, [T^2] \rangle = 1, \quad i_*(\beta) = 0 \quad \text{in } K_0(C_r^*(G_{\text{inj}})).$$

It remains to detect the corresponding geometric class and identify its assembly image. We use compactly supported K -homology throughout. On finite CW complexes, the natural comparison of geometric cycles with analytic K -homology sends (M, E, f) to $f_*[D_E]$ [7, Theorems 6.1–6.2]; passage to finite-subcomplex colimits gives the convention used for BG_{inj} . In particular the compact torus cycle used here defines a class without any compactness assumption on BG_{inj} .

Let L be the line bundle on BG_{inj} classified by c . The naturality of the index pairing follows at a finite stage from pullback of vector bundles and associativity of the Kasparov product:

$$[L] \otimes f_*[D] = [f^*L] \otimes [D], \quad f = Bi.$$

The same identity holds for the virtual bundle $[L] - [1]$. Applying the twisted spin Dirac index

theorem [3], in the dimension-two form of [8, Theorem 1], gives

$$\begin{aligned}\langle [L] - [1], (Bi)_*[D_{\mathbb{T}^2}] \rangle &= \text{index}(D_{\mathbb{T}^2} \otimes (Bi)^*L) - \text{index}(D_{\mathbb{T}^2}) \\ &= \langle (Bi)^*c, [\mathbb{T}^2] \rangle = 1.\end{aligned}$$

The torus tangent bundle is trivial, its $\widehat{A}$ class is one, and its untwisted complex Dirac index is zero. Thus the pushed-forward K -homology class has infinite order and is nonzero after tensoring with $\mathbb{Q}$.

Under Fourier transform $C_r^*(\mathbb{Z}^2) \cong C(\mathbb{T}^2)$, the assembly image of the torus spin Dirac class is the rank-zero Bott generator, up to orientation. This is the full-torus case of Emerson–Hudson’s assembly formula for a K -oriented subtorus [12, Theorem 1.3]: the dual subtorus is a point, whose shriek class is the compact Bott class.

For completeness we verify the required subgroup naturality at this particular cycle. Put $A = C_r^*(\mathbb{Z}^2)$, $B = C_r^*(G_{\text{inj}})$, and write $j : A \rightarrow B$ for the isometric inclusion. If $P \rightarrow T^2$ is the principal $\mathbb{Z}^2$ -bundle classified by the torus map, its Miščenko bundle is $E_A = P \times_{\mathbb{Z}^2} A$. The bundle obtained after the subgroup map is canonically

$$(P \times_{\mathbb{Z}^2} G_{\text{inj}}) \times_{G_{\text{inj}}} B \cong P \times_{\mathbb{Z}^2} B \cong E_A \otimes_j B.$$

The last identification sends $([p, a], b)$ to $[p, j(a)b]$; the balancing relations and fiber inner products agree because j sends each subgroup unitary to the same group unitary. Extension of coefficients therefore commutes with the index Kasparov product. In explicit notation,

$$([E_A] \otimes_{C(T^2) \otimes A} ([D_{T^2}] \boxtimes 1_A)) \otimes_A [j] = [E_A \otimes_j B] \otimes_{C(T^2) \otimes B} ([D_{T^2}] \boxtimes 1_B).$$

These indices are the assembly classes by the Miščenko comparison [26, Corollary 1.4 and diagram (1.6)]; see also [28, Proposition 3.7 and Theorem 4.1]. The comparison applies to the compact torus cycle, and its finite-subcomplex colimit handles the target BG_{inj} . Consequently

$$\mu_{G_{\text{inj}}}^r((Bi)_*[D_{T^2}]) = j_*\mu_{\mathbb{Z}^2}^r([D_{T^2}]) = 0$$

by Proposition 8.8 and Lemma 8.9. This is integral vanishing, whereas the index pairing above proves infinite order of the source class. $\square$

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